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On Zero-Initialized Attention: Optimal Prompt and Gating Factor Estimation

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Abstract

LLaMA-Adapter has recently emerged as an efficient fine-tuning technique for LLaMA models, leveraging zero-initialized attention to stabilize training and enhance performance. However, despite its empirical success, the theoretical foundations of zero-initialized attention remain largely unexplored. In this paper, we provide a rigorous theoretical analysis, establishing a connection between zero-initialized attention and mixture-of-expert models. We prove that both linear and non-linear prompts, along with gating functions, can be optimally estimated, with non-linear prompts offering greater flexibility for future applications. Empirically, we validate our findings on the open LLM benchmarks, demonstrating that non-linear prompts outperform linear ones. Notably, even with limited training data, both prompt types consistently surpass vanilla attention, highlighting the robustness and adaptability of zero-initialized attention. Our implementation is publicly available on GitHub.

1. Introduction

Large Language Models (LLMs) have revolutionized natural language processing (Radford et al., 2019; Raffel et al., 2020; Anil et al., 2023; Minaee et al., 2024), demonstrating remarkable understanding and generative capabilities across various tasks. While proprietary models like ChatGPT (OpenAI, 2025) and GPT-4 (OpenAI et al., 2024) set new records in instruction-following performance (Peng et al., 2023), their closed-source nature and high development costs limit accessibility. To address this, several efforts have explored fine-tuning open-source LLMs, e.g., LLaMA (Touvron et al., 2023a), on large-scale instruction datasets where small human-annotated samples are expanded into massive training corpora using self-instruct methods (Taori et al., 2023). This approach has enabled models like LLaMA to achieve instruction-following performance comparable to GPT-3.5 and, with further advancements, shows promise in approaching GPT-4's capabilities. However, full fine-tuning of large-scale LLMs remains computationally intensive, highlighting the need for more efficient adaptation techniques to unlock their full potential across diverse downstream applications.

To address this challenge, researchers have increasingly turned to parameter-efficient fine-tuning (PEFT) techniques, which update only a small subset of parameters while keeping most of the backbone frozen, enabling efficient and scalable adaptation (Houlsby et al., 2019; Lester et al., 2021; Hu et al., 2021). Among these approaches, the LLaMA-Adapter (Zhang et al., 2024) has emerged as a specialized PEFT method for transforming LLaMA into an instruction-following model, demonstrating strong performance across multiple benchmarks. Beyond adaptation prompts, its key innovation lies in the *introduction of zero-initialized attention with zero gating*, which facilitates the seamless integration of new instructional prompts while preserving the model's pre-existing knowledge. However, despite its empirical success, the theoretical foundations of zero-initialized attention remain largely unexplored, limiting a deeper understanding of its underlying mechanisms and potential for further advancements.

In this work, we conduct a rigorous theoretical and empirical investigation into zero-initialized attention, showing that it is not merely an engineering trick. Building on recent findings (Le et al., 2024; 2025) that draw connections between the attention mechanism (Vaswani, 2017) and the mixture of experts (MoE) model (Jacobs et al., 1991; Jordan & Jacobs, 1994; Shazeer et al., 2017), we investigate how zero-initialized attention in the LLaMA-Adapter (Zhang et al., 2024) can be interpreted within this framework. Specifically, we show that *zero-initialized attention can be viewed as a special formulation of the MoE model* and that under lin-

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关于零初始化 zed Attention：最优提示词与门控因子估算

Nghiem T. Diep [*1 2 3] **Huy Nguyen** [*4] **Chau Nguyen** [*5 6] **Minh Le** [5] **Duy M. H. Nguyen** [1 7 8] **Daniel Sonntag** [1 9] **Mathias Niepert** [7 8] **Nhat Ho** [4]

摘要

LLaMA-Adapter has recently emerged as an efficient fine-tuning technique for LLaMA models, leveraging zero-initialized attention to stabilize training and enhance performance. However, despite its empirical success, the theoretical foundations of zero-initialized attention remain largely unexplored. In this paper, we provide a rigorous theoretical analysis, establishing a connection between zero-initialized attention and mixture-of-expert models. We prove that both linear and non-linear prompts, along with gating functions, can be optimally estimated, with non-linear prompts offering greater flexibility for future applications. Empirically, we validate our findings on the open LLM benchmarks, demonstrating that non-linear prompts outperform linear ones. Notably, even with limited training data, both prompt types consistently surpass vanilla attention, highlighting the robustness and adaptability of zero-initialized attention. Our implementation is publicly available on GitHub.

1. 引言

Large Language Models (LLMs) have revolutionized natural language processing (Radford et al., 2019; Raffel et al., 2020; Anil et al., 2023; Minaee et al., 2024), demonstrating remarkable understanding and generative capabilities across various tasks. While proprietary models like ChatGPT (OpenAI, 2025) and GPT-4 (OpenAI et al., 2024) set new records in instruction-following performance (Peng et al., 2023), their closed-source nature and high development costs limit accessibility. To address this, several efforts have explored fine-tuning open-source LLMs, e.g., LLaMA (Touvron et al., 2023a), on large-scale instruction datasets where small human-annotated samples are expanded into massive training corpora using self-instruct methods (Taori et al., 2023). This approach has enabled models like LLaMA to achieve instruction-following performance comparable to GPT-3.5 and, with further advancements, shows promise in approaching GPT-4's capabilities. However, full fine-tuning of large-scale LLMs remains computationally intensive, highlighting the need for more efficient adaptation techniques to unlock their full potential across diverse downstream applications.

To address this challenge, researchers have increasingly turned to parameter-efficient fine-tuning (PEFT) techniques, which update only a small subset of parameters while keeping most of the backbone frozen, enabling efficient and scalable adaptation (Houlsby et al., 2019; Lester et al., 2021; Hu et al., 2021). Among these approaches, the LLaMA-Adapter (Zhang et al., 2024) has emerged as a specialized PEFT method for transforming LLaMA into an instruction-following model, demonstrating strong performance across multiple benchmarks. Beyond adaptation prompts, its key innovation lies in the *introduction of zero-initialized attention with zero gating*, which facilitates the seamless integration of new instructional prompts while preserving the model's pre-existing knowledge. However, despite its empirical success, the theoretical foundations of zero-initialized attention remain largely unexplored, limiting a deeper understanding of its underlying mechanisms and potential for further advancements.

In this work, we conduct a rigorous theoretical and empirical investigation into zero-initialized attention, showing that it is not merely an engineering trick. Building on recent findings (Le et al., 2024; 2025) that draw connections between the attention mechanism (Vaswani, 2017) and the mixture of experts (MoE) model (Jacobs et al., 1991; Jordan & Jacobs, 1994; Shazeer et al., 2017), we investigate how zero-initialized attention in the LLaMA-Adapter (Zhang et al., 2024) can be interpreted within this framework. Specifically, we show that *zero-initialized attention can be viewed as a special formulation of the MoE model* and that under lin-

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ear prompt settings, both the prompts and the gating factor can be optimally estimated, offering significant statistical benefits. Furthermore, we *extend this analysis to scenarios in which the trainable prompts are nonlinear (e.g., implemented via an MLP)* and prove that the optimal estimation of the prompts and gating parameters remains achievable, thus enhancing the flexibility for future applications.

Our empirical evaluations further substantiate these theoretical findings. Across multiple datasets in the open LLM benchmark (Beeching et al., 2024), zero-initialized attention consistently demonstrates better performance than random-initialized attention. Additionally, non-linear prompts exhibit improved results compared to their linear counterparts, thereby offering the potential to boost the LLaMA-Adapter's capabilities. Notably, on various small rates of limited training data, both linear and non-linear prompts combined with zero-initialized attention substantially outperform standard attention, reinforcing our theoretical findings and demonstrating the robustness of the proposed approach.

Contribution. Our contributions are four-fold can be summarized as follows: **(i)** We develop a theoretical framework that examines the connection between zero-initialized attention and MoE models; **(ii)** We demonstrate the statistical advantages of zero-initialized attention over conventional attention, enabling optimal estimation of prompt parameters and gating factors; **(iii)** We extend our analysis to accommodate non-linear prompts, offering increased flexibility for a broader range of applications; **(iv)** Finally, extensive experiments on multiple question-answering datasets validate our theoretical insights, highlighting the robustness and significance of our findings.

Organization. The remainder of this paper is organized as follows: Section 2 reviews related work. In Section 3, we establish the connection between zero-initialized attention and the mixture-of-experts model. Section 4 presents our theoretical results. In Section 4.2, we extend the analysis for the non-linear setting and outline the main algorithm for the non-linear case, followed by empirical evaluations in Section 5. Finally, Section 6 discusses limitations and future research directions. Detailed proofs and additional experimental results are provided in the Appendix.

Notation. For any $n \in \mathbb{N}$, we denote $[n]$ as the set $\{1, 2, \ldots, n\}$. For any set S, we let $|S|$ be its cardinality. For any vectors $u := (u_1, u_2, \ldots, u_d) \in \mathbb{R}^d$ and $\alpha := (\alpha_1, \alpha_2, \ldots, \alpha_d) \in \mathbb{N}^d$, we let $u^\alpha = u_1^{\alpha_1} u_2^{\alpha_2} \ldots u_d^{\alpha_d}$, $|u| := u_1 + u_2 + \ldots + u_d$, and $\alpha! := \alpha_1! \alpha_2! \ldots \alpha_d!$, while $\|u\|$ denotes the ℓ_2-norm value of u. Lastly, for any two positive sequences $(a_n)_{n\geq 1}$ and $(b_n)_{n\geq 1}$, we write $a_n = \mathcal{O}(b_n)$ or $a_n \lesssim b_n$ if $a_n \leq C b_n$ for all $n \in \mathbb{N}$, where $C > 0$ is some constant. Finally, $a_n = \mathcal{O}_P(b_n)$ means a_n / b_n is stochastically bounded.

2. Related Work

Parameter Efficient Fine-Tuning for LLMs. Large foundational models have demonstrated remarkable generalization capabilities across a wide range of tasks (Devlin, 2018; Brown et al., 2020; Touvron et al., 2023b;c). However, as model sizes grow exponentially, fine-tuning all parameters in a large-scale model becomes increasingly impractical. In contrast, PEFT techniques (Hu et al., 2021; Karimi Mahabadi et al., 2021; Gao et al., 2024; Houlsby et al., 2019; Li et al., 2024c; Li & Liang, 2021; Zhou et al., 2022; Mangrulkar et al., 2022) have emerged as a promising strategy to adapt these models to downstream tasks while freezing most of the backbone's parameters. These approaches can be categorized into three directions, namely (i) *low-rank decomposition* like LoRA (Hu et al., 2021) and its advanced versions (Karimi Mahabadi et al., 2021; Zhang et al., 2023); (ii) *adapters* which insert lightweight adaptation modules into each block of the transformer and have been applied across numerous domains (Gao et al., 2024; Houlsby et al., 2019; Li et al., 2024c); and (iii) *prompt tuning* where trainable tokens are appended to the input embeddings (Lester et al., 2021; Nguyen et al., 2024b) or at intermediate layers, i.e., pre-fixed tuning (Liu et al., 2021; Li & Liang, 2021; Shi & Lipani, 2023).

Unlike the aforementioned PEFT methods, the LLaMA-Adapter (Zhang et al., 2024) is specifically designed to enhance instruction-following capabilities, where the model learns to generate contextually relevant responses based on natural language instructions. This is done by introducing a concept of zero-initialized attention to integrate new instructional prompts while preserving the model's existing knowledge. Through this mechanism, the algorithm starts with minimal impact and prevents training instability and catastrophic forgetting by selectively activating relevant information while allowing the model to incorporate instructions incrementally. In this work, we conduct a comprehensive theoretical and empirical investigation into zero-initialized attention, demonstrating that it is more than just an engineering trick and uncovering its fundamental properties and advantages.

Mixture of Experts (MoE) in PEFT. Recent research has explored MoE in PEFT to enhance the adaptability of large pre-trained models while minimizing computational costs. MoE-based approaches, such as Switch Transformers (Fedus et al., 2022) leveraged sparse activation of expert networks to achieve efficient scaling. In the context of PEFT, techniques like AdapterDrop (Rücklé et al., 2020) and LoRA (Hu et al., 2021) have been combined with MoE to dynamically allocate resources to task-specific experts, reducing the number of trainable parameters (Li et al., 2024a; Chen et al., 2024; 2023). These works demonstrate that MoE can significantly enhance parameter efficiency without compro-

在ear promptsettings框架下，既可以对提示词本身也可以对门控因子进行最优估计，从而带来显著的统计优势。此外，我们将这一分析扩展到可训练提示词为非线性形式的情况（例如通过多层感知机来实现），并证明了仍然能够实现对提示词及门控参数的最优估计，进而提升了其在未来应用中的灵活性。

我们的实证评估进一步印证了这些理论研究结果。在开放型大语言模型基准测试中的多个数据集上（Beeching等人，2024年），采用零初始化注意力机制的模型始终展现出优于随机初始化注意力机制的性能。此外，非线性提示词相较于其线性对应物也能带来更好的效果，从而为提升LLaMA-Adapter的性能潜力提供了可能。值得注意的是，在训练数据量较少的各种场景下，无论是线性提示词还是非线性提示词，与零初始化注意力机制结合使用，其性能都远超传统注意力机制，这不仅进一步巩固了我们的理论发现，也体现了所提出方法的稳健性。

主要贡献。我们的贡献共有四点，可概括如下：**(i)** 我们构建了一个理论框架，用于研究零初始化注意力机制与MoE模型之间的关联；**(ii)** 我们证明了零初始化注意力机制相较于传统注意力机制在统计上的优势，这使得提示词参数和门控因子的最优估计成为可能；**(iii)** 我们将分析范围扩展到非线性提示词，从而为更广泛的应用场景提供更高的灵活性；**(iv)** 最后，通过在多个问答数据集上进行的大量实验，我们的理论见解得到了验证，进一步凸显了这些研究成果的稳健性与重要性。

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2. 相关工作

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与前述的PEFT方法不同，LLaMA-Adapter（张等人，2024年）是专为提升模型的指令遵循能力而设计的，该模型能够根据自然语言指令学习生成与上下文相关的响应。实现这一目标的方式是引入零初始化注意力机制的概念，从而在保留模型现有知识的同时整合新的指令提示词。通过这种机制，算法以最小的影响启动训练过程，通过选择性激活相关信息来避免训练不稳定以及灾难性遗忘问题，让模型能够逐步融入新指令。在本研究中，我们对零初始化注意力机制进行了全面的理论与实证分析，证明了它并非仅仅是一种工程技巧，同时揭示了其内在的本质特性与优势。

Mixture of Experts (MoE) in PEFT. Recent research has explored MoE in PEFT to enhance the adaptability of large pre-trained models while minimizing computational costs. MoE-based approaches, such as Switch Transformers (Fedus et al., 2022) leveraged sparse activation of expert networks to achieve efficient scaling. In the context of PEFT, techniques like AdapterDrop (Rücklé et al., 2020) and LoRA (Hu et al., 2021) have been combined with MoE to dynamically allocate resources to task-specific experts, reducing the number of trainable parameters (Li et al., 2024a; Chen et al., 2024; 2023). These works demonstrate that MoE can significantly enhance parameter efficiency without compro-

mising performance, making it a promising direction for fine-tuning large-scale models.

In another line of research, the MoE framework has been leveraged in (Le et al., 2024; 2025) to investigate the convergence behavior of learnable prompt vectors in the context of prefix tuning method, which are attached to the key and value matrices of self-attention mechanism to learn downstream tasks. In particular, by showing rigorously that each row of an attention head can be represented as an MoE, they demonstrate theoretically and empirically that the prompt convergence will be significantly accelerated if there exists a shared structure among the prompt vectors. However, although the zero-initialized attention has been widely used as an PEFT method, its theoretical understanding has remained missing in the literature. To close this gap, we provide a comprehensive study on the convergence of prompt vectors within the zero-initialized attention by establishing a connection between this model and MoE in Section 3. Our theory indicates that linear prompts and non-linear prompts share the same convergence behavior and can be both optimally estimated. On the empirical side, we observe that the non-linear prompts in zero-initalized attention work favorably compared to linear prompts in several benchmark datasets.

Additional discussion on related work of the theory of mixture of experts is in Appendix B.

3. Zero-initialized Attention meets Mixture of Experts

Zero-initialized attention. We revisit the zero-initialized attention formulation introduced by Zhang et al. (2024). Let $\mathbf{X}_l = [\boldsymbol{x}_{l,1}, \ldots, \boldsymbol{x}_{l,N}]^\top \in \mathbb{R}^{N\times d}$ denote the input tokens at the l-th layer of LLaMA's transformer model, where N is the input sequence length, and d represents the feature dimensionality. Similarly, let $\mathbf{P}_l = [\boldsymbol{p}_{l,1}, \ldots, \boldsymbol{p}_{l,L}]^\top \in \mathbb{R}^{L\times d}$ denote the learnable adaptation prompt used to fine-tune the model. This prompt is concatenated with $\mathbf{X}_l$ along the token dimension, serving as a prefix, where L denotes the prompt length. For simplicity, we omit the subscript l in subsequent notations.

Suppose the model is generating the $(N+1)$-th word based on $(\mathbf{P}, \mathbf{X})$ at the l-th layer. We denote the corresponding $(N+1)$-th word token as $\boldsymbol{t} \in \mathbb{R}^d$. Within the attention mechanism, linear projection layers are applied to the input tokens, transforming them into queries, keys, and values, defined as follows:

$$\boldsymbol{Q} = \boldsymbol{t}^\top W^Q \in \mathbb{R}^{1\times d_k}, \tag{1}$$

$$\boldsymbol{K} = \left[\mathbf{P}^\top, \mathbf{X}^\top, \boldsymbol{t}\right]^\top W^K \in \mathbb{R}^{(L+N+1)\times d_k}, \tag{2}$$

$$\boldsymbol{V} = \left[\mathbf{P}^\top, \mathbf{X}^\top, \boldsymbol{t}\right]^\top W^V \in \mathbb{R}^{(L+N+1)\times d_v}, \tag{3}$$

where $W^Q \in \mathbb{R}^{d\times d_k}$, $W^K \in \mathbb{R}^{d\times d_k}$, and $W^V \in \mathbb{R}^{d\times d_v}$ are pre-trained projection matrices. The attention scores computed between $\boldsymbol{Q}$ and $\boldsymbol{K}$ before applying the softmax function are given by:

$$\boldsymbol{S} = \frac{\boldsymbol{Q}\boldsymbol{K}^\top}{\sqrt{d_k}} = [\boldsymbol{S}_P,\ \boldsymbol{S}_X] \in \mathbb{R}^{1\times(L+N+1)}, \tag{4}$$

where $\boldsymbol{S}_P \in \mathbb{R}^{1\times L}$ and $\boldsymbol{S}_X \in \mathbb{R}^{1\times(N+1)}$ denote the attention scores of L adaption prompts and $N+1$ word tokens, respectively. Rather than applying the softmax function directly, Zhang et al. (2024) propose an alternative approach. They suggest computing the softmax independently over the two components $\boldsymbol{S}_P$ and $\boldsymbol{S}_X$, and incorporating a learnable gating factor $\alpha \in \mathbb{R}$ as follows:

$$\boldsymbol{S}_g = [\mathrm{softmax}(\boldsymbol{S}_P)\cdot\tanh(\alpha),\ \mathrm{softmax}(\boldsymbol{S}_X)]\,. \tag{5}$$

This approach aims to decouple the knowledge contained within the pre-trained model and the adaptation prompts, thereby preserving the pre-trained model's original knowledge. The activation function $\tanh(\cdot)$ is used to regulate the scale of α within the range $[-1, 1]$. Consequently, the output of the zero-initialized attention mechanism can be expressed as:

$$\mathbf{y} = \boldsymbol{S}_g \cdot \boldsymbol{V} \in \mathbb{R}^{d_v}. \tag{6}$$

During training, only the prompt parameters $\mathbf{P}$ are optimized, while all other parameters of the pre-trained model remain frozen. Next, we examine how zero-initialized attention can be interpreted through the lens of the mixture of experts framework.

Connection to mixture of experts. Recent studies (Le et al., 2024; 2025) have revealed a notable connection between the attention mechanisms and the mixture of experts (MoE) architectures (Jacobs et al., 1991; Jordan & Jacobs, 1994), showing that attention can be seen as a form of MoE, where attention weights act as gating functions over token interactions. This MoE perspective provides a useful framework for analyzing zero-initialized attention by viewing its components as gates and expert responses.

Specifically, let $\boldsymbol{X} = \left[\boldsymbol{x}_1^\top, \ldots, \boldsymbol{x}_N^\top, \boldsymbol{t}^\top\right]^\top \in \mathbb{R}^{(N+1)d}$, which is the concatenation of input tokens. For $i \in [N+1]$, define $E_i \in \mathbb{R}^{d\times(N+1)d}$ such that $E_i\boldsymbol{X} = \boldsymbol{x}_i$ for $i = 1, \ldots, N$, and $E_{N+1}\boldsymbol{X} = \boldsymbol{t}$. We then introduce a set of $L+N+1$ experts $f_j : \mathbb{R}^{(N+1)d} \to \mathbb{R}^{d_v}$, defined as:

$$f_j(\boldsymbol{X}) = {W^V}^\top E_j \boldsymbol{X} = {W^V}^\top \boldsymbol{x}_j,\ j \in [N] \tag{7}$$

$$f_{N+1}(\boldsymbol{X}) = {W^V}^\top E_{N+1}\boldsymbol{X} = {W^V}^\top \boldsymbol{t} \tag{8}$$

$$f_{N+1+j'}(\boldsymbol{X}) = {W^V}^\top \boldsymbol{p}_{j'},\ j' \in [L]. \tag{9}$$

Based on equation (5), the weights $G_j : \mathbb{R}^{(N+1)d} \to \mathbb{R}$ associated with each expert are defined as follows:

性能表现优异，使其成为微调大规模模型的极具前景的方向。

In another line of research, the MoE framework has been leveraged in (Le et al., 2024; 2025) to investigate the convergence behavior of learnable prompt vectors in the context of prefix tuning method, which are attached to the key and value matrices of self-attention mechanism to learn downstream tasks. In particular, by showing rigorously that each row of an attention head can be represented as an MoE, they demonstrate theoretically and empirically that the prompt convergence will be significantly accelerated if there exists a shared structure among the prompt vectors. However, although the zero-initialized attention has been widely used as an PEFT method, its theoretical understanding has remained missing in the literature. To close this gap, we provide a comprehensive study on the convergence of prompt vectors within the zero-initialized attention by establishing a connection between this model and MoE in Section 3. Our theory indicates that linear prompts and non-linear prompts share the same convergence behavior and can be both optimally estimated. On the empirical side, we observe that the non-linear prompts in zero-initalized attention work favorably compared to linear prompts in several benchmark datasets.

关于专家混合理论相关研究的进一步讨论见附录 B。

3. 零初始化注意力机制遇上专家混合模型

Zero-initialized attention. We revisit the zero-initialized attention formulation introduced by Zhang et al. (2024). Let $\mathbf{X}_l = [\boldsymbol{x}_{l,1}, \ldots, \boldsymbol{x}_{l,N}]^\top \in \mathbb{R}^{N\times d}$ denote the input tokens at the l-th layer of LLaMA's transformer model, where N is the input sequence length, and d represents the feature dimensionality. Similarly, let $\mathbf{P}_l = [\boldsymbol{p}_{l,1}, \ldots, \boldsymbol{p}_{l,L}]^\top \in \mathbb{R}^{L\times d}$ denote the learnable adaptation prompt used to fine-tune the model. This prompt is concatenated with $\mathbf{X}_l$ along the token dimension, serving as a prefix, where L denotes the prompt length. For simplicity, we omit the subscript l in subsequent notations.

假设模型正在基于第 l层的($\mathbf{P}$、$\mathbf{X}$)来生成第$(N+1)$个单词。我们将对应的第$(N+1)$个单词标记记为 $\boldsymbol{t} \in \mathbb{R}^d$。在注意力机制中，会对输入标记应用线性投影层，将其转换为查询项、键项和值项，其定义如下：

$$\boldsymbol{Q} = \boldsymbol{t}^\top W^Q \in \mathbb{R}^{1\times d_k}, \tag{1}$$

$$\boldsymbol{K} = \left[\mathbf{P}^\top, \mathbf{X}^\top, \boldsymbol{t}\right]^\top W^K \in \mathbb{R}^{(L+N+1)\times d_k}, \tag{2}$$

$$\boldsymbol{V} = \left[\mathbf{P}^\top, \mathbf{X}^\top, \boldsymbol{t}\right]^\top W^V \in \mathbb{R}^{(L+N+1)\times d_v}, \tag{3}$$

其中 $W^Q \in \mathbb{R}^{d\times d_k}$、$W^K \in \mathbb{R}^{d\times d_k}$和 $W^V \in \mathbb{R}^{d\times d_v}$均为预训练的投影矩阵。

在应用softmax函数之前， $\boldsymbol{Q}$ 与 $\boldsymbol{K}$ 之间计算得到的注意力分数公式为：

$$\boldsymbol{S} = \frac{\boldsymbol{Q}\boldsymbol{K}^\top}{\sqrt{d_k}} = [\boldsymbol{S}_P,\ \boldsymbol{S}_X] \in \mathbb{R}^{1\times(L+N+1)}, \tag{4}$$

其中 $\boldsymbol{S}_P \in \mathbb{R}^{1\times L}$ 和 $\boldsymbol{S}_X \in \mathbb{R}^{1\times(N+1)}$ 分别表示 L 适配提示词以及 $N+1$ 词元Token的注意力分数。张等人并未直接应用softmax函数，张等人(2024年)提出了一种替代方法。他们建议分别针对 $\boldsymbol{S}_P$和 $\boldsymbol{S}_X$这两个组成部分独立计算softmax值，并按如下方式引入一个可学习的门控因子 $\alpha \in \mathbb{R}$：

$$\boldsymbol{S}_g = [\mathrm{softmax}(\boldsymbol{S}_P)\cdot\tanh(\alpha),\ \mathrm{softmax}(\boldsymbol{S}_X)]\,. \tag{5}$$

该方法旨在将预训练模型中蕴含的知识与适配提示词分离开来，从而保留预训练模型原有的知识结构。通过使用激活函数$\tanh(\cdot)$，可将 α 的数值控制在 $[-1, 1]$ 范围内。因此，零初始化注意力机制的输出可表示为：

$$\mathbf{y} = \boldsymbol{S}_g \cdot \boldsymbol{V} \in \mathbb{R}^{d_v}. \tag{6}$$

在训练过程中，只有提示词参数 $\mathbf{P}$ 会经过优化，而预训练模型中的其他所有参数则保持不变。接下来，我们将从专家混合模型的视角来探讨如何理解零初始化注意力机制。

Connection to mixture of experts. Recent studies (Le et al., 2024; 2025) have revealed a notable connection between the attention mechanisms and the mixture of experts (MoE) architectures (Jacobs et al., 1991; Jordan & Jacobs, 1994), showing that attention can be seen as a form of MoE, where attention weights act as gating functions over token interactions. This MoE perspective provides a useful framework for analyzing zero-initialized attention by viewing its components as gates and expert responses.

具体而言，令 $\boldsymbol{X} = \left[\boldsymbol{x}_1^\top, \ldots, \boldsymbol{x}_N^\top, \boldsymbol{t}^\top\right]^\top \in \mathbb{R}^{(N+1)d}$为输入标记的连接结果。对于 $i \in [N+1]$，定义 $E_i \in \mathbb{R}^{d\times(N+1)d}$ 使得 $E_i\boldsymbol{X} = \boldsymbol{x}_i$ 对 i =1、.....、N以及 $E_{N+1}\boldsymbol{X} = \boldsymbol{t}$均成立。随后我们引入一组$L+N+1$ 专家 f_j： $\mathbb{R}^{(N+1)d} \to \mathbb{R}^{d_v}$，其定义为：

$$f_j(\boldsymbol{X}) = {W^V}^\top E_j \boldsymbol{X} = {W^V}^\top \boldsymbol{x}_j,\ j \in [N] \tag{7}$$

$$f_{N+1}(\boldsymbol{X}) = {W^V}^\top E_{N+1}\boldsymbol{X} = {W^V}^\top \boldsymbol{t} \tag{8}$$

$$f_{N+1+j'}(\boldsymbol{X}) = {W^V}^\top \boldsymbol{p}_{j'},\ j' \in [L]. \tag{9}$$

根据方程（5），与各专家相关的权重 G_j 如下所示：

$$G_j(\boldsymbol{X}) = \frac{\exp(\frac{\boldsymbol{X}^\top E_{N+1}^\top W^Q W^{K\top} E_j \boldsymbol{X}}{\sqrt{d_k}})}{\sum_{k=1}^{N+1}\exp(\frac{\boldsymbol{X}^\top E_{N+1}^\top W^Q W^{K\top} E_k \boldsymbol{X}}{\sqrt{d_k}})},$$

$$G_{N+1+j'}(\boldsymbol{X}) = \frac{\exp(\frac{\boldsymbol{X}^\top E_{N+1}^\top W^Q W^{K\top} \boldsymbol{p}_{j'}}{\sqrt{d_k}})}{\sum_{k'=1}^{L}\exp(\frac{\boldsymbol{X}^\top E_{N+1}^\top W^Q W^{K\top} \boldsymbol{p}_{k'}}{\sqrt{d_k}})},$$

for $j' \in [L]$ and $j \in [N+1]$. With these formulations, the output of zero-initialized attention, as described in equation (6), can be expressed as:

$$\mathbf{y} = \sum_{j=1}^{N+1} G_j(\boldsymbol{X})\cdot f_j(\boldsymbol{X}) + \tanh(\alpha) \times \left(\sum_{j'=1}^{L} G_{N+1+j'}(\boldsymbol{X})\cdot f_{N+1+j'}(\boldsymbol{X})\right). \quad (10)$$

From this formulation, zero-initialized attention can be interpreted as a specialized form of a mixture of experts model. The set of experts $f_1, \ldots, f_{N+1}$, along with their associated weight functions, are pre-trained and require no additional training, as their parameters are encoded within the pre-trained transformer model, representing existing knowledge. In contrast, the prompt experts $f_{N+2}, \ldots, f_{N+1+L}$ and their weight functions work in conjunction with the pre-trained experts, effectively integrating newly acquired information into the model through learnable prompts. Viewing zero-initialized attention through this lens as a specialized mixture of experts model provides the foundation for further theoretical analysis, as demonstrated in the next section.

4. Convergence Analysis of Prompt and Gating Factor Estimations

In this section, we provide a theoretical analysis for linear prompts and non-linear prompts through the connection between the zero-initialized attention and the mixture of experts in equation (10), demonstrating the sample-efficiency of the zero-initalized attention over the random-initalized attention. In addition, the theoretical benefits of non-linear prompts offer great flexibility in improving the practical performance of the zero-initialized attention, which will be investigated extensively with several benchmark datasets and tasks in Section 5.

4.1. Analytics for Linear Prompts

We first consider the original setting of zero-initialized attention when the prompts are linear.

Problem setting. Assume that $(\boldsymbol{X}_1, Y_1), (\boldsymbol{X}_2, Y_2), \ldots, (\boldsymbol{X}_n, Y_n) \in \mathbb{R}^d \times \mathbb{R}^{d'}$ are i.i.d. samples of size n generated from the following regression model:

$$Y_i = f_{G_*,\alpha_*}(\boldsymbol{X}_i) + \varepsilon_i, \quad i = 1, 2, \ldots, n, \quad (11)$$

where the variables $\varepsilon_1, \ldots, \varepsilon_n$ are independent Gaussian noise satisfying $\mathbb{E}[\varepsilon_i|\boldsymbol{X}_i] = 0$ and $\mathrm{Var}(\varepsilon_i|\boldsymbol{X}_i) = \sigma^2 I_{d'}$ for all $i \in [n]$. Additionally, we assume that $\boldsymbol{X}_1, \boldsymbol{X}_2, \ldots, \boldsymbol{X}_n$ are i.i.d. samples from some probability distribution μ. The regression function $f_{G_*,\alpha_*}(\cdot)$ in equation (11) takes the form of the MoE model with N pre-trained experts and L unknown experts, which is given by:

$$f_{G_*,\alpha_*}(\boldsymbol{X}) := \sum_{j=1}^{N} \frac{\exp(\boldsymbol{X}^\top \bar{A}_j^0 \boldsymbol{X} + \bar{a}_j^0)}{\sum_{k=1}^{N}\exp(\boldsymbol{X}^\top \bar{A}_k^0 \boldsymbol{X} + \bar{a}_k^0)} \cdot h(\boldsymbol{X}, \bar{\eta}_j^0) + \tanh(\alpha_*)\cdot \sum_{j'=1}^{L} \frac{\exp((\bar{B}\boldsymbol{p}_{*,j'})^\top \boldsymbol{X} + \bar{b}_{*,j'})}{\sum_{k'=1}^{L}\exp((\bar{B}\boldsymbol{p}_{*,k'})^\top \boldsymbol{X} + \bar{b}_{*,k'})} \cdot \bar{C}\boldsymbol{p}_{*,j'}, \quad (12)$$

where $G_* := \sum_{j'=1}^{L}\exp(\bar{b}_{*,j'})\delta_{\boldsymbol{p}_{*,j'}}$ denotes a true but unknown *mixing measure*, which is a weighted sum of Dirac measures δ, associated with unknown biases and prompts $(\bar{b}_{*,j'}, \boldsymbol{p}_{*,j'})_{j'=1}^{L}$ in the parameter space $\Theta \subset \mathbb{R}\times\mathbb{R}^d$. Furthermore, the *gating factor* α^* is unknown and belongs to the parameter space $\Omega \subset \mathbb{R}$. In the model (12), the matrices $\bar{A}_j^0$, the expert parameters $\bar{\eta}_j^0$, and the bias parameters $\bar{a}_j^0$ are known for all $j \in [N]$. Finally, $\bar{B} \in \mathbb{R}^{d\times d}$ and $\bar{C} \in \mathbb{R}^{d'\times d}$ are given and they play the role of pre-trained projection matrices in the context of zero-initialized attention.

Least-square estimator. We can estimate the unknown prompts and gating factor in the regression model (11) via estimating the mixing measure G_* using least-square method as follows:

$$(\widehat{G}_n, \widehat{\alpha}_n) := \operatorname*{arg\,min}_{G\in\mathcal{G}_{L'}(\Theta),\alpha\in\Omega} \sum_{i=1}^{n} \|Y_i - f_{G,\alpha}(\boldsymbol{X}_i)\|^2, \quad (13)$$

where $\mathcal{G}_{L'}(\Theta) := \{G = \sum_{i=1}^{\ell}\exp(\bar{b}_i)\delta_{\boldsymbol{p}_i} : 1 \le \ell \le L', (b_i, \boldsymbol{p}_i) \in \Theta\}$ denotes the set of all mixing measures with at most L' prompts. For practical purpose, the number of chosen prompts L' is generally larger than the number of true prompts L, i.e., $L' \ge L$, to guarantee that the estimated prompts and gating factor from the least-square method converge to the true prompts and gating factor.

Convergence rate of regression estimator. We first show that the regression estimator $f_{\widehat{G}_n,\widehat{\alpha}_n}$ can still estimate the true regression function f_{G_*,α_*} at the standard parametric rate in terms of the sample size n though we overspecify the number of prompts, i.e., $L' > L$.

Proposition 4.1. *The convergence rate of the regression estimator $f_{\widehat{G}_n,\widehat{\alpha}_n}(\cdot)$ to the true regression function $f_{G_*,\alpha_*}(\cdot)$ under the $L^2(\mu)$ norm is of parametric order, that is,*

$$\|f_{\widehat{G}_n,\widehat{\alpha}_n} - f_{G_*,\alpha_*}\|_{L^2(\mu)} = \mathcal{O}_P(\sqrt{\log(n)/n}). \quad (14)$$

$$G_j(\boldsymbol{X}) = \frac{\exp(\frac{\boldsymbol{X}^\top E_{N+1}^\top W^Q W^{K\top} E_j \boldsymbol{X}}{\sqrt{d_k}})}{\sum_{k=1}^{N+1}\exp(\frac{\boldsymbol{X}^\top E_{N+1}^\top W^Q W^{K\top} E_k \boldsymbol{X}}{\sqrt{d_k}})},$$

$$G_{N+1+j'}(\boldsymbol{X}) = \frac{\exp(\frac{\boldsymbol{X}^\top E_{N+1}^\top W^Q W^{K\top} \boldsymbol{p}_{j'}}{\sqrt{d_k}})}{\sum_{k'=1}^{L}\exp(\frac{\boldsymbol{X}^\top E_{N+1}^\top W^Q W^{K\top} \boldsymbol{p}_{k'}}{\sqrt{d_k}})},$$

适用于 $j' \in [L]$ 和 $j \in [N+1]$。借助这些公式，如方程（6）中所述，零初始化注意力机制的输出可表示为：

$$\mathbf{y} = \sum_{j=1}^{N+1} G_j(\boldsymbol{X})\cdot f_j(\boldsymbol{X}) + \tanh(\alpha) \times \left(\sum_{j'=1}^{L} G_{N+1+j'}(\boldsymbol{X})\cdot f_{N+1+j'}(\boldsymbol{X})\right). \quad (10)$$

通过这一表述，零初始化注意力机制可以被视作专家混合模型的一种特殊形式。由 f_1、……、f_{N+1}构成的专家集合及其对应的权重函数早已完成预训练，无需进一步训练，因为这些参数已编码在预训练的变换器模型中，代表着现有的知识体系。相比之下，提示词专家 f_{N+2}、……、f_{N+1+L} 及其权重函数则与预训练的专家们协同工作，通过可学习的提示词将新获取的信息有效整合到模型中。从这一视角将零初始化注意力机制视为一种特殊的专家混合模型，为后续的理论分析奠定了基础，具体内容将在下一节中阐述。

4. 提示词与门控因子估计值的收敛性分析

在本节中，我们通过方程式（10）中所体现的零初始化注意力机制与专家混合模型之间的关联，对线性提示词与非线性提示词进行了理论分析，从而证明了零初始化注意力机制相较于随机初始化注意力机制在样本效率上的优势。此外，非线性提示词所带来的理论优势为提升零初始化注意力机制的实际性能提供了极大的灵活性，这一内容将在第5节中借助多个基准数据集和任务进行深入探讨。

4.1. 线性提示词的分析方法

我们首先研究在提示词采用线性策略时的原始零初始化注意力机制设置。

问题设定。假设 $(\boldsymbol{X}_1, Y_1)$、$(\boldsymbol{X}_2, Y_2)$……、$(\boldsymbol{X}_n, Y_n) \in \mathbb{R}^d \times \mathbb{R}^{d'}$ 是从以下回归模型中生成的尺寸为 n 的独立同分布样本：

$$Y_i = f_{G_*,\alpha_*}(\boldsymbol{X}_i) + \varepsilon_i, \quad i = 1, 2, \ldots, n, \quad (11)$$

where the variables $\varepsilon_1, \ldots, \varepsilon_n$ are independent Gaussian noise satisfying $\mathbb{E}[\varepsilon_i|\boldsymbol{X}_i] = 0$ and $\mathrm{Var}(\varepsilon_i|\boldsymbol{X}_i) = \sigma^2 I_{d'}$ for all $i \in [n]$. Additionally, we assume that $\boldsymbol{X}_1, \boldsymbol{X}_2, \ldots, \boldsymbol{X}_n$ are i.i.d. samples from some probability distribution μ. The regression function $f_{G_*,\alpha_*}(\cdot)$ in equation (11) takes the form of the MoE model with N pre-trained experts and L unknown experts, which is given by:

$$f_{G_*,\alpha_*}(\boldsymbol{X}) := \sum_{j=1}^{N} \frac{\exp(\boldsymbol{X}^\top \bar{A}_j^0 \boldsymbol{X} + \bar{a}_j^0)}{\sum_{k=1}^{N}\exp(\boldsymbol{X}^\top \bar{A}_k^0 \boldsymbol{X} + \bar{a}_k^0)} \cdot h(\boldsymbol{X}, \bar{\eta}_j^0) + \tanh(\alpha_*)\cdot \sum_{j'=1}^{L} \frac{\exp((\bar{B}\boldsymbol{p}_{*,j'})^\top \boldsymbol{X} + \bar{b}_{*,j'})}{\sum_{k'=1}^{L}\exp((\bar{B}\boldsymbol{p}_{*,k'})^\top \boldsymbol{X} + \bar{b}_{*,k'})} \cdot \bar{C}\boldsymbol{p}_{*,j'}, \quad (12)$$

其中 $G_* := \sum_{j'=1}^{L}\exp(\bar{b}_{*,j'})\delta_{\boldsymbol{p}_{*,j'}}$ 表示一个真实但未知的混合度量，它是与参数空间 $\Theta \subset \mathbb{R}\times\mathbb{R}^d$ 中的未知偏差及提示词 $(\bar{b}_{*,j'}, \boldsymbol{p}_{*,j'})_{j'=1}^{L}$ 相关联的狄拉克度量 δ 的加权之和。此外，门控因子 α^* 也是未知的，属于参数空间 $\Omega \subset \mathbb{R}$。在模型 (12) 中，矩阵 $\bar{A}_j^0$、专家参数 $\bar{\eta}_j^0$ 以及偏差参数 $\bar{a}_j^0$ 对所有的 $j \in [N]$ 来说都是已知的。最后，$\bar{B} \in \mathbb{R}^{d\times d}$ 和 $\bar{C} \in \mathbb{R}^{d'\times d}$ 是给定的值，在零初始化注意力机制的背景下它们起着预训练投影矩阵的作用。

最小二乘估计器。我们可以通过如下方式，利用最小二乘法来估算混合度量 G_*，进而估计回归模型（11）中的未知提示词与门控因子：

$$(\widehat{G}_n, \widehat{\alpha}_n) := \operatorname*{arg\,min}_{G\in\mathcal{G}_{L'}(\Theta),\alpha\in\Omega} \sum_{i=1}^{n} \|Y_i - f_{G,\alpha}(\boldsymbol{X}_i)\|^2, \quad (13)$$

其中 $\mathcal{G}_{L'}(\Theta) := \{G = \sum_{i=1}^{\ell}\exp(\bar{b}_i)\delta_{\boldsymbol{p}_i} : 1 \le \ell \le L', (b_i, \boldsymbol{p}_i) \in \Theta\}$ 表示所有包含至多 L' 个提示词的混合度量集合。出于实际应用考虑，所选取的提示词数量 L' 通常会大于真实提示词的数量 L，即 $L' \ge L$，以此确保通过最小二乘法估算得到的提示词与门控因子能够收敛到真实的提示词与门控因子。

回归估计器的收敛速率。我们首先证明，即便过度规定了提示词的数量，即 $L' > L$，回归估计器$f_{\widehat{G}_n,\widehat{\alpha}_n}$依然能够以与样本量 n 相关的标准参数速率来估计真实的回归函数f_{G_*,α_*}。

命题 4.1.在 $L^2(\mu)$ 范数下，回归估计器$f_{\widehat{G}_n,\widehat{\alpha}_n}(\cdot)$向真实回归函数$f_{G_*,\alpha_*}(\cdot)$收敛的收敛速率具有参数阶数。

$$\|f_{\widehat{G}_n,\widehat{\alpha}_n} - f_{G_*,\alpha_*}\|_{L^2(\mu)} = \mathcal{O}_P(\sqrt{\log(n)/n}). \quad (14)$$

Proof of Proposition 4.1 is in Appendix A.3. Given the above convergence rate of the regression estimator, we aim to construct a loss function among parameters, denoted by $\mathcal{D}(\widehat{G}_n, G_*)$, such that $\|f_{\widehat{G}_n,\widehat{\alpha}_n} - f_{G_*,\alpha_*}\|_{L^2(\mu)} \gtrsim [\mathcal{D}(\widehat{G}_n, G_*) + |\widehat{\alpha}_n - \alpha_*|]$. Then, this lower bound together with the bound (14) will lead to our desired prompt convergence rates. To this end, we will build a loss function based on the concept of Voronoi cells as in (Manole & Ho, 2022).

Voronoi loss function. For a mixing measure $G \in \mathcal{G}_{L'}(\Theta)$, we distribute its atoms across the Voronoi cells $\{\mathcal{C}_j \equiv \mathcal{C}_j(G), j \in [L]\}$ generated by the atoms of G_*, where

$$\mathcal{C}_j := \{i \in [L'] : \|\boldsymbol{p}_i - \boldsymbol{p}_{*,j}\| \leq \|\boldsymbol{p}_i - \boldsymbol{p}_{*,\ell}\|, \forall \ell \neq j\}.$$

Then, the Voronoi loss function is given by

$$\begin{aligned}\mathcal{D}(G, G_*) &:= \sum_{j'=1}^{L} \Big| \sum_{i \in \mathcal{C}_{j'}} \exp(b_i) - \exp(b_{*,j'}) \Big| \\ &+ \sum_{j' \in [L]: |\mathcal{C}_{j'}| = 1} \sum_{i \in \mathcal{C}_{j'}} \exp(b_i) \|\Delta \boldsymbol{p}_{ij'}\| \\ &+ \sum_{j' \in [L]: |\mathcal{C}_{j'}| > 1} \sum_{i \in \mathcal{C}_{j'}} \exp(b_i) \|\Delta \boldsymbol{p}_{ij'}\|^2, \end{aligned} \tag{15}$$

where $\Delta \boldsymbol{p}_{ij'} = \boldsymbol{p}_i - \boldsymbol{p}_{*,j'}$ for all $i \in \mathcal{C}_{j'}$ and $j' \in [L]$. Given the above loss function, we are now ready to capture the convergence behavior of linear prompts in Theorem 4.2.

Theorem 4.2. *Assume that $L' > L$, i.e., the number of prompts is unknown and is overspecified by L' prompts. Then, the least square estimator $(\widehat{G}_n, \widehat{\alpha}_n)$ defined in equation (13) satisfies*

$$\mathcal{D}(\widehat{G}_n, G_*) = \mathcal{O}_P(\sqrt{\log(n)/n}),$$
$$|\widehat{\alpha}_n - \alpha_*| = \mathcal{O}_P(\sqrt{\log(n)/n}).$$

Proof of Theorem 4.2 is in Appendix A.1. Putting the first bound and the formulation of the Voronoi loss function $\mathcal{D}$ together, we observe that the convergence rates of estimating linear prompts range from the order $\mathcal{O}_P([\log(n)/n]^{\frac{1}{2}})$ to $\mathcal{O}_P([\log(n)/n]^{\frac{1}{4}})$, which are optimal. Therefore, we need a polynomial number of data, either $\mathcal{O}(\epsilon^{-2})$ or $\mathcal{O}(\epsilon^{-4})$, to approximate the linear prompts with a given error ϵ.

Sample complexity of random-initialized attention v.s. zero-initialized attention: From the above the convergence analysis of prompt estimation under the zero-initialized attention and that under the random-initialized attention in (Akbarian et al., 2024; Le et al., 2024), we claim that using the zero-initialized attention is more sample-efficient than using the random-initialized attention in terms of prompt convergence due to the following reasons:

(i) Prompt convergence in random-initialized attention: The convergence rates of prompt estimation are significantly slow, standing at the order of $\mathcal{O}_P(1/\log^{\tau}(n))$ for some constant $\tau > 0$, where n is the sample size. Thus, to approximate prompts with a given error ϵ, we need exponentially many data points $\mathcal{O}(\exp(\epsilon^{-1/\tau}))$, which is not sample-efficient.

(ii) Prompt convergence in zero-initialized attention: As shown in Theorem 4.2, the convergence rates of linear prompt estimations are of polynomial orders, ranging from $\mathcal{O}_P(n^{-1/2})$ to $\mathcal{O}_P(n^{-1/4})$ (up to some logarithmic term), which are substantially faster than those under the random-initialized attention. Therefore, we only need polynomially many data points $\mathcal{O}(\epsilon^{-2})$ or $\mathcal{O}(\epsilon^{-4})$ to approximate the prompts with a given error ϵ.

4.2. Theoretical Benefits of Non-Linear Prompts

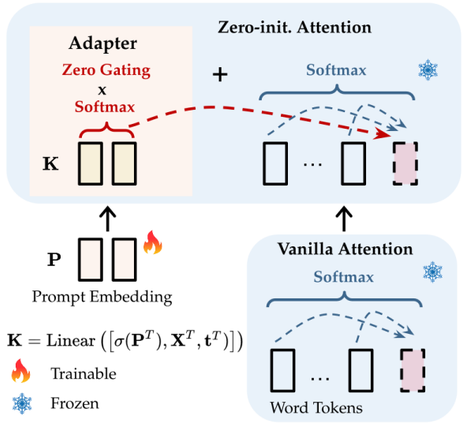


Figure 1. LLaMA-Adapter with non-linear prompt structures. Trainable prompts are integrated into the final layers of the LLaMA model, where a zero-gating mechanism modulates the added prompts. This approach enables progressive learning of instructional knowledge while keeping the remaining model parameters frozen.

While the original zero-initialization approach considered only linear prompts (Zhang et al., 2024), most current prompt-based techniques commonly reparameterize the prompt parameters with an MLP rather than optimizing them directly, in order to enhance training stability (Li & Liang, 2021; Liu et al., 2021; Le et al., 2025). To increase both the flexibility and practical relevance of our results, we extend our analysis to zero-initialized attention equipped with non-linear prompts.

Problem setting. Suppose that the data $(\boldsymbol{X}_1, Y_1), (\boldsymbol{X}_2, Y_2), \ldots, (\boldsymbol{X}_n, Y_n) \in \mathbb{R}^d \times \mathbb{R}^{d'}$ are

Proof of Proposition 4.1 is in Appendix A.3. Given the above convergence rate of the regression estimator, we aim to construct a loss function among parameters, denoted by $\mathcal{D}(\widehat{G}_n, G_*)$, such that $\|f_{\widehat{G}_n,\widehat{\alpha}_n} - f_{G_*,\alpha_*}\|_{L^2(\mu)} \gtrsim [\mathcal{D}(\widehat{G}_n, G_*) + |\widehat{\alpha}_n - \alpha_*|]$. Then, this lower bound together with the bound (14) will lead to our desired prompt convergence rates. To this end, we will build a loss function based on the concept of Voronoi cells as in (Manole & Ho, 2022).

Voronoi损失函数。对于混合度量$G \in \mathcal{G}_{L'}(\Theta)$，我们会将其原子分配到由 G_*的原子所生成的Voronoi单元 $\{\mathcal{C}_j \equiv \mathcal{C}_j(G)$ $j \in [L]\}$ 中，具体分布方式为

$$\mathcal{C}_j := \{i \in [L'] : \|\boldsymbol{p}_i - \boldsymbol{p}_{*,j}\| \leq \|\boldsymbol{p}_i - \boldsymbol{p}_{*,\ell}\|, \forall \ell \neq j\}.$$

此时，Voronoi损失函数的表达式为

$$\begin{aligned}\mathcal{D}(G, G_*) &:= \sum_{j'=1}^{L} \Big| \sum_{i \in \mathcal{C}_{j'}} \exp(b_i) - \exp(b_{*,j'}) \Big| \\ &+ \sum_{j' \in [L]: |\mathcal{C}_{j'}| = 1} \sum_{i \in \mathcal{C}_{j'}} \exp(b_i) \|\Delta \boldsymbol{p}_{ij'}\| \\ &+ \sum_{j' \in [L]: |\mathcal{C}_{j'}| > 1} \sum_{i \in \mathcal{C}_{j'}} \exp(b_i) \|\Delta \boldsymbol{p}_{ij'}\|^2, \end{aligned} \tag{15}$$

其中 $\Delta \boldsymbol{p}_{ij'} = \boldsymbol{p}_i - \boldsymbol{p}_{*,\ j'}$ 对所有的 $i \in \mathcal{C}_{j'}$ 以及 $j' \in [L]$ 而言。基于上述损失函数，我们现在可以开始在定理4.2中探讨线性提示词的收敛行为。

定理4.2。假设 $L' > L$，即提示词的数量是未知的，并且被 L' 个提示词过度指定。那么，在方程(<style '26'>13<style '28'>)中定义的最小二乘估计量$(G_n,\ \widehat{\alpha}_n)$ 满足

$$\mathcal{D}(\widehat{G}_n, G_*) = \mathcal{O}_P(\sqrt{\log(n)/n}),$$
$$|\widehat{\alpha}_n - \alpha_*| = \mathcal{O}_P(\sqrt{\log(n)/n}).$$

Proof of Theorem 4.2 is in Appendix A.1. Putting the first bound and the formulation of the Voronoi loss function $\mathcal{D}$ together, we observe that the convergence rates of estimating linear prompts range from the order $\mathcal{O}_P([\log(n)/n]^{\frac{1}{2}})$ to $\mathcal{O}_P([\log(n)/n]^{\frac{1}{4}})$, which are optimal. Therefore, we need a polynomial number of data, either $\mathcal{O}(\epsilon^{-2})$ or $\mathcal{O}(\epsilon^{-4})$, to approximate the linear prompts with a given error ϵ.

随机初始化的注意力机制与零初始化注意力机制的样本复杂度对比：基于上述关于在零初始化注意力机制以及随机初始化的注意力机制下提示词估计的收敛性分析（Akbarian等人，2024年；Le等人，2024年），我们认为由于以下原因，在提示词收敛性方面，使用零初始化注意力机制比使用随机初始化的注意力机制更具样本效率：

*(i)*随机初始化的注意力机制中提示词收敛情况：提示词估计的收敛速率存在显著差异

也就是说，当采用随机初始化的注意力机制时，提示词估计的收敛速率极为缓慢，具体表现形式为 $\mathcal{O}_P(1/\log^{\tau}(n))$，这里的 $\tau > 0$是固定常数， n 代表样本总数。要想以特定的误差 ϵ精准生成提示词，就必须使用数量呈指数级增长的数据点 $\mathcal{O}(\exp(\epsilon^{-1/\tau}))$，这样的处理方式显然无法高效利用样本资源。

(ii) Prompt convergence in zero-initialized attention: As shown in Theorem 4.2, the convergence rates of linear prompt estimations are of polynomial orders, ranging from $\mathcal{O}_P(n^{-1/2})$ to $\mathcal{O}_P(n^{-1/4})$ (up to some logarithmic term), which are substantially faster than those under the random-initialized attention. Therefore, we only need polynomially many data points $\mathcal{O}(\epsilon^{-2})$ or $\mathcal{O}(\epsilon^{-4})$ to approximate the prompts with a given error ϵ.

4.2. 非线性提示词的理论优势

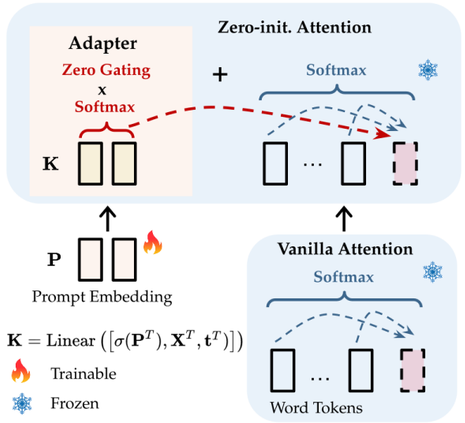


图*1*。采用非线性提示词结构的LLaMA-Adapter。可训练的提示词被整合到LLaMA模型的最底层中，此时一个零门控机制会用于调节这些新增的提示词。该方法能够在冻结其余模型参数的同时，逐步学习指令相关的知识。

While the original zero-initialization approach considered only linear prompts (Zhang et al., 2024), most current prompt-based techniques commonly reparameterize the prompt parameters with an MLP rather than optimizing them directly, in order to enhance training stability (Li & Liang, 2021; Liu et al., 2021; Le et al., 2025). To increase both the flexibility and practical relevance of our results, we extend our analysis to zero-initialized attention equipped with non-linear prompts.

问题设定。假设数据(、Y_1)、($\boldsymbol{X}_2$、Y_2).....、($\boldsymbol{X}_n$、Y_n) $\in \mathbb{R}^d \times \mathbb{R}^{d'}$均为

i.i.d. samples of size n generated from the model:

$$Y_i = g_{G_*,\alpha_*}(\boldsymbol{X}_i) + \varepsilon_i, \quad i = 1, 2, \ldots, n. \tag{16}$$

Here, we impose the same assumptions on the noise variables ε_i and the input $\boldsymbol{X}_i$ as in Section 4.1. Nevertheless, the regression function $g_{G_*,\alpha_*}(\cdot)$ in equation (16) now takes the form of a prefix MoE model with N pre-trained experts and L unknown experts,

$$g_{G_*,\alpha_*}(\boldsymbol{X}) := \sum_{j=1}^{N} \frac{\exp(\boldsymbol{X}^\top \bar{A}_j^0 \boldsymbol{X} + \bar{a}_j^0)}{\sum_{k=1}^{N} \exp(\boldsymbol{X}^\top \bar{A}_k^0 \boldsymbol{X} + \bar{a}_k^0)} \cdot h(\boldsymbol{X}, \bar{\eta}_j^0) + \tanh(\alpha_*) \times \left(\sum_{j'=1}^{L} \frac{\exp((\bar{B}\sigma(\boldsymbol{p}_{*,j'}))^\top \boldsymbol{X} + \bar{b}_{*,j'})}{\sum_{k'=1}^{L} \exp((\bar{B}\sigma(\boldsymbol{p}_{*,k'}))^\top \boldsymbol{X} + \bar{b}_{*,k'})} \cdot \bar{C}\sigma(\boldsymbol{p}_{*,j'}) \right), \tag{17}$$

where $\sigma : \mathbb{R}^d \to \mathbb{R}^{d'}$ is some activation function applied element-wise to the prompts $\boldsymbol{p}_{*,j'}$. According to the change of the regression function, the least-square estimator of the true mixing measure G_* under this setting becomes

$$(\widetilde{G}_n, \widetilde{\alpha}_n) := \operatorname*{arg\,min}_{G \in \mathcal{G}_{L'}(\Theta), \alpha \in \Omega} \sum_{i=1}^{n} \Big(Y_i - g_{G,\alpha}(\boldsymbol{X}_i) \Big)^2. \tag{18}$$

In the following proposition, we will illustrate that the parametric convergence rate of the regression function estimator still holds true under the setting of non-linear prompts.

Proposition 4.3. *The convergence rate of the model estimation $g_{\widetilde{G}_n,\widetilde{\alpha}_n}(\cdot)$ to the true model $g_{G_*,\alpha_*}(\cdot)$ under the $L^2(\mu)$ norm is parametric on the sample size, that is,*

$$\|g_{\widetilde{G}_n,\widetilde{\alpha}_n} - g_{G_*,\alpha_*}\|_{L^2(\mu)} = \mathcal{O}_P(\sqrt{\log(n)/n}). \tag{19}$$

Similar to Section 4.1, by utilizing the Voronoi loss function $\mathcal{D}$ defined in Eq.(15), we are able to investigate the convergence behavior of non-linear prompts in the zero-initialized attention in Theorem 4.4 whose proof can be found in Appendix A.2.

Theorem 4.4. *Assume that the activation function σ satisfies the Assumptions (A.1)-(A.2) specified in Appendix A.2. Then, the least square estimator $(\widetilde{G}_n, \widetilde{\alpha}_n)$ defined in equation (18) satisfies*

$$\mathcal{D}(\widetilde{G}_n, G_*) = \mathcal{O}_P(\sqrt{\log(n)/n}),$$
$$|\widetilde{\alpha}_n - \alpha_*| = \mathcal{O}_P(\sqrt{\log(n)/n}).$$

It can be seen that the convergence of prompt parameters under this setting behaves analogously to that in Theorem 4.2. In particular, the prompt parameters $\boldsymbol{p}_{*,j}$ still admit the estimation rates of order $\mathcal{O}_P([\log(n)/n]^{\frac{1}{2}})$ or $\mathcal{O}_P([\log(n)/n]^{\frac{1}{4}})$. Thus, a polynomial number of data points ranging from $\mathcal{O}(\varepsilon^{-2})$ to $\mathcal{O}(\varepsilon^{-4})$ is required to achieve the prompt approximation with a given error ϵ. As a result, we claim that the zero-initialized attention with non-linear prompts is also more sample-efficient than the random-initialized attention in terms of prompt convergence. Moreover, although sharing the same sample complexity as when using linear prompts, the zero-initialized attention with non-linear prompts will be shown to offer greater flexibility in practical applications, which will be discussed further in Section 5.

Non-Linear prompt optimization in practice. Motivated by the theoretical benefits of using non-linear prompts in zero-initialized attention in Eq.(17), one replaces the linear prompts $\mathbf{P}$ in zero-initialized attention with non-linear prompts (Figure 1), which is given by:

$$\tilde{\mathbf{P}} = \sigma(\mathbf{P}) \in \mathbb{R}^{L \times d}, \tag{20}$$

where $\sigma(\cdot)$ is a non-linear activation function or a lightweight MLP, in line with common practice in most current prompt-based techniques, and $\mathbf{P}$ can be a layer embedding vector or a set of embedding vectors equal to the length of prompt for each layer. For instance, we can choose σ as a lightweight MLP with 2 layers combined with non-linear activation such as Tanh, ReLU, and Leaky-ReLU:

$$\sigma(\mathbf{P}) = f_2(\phi((f_1(\mathbf{P})))), \tag{21}$$

where $f_1(.), f_2(.)$ are separate linear transformations, $\phi(.)$ represents the non-linear activation function, e.g., ReLU or Leaky-ReLU, and $\mathbf{P}$ is defined as a layer embedding vector. To ensure parameter efficiency and facilitate knowledge sharing across layers, this MLP can be shared among the layers that utilize the prompts.

As established in Theorem 4.4, this non-linear formulation retains the same estimation rates as the linear variant, thus offering greater flexibility for practical applications.

5. Experiments

To highlight the statistical advantages of zero-initialized attention and explore the potential of non-linear prompts, we conduct a series of question-answering experiments on LLM tasks. Section 5.1 provides an overview of our experimental setup, while the main results are presented in Section 5.2. Additional details and prompt templates are included in Appendix C.

5.1. Experimental Setup

Datasets and Evaluations. We use the Open LLM benchmarks as in Beeching et al. (2024). These benchmarks evaluate the generative abilities of LLMs in four different tasks, including `(i) AI2 Reasoning Challenge` (ARC) with Easy (eas) and Challenge (cha) types (Clark et al., 2018), `(ii) HellaSwag` (Zellers et al., 2019), `(iii)`

从该模型中生成的尺寸为 n 的独立同分布样本 del:

$$Y_i = g_{G_*,\alpha_*}(\boldsymbol{X}_i) + \varepsilon_i, \quad i = 1, 2, \ldots, n. \tag{16}$$

在此，我们对噪声变量 ε_i 以及输入数据 $\boldsymbol{X}_i$ 采用了与4.1节中相同的假设。不过，方程（16）中的回归函数 $g_{G_*,\alpha_*}(\cdot)$ 现在表现为一种带有 N 预训练专家模型和未知专家模型的前缀MoE模型形式。

$$g_{G_*,\alpha_*}(\boldsymbol{X}) := \sum_{j=1}^{N} \frac{\exp(\boldsymbol{X}^\top \bar{A}_j^0 \boldsymbol{X} + \bar{a}_j^0)}{\sum_{k=1}^{N} \exp(\boldsymbol{X}^\top \bar{A}_k^0 \boldsymbol{X} + \bar{a}_k^0)} \cdot h(\boldsymbol{X}, \bar{\eta}_j^0) + \tanh(\alpha_*) \times \left(\sum_{j'=1}^{L} \frac{\exp((\bar{B}\sigma(\boldsymbol{p}_{*,j'}))^\top \boldsymbol{X} + \bar{b}_{*,j'})}{\sum_{k'=1}^{L} \exp((\bar{B}\sigma(\boldsymbol{p}_{*,k'}))^\top \boldsymbol{X} + \bar{b}_{*,k'})} \cdot \bar{C}\sigma(\boldsymbol{p}_{*,j'}) \right), \tag{17}$$

其中 $\sigma: \mathbb{R}^d \to \mathbb{R}^{d'}$ 是某种激活函数，它被逐元素地应用于提示词 $\boldsymbol{p}_{*,\ j'}$。根据回归函数的变化，在此设定下真实混合度量 G_* 的最小二乘估计器会随之改变

$$(\widetilde{G}_n, \widetilde{\alpha}_n) := \operatorname*{arg\,min}_{G \in \mathcal{G}_{L'}(\Theta), \alpha \in \Omega} \sum_{i=1}^{n} \Big(Y_i - g_{G,\alpha}(\boldsymbol{X}_i) \Big)^2. \tag{18}$$

在接下来的论述中，我们将证明：即使在非线性提示向量的设定下，回归函数估计器的参数收敛速率依然成立。

命题 4.3. 在 $L^2(\mu)$ 范数下，模型估计 $g_{\widetilde{G}_n,\ \widetilde{\alpha}_n}(\cdot)$ 向真实模型 $g_{G_*,\ \alpha_*}(\cdot)$ 收敛的速率是关于样本量呈参数依赖关系的，也就是说

$$\|g_{\widetilde{G}_n,\widetilde{\alpha}_n} - g_{G_*,\alpha_*}\|_{L^2(\mu)} = \mathcal{O}_P(\sqrt{\log(n)/n}). \tag{19}$$

与Section 4.1类似，通过运用在方程（15）中定义的Voronoi损失函数$\mathcal{D}$，我们能够在定理4.4 中研究零初始化注意力机制下非线性提示向量的收敛行为，该定理的证明可见于附录A.2。

定理 4.4. 假设激活函数 σ 满足附录中规定的假设 *(A.1)-(A.2)*A.2。~那么，在方程（18）中定义的最小二乘估计量（G_n，$\widetilde{\alpha}_n$）满足

$$\mathcal{D}(\widetilde{G}_n, G_*) = \mathcal{O}_P(\sqrt{\log(n)/n}),$$
$$|\widetilde{\alpha}_n - \alpha_*| = \mathcal{O}_P(\sqrt{\log(n)/n}).$$

可以看出，在此设定下提示词参数的收敛行为与定理4.2中的情况类似。具体而言，提示词参数$\boldsymbol{p}_{*,\ j}$ 仍然存在阶数为 $\mathcal{O}_P([\log(n)/n]^{\frac{1}{2}})$或$\mathcal{O}_P([\log(n)/n]^{\frac{1}{4}})$的估计速率。因此,_ 需要 $\mathcal{O}(\varepsilon^{-2})$到 $\mathcal{O}(\varepsilon^{-4})$数量级的多项式个数的数据点才能实现该收敛。

以给定的误差范围实现提示词近似 ϵ。因此，我们认为采用非线性提示词的零初始化注意力机制在提示词收敛速度方面也优于随机初始化的注意力机制，且更具样本效率。此外，尽管其样本复杂度与使用线性提示词时相同，但实验证明采用非线性提示词的零初始化注意力机制在实际应用中具有更高的灵活性，这一内容将在第5节中进一步阐述。

非线性提示词在实际应用中的优化方法。受方程（17）中关于在零初始化注意力机制中使用非线性提示词所带来的理论优势的启发，人们用非线性提示词（见图1）来替代零初始化注意力机制中的线性提示词$\mathbf{P}$，其表达式为：

$$\tilde{\mathbf{P}} = \sigma(\mathbf{P}) \in \mathbb{R}^{L \times d}, \tag{20}$$

其中 $\sigma(\cdot)$为非线性激活函数或轻量级多层感知机，这一设计符合当前大多数基于提示词的技术中的常见做法；而$\mathbf{P}$ 则可以是每层对应的嵌入向量，或者是一组数量与提示词长度相等的嵌入向量。例如，我们可以选择由2层构成的轻量级多层感知机，并结合Tanh、ReLU以及Leaky-ReLU等非线性激活函数作为σ：

$$\sigma(\mathbf{P}) = f_2(\phi((f_1(\mathbf{P})))), \tag{21}$$

其中 $f_1(.)$、 $f_2(.)$为独立的线性变换， $\phi(.)$代表非线性激活函数，例如ReLU或Leaky-ReLU，而$\mathbf{P}$ 则被定义为层嵌入向量。为提升参数效率并便于各层之间共享知识，该多层感知机可在使用这些提示词的各层之间相互复用。

正如定理4.4中所阐述的，这种非线性表述方式与线性版本具有相同的估计速率，因此能为实际应用提供更大的灵活性。

5. 实验部分

为凸显零初始化注意力机制的统计优势并探索非线性提示向量的潜力，我们在大语言模型任务上开展了一系列问答式实验。第5.1 节概述了我们的实验设置，主要结果则呈现在第5.2节。更多细节及提示词模板详见附录C节。

5.1. 实验环境配置

Datasets and Evaluations. We use the Open LLM benchmarks as in Beeching et al. (2024). These benchmarks evaluate the generative abilities of LLMs in four different tasks, including `(i) AI2 Reasoning Challenge` (ARC) with Easy (eas) and Challenge (cha) types (Clark et al., 2018), `(ii) HellaSwag` (Zellers et al., 2019), `(iii)`

Table 1. Commparison between **Linear prompt** (zero-initialized mechanism) and **Random-Init** prompt on 4 LLM tasks using LLaMA-7B and LLaMA-13B models.

Method	ARC			MMLU	Hellaswag	TruthfullQA	Average
	Acc (eas)	*Acc (cha)*	*Acc (aver)*	*Acc*	*Acc*	*Acc*	
LLaMA-7B + zero-init	62.29 ↑1.64	43.17 ↑2.47	52.73 ↑2.06	36.28 ↑1.16	76.79 ↑4.17	45.53 ↑7.71	52.83 ↑3.77
LLaMA-7B + rand-init	60.65	40.7	50.67	35.12	72.62	37.82	49.06
LLaMA-13B + zero-init	81.78 ↑0.17	64.33 ↑0.42	73.06 ↑0.3	49.64 ↑1.62	81.21 ↑0.05	34.88 ↑0.36	59.70 ↑0.58
LLaMA-13B + rand-init	81.61	63.91	72.76	48.02	81.16	34.52	59.12

`MMLU` (Hendrycks et al., 2020), and `(iv) TruthfulQA` (Lin et al., 2021). All these tasks evaluate the model through multiple-choice questions, where ARC and MMLU test the LLM's knowledge, HellaSwag tests the model's ability to finish sentences, and TruthfulQA measures whether an LLM is truthful in generating answers to given questions.

We follow the experimental setup of LLaMA-Adapter (Zhang et al., 2024) by fine-tuning LLaMA on the Alpaca dataset (Taori et al., 2023). The model performance is evaluated on the test set by conducting a zero-shot evaluation for ARC, MMLU, and TruthfulQA while using a 10-shot setting for HellaSwag. Here, n -shot refers to incorporating n instruction-following samples into the prompt question.

Architectures Training. We employ experiments on two LLaMA versions, LLaMA-7B with 32 Transformer layers and LLaMA-13B with 40 Transformer layers (Touvron et al., 2023b;c). The models are trained with 4 A100-GPUs for 5 epochs. The training configuration includes a warmup period of 2 epochs, a total batch size of 64, a learning rate of 0.009, and a weight decay of 0.02. With LLaMA-7B, we use a prompt with length $L = 10$ and integrate adaptation prompts into the last $K = 30$ layers. On LLaMA-13B, we use $L = 10$ and insert prompts at the last $K = 38$ layers.

Baselines. To access the effectiveness of zero-initialized attention and demonstrate the potential of integrating it with the proposed non-linear prompt, we conduct experiments using the following configurations:

- (1) **Linear prompt**: i.e., the default setting as LLaMA-Adapter (Zhang et al., 2024), where prompt vectors $\mathbf{P}$ are zero-initialized and used directly.
- (2) **Non-Linear prompt**: use zero-initialized mechanism and apply a nonlinear MLP on input prompts $\mathbf{P}$ to generate prompt vectors $\tilde{\mathbf{P}} = \sigma(\mathbf{P})$ and the MLP layers are shared among layers.
- (3) **Random-Init prompt**: use the input prompts with conventional randomly-initialization, i.e., forms in Eq.(4) rather than the zero-initialized mechanism.
- (4) **Finetuning & low-rank decomposition**: We compare with a fully fine-tuned LLaMA model trained on the Alpaca dataset (Taori et al., 2023), where all model parameters are updated. Additionally, we benchmark against `LoRA` (Hu et al., 2021), a PEFT method that inserts trainable low-rank decomposition matrices into each layer's weights, and `VeRA` (Kopiczko et al., 2023), another PEFT method that uses a single random pair of low-rank matrices shared across all layers, while learning separate scaling vectors for each layer.
- (5) **Other Prompt Tuning Methods**: including (Lester et al., 2021) which optimize continuous prompts added to input text; `IA3` (Liu et al., 2022) a method using three learnable vectors to rescale the keys, values, and position-wise feed-forward activations in the attention layers.

5.2. Main Results

I. Zero-initialized attention is essential in prompt-tuning, enhancing both *robustness* and *effectiveness* compared to random-initialized attention.

We begin by investigating the impact of zero-initialized prompt-tuning on LLaMA-7B and LLaMA-13B using the `linear prompt` setting, comparing its performance against conventional random-initialization strategies.

As shown in Table 1, the zero-initialized mechanism enhances stability in LLaMA-13B and significantly boosts the performance of LLaMA-Adapter when using LLaMA-7B, compared to conventional random-initialization strategies. For instance, with LLaMA-7B on Hellaswag and TruthfulQA, the Linear prompt surpasses the Random-Init prompt by 4.17% and 7.71%, respectively. The varying impact of zero-initialization between LLaMA-7B and LLaMA-13B can be attributed to differences in model capacity and expressiveness. LLaMA-13B, with its larger parameter space, naturally generalizes better even with randomly initialized prompts, reducing the relative advantage of zero-initialization. In contrast, LLaMA-7B, with its lower capacity, benefits more from zero-initialization, as it relies heavily on efficient adaptation mechanisms to optimize learning.

In summary, those results align with our theoretical findings in Section 4, which demonstrate how zero-initialized

表*1.*使用LLaMA-7B模型和LLaMA-13B模型在4项大语言模型任务上，对比**线性提示词**（采用零初始化机制）与**随机初始化提示词**的表现。

方法	ARC			MMLU	HellaSwag	TruthfullQA	平均值
	准确率（简单模式）	准确率（混乱模式）	准确率（平均模式）	*Acc*	*Acc*	*Acc*	
LLaMA-7B +零初始化	62.29 ↑1.64	43.17 ↑2.47	52.73 ↑2.06	36.28 ↑1.16	76.79 ↑4.17	45.53 ↑7.71	52.83 ↑3.77
LLaMA-7B + 随机初始化	60.65	40.7	50.67	35.12	72.62	37.82	49.06
LLaMA-13B模型采用 +零初始化	81.78 ↑0.17	64.33 ↑0.42	73.06 ↑0.3	49.64 ↑1.62	81.21 ↑0.05	34.88 ↑0.36	59.70 ↑0.58
LLaMA-13B + 的随机初始化	81.61	63.91	72.76	48.02	81.16	34.52	59.12

`MMLU` (Hendrycks等人，2020年)，以及(iv) TruthfulQA测试(Lin等人，2021年)。这些任务均通过选择题来评估模型性能：ARC测试和MMLU测试大语言模型的知识储备，HellaSwag测试模型完成句子的能力，而TruthfulQA测试大语言模型在回答给定问题时是否诚实。

We follow the experimental setup of LLaMA-Adapter (Zhang et al., 2024) by fine-tuning LLaMA on the Alpaca dataset (Taori et al., 2023). The model performance is evaluated on the test set by conducting a zero-shot evaluation for ARC, MMLU, and TruthfulQA while using a 10-shot setting for HellaSwag. Here, n -shot refers to incorporating n instruction-following samples into the prompt question.

Architectures Training. We employ experiments on two LLaMA versions, LLaMA-7B with 32 Transformer layers and LLaMA-13B with 40 Transformer layers (Touvron et al., 2023b;c). The models are trained with 4 A100-GPUs for 5 epochs. The training configuration includes a warmup period of 2 epochs, a total batch size of 64, a learning rate of 0.009, and a weight decay of 0.02. With LLaMA-7B, we use a prompt with length $L = 10$ and integrate adaptation prompts into the last $K = 30$ layers. On LLaMA-13B, we use $L = 10$ and insert prompts at the last $K = 38$ layers.

基准方法。为验证零初始化注意力机制的效率，并展示将其与所提出的非线性提示词结合的潜力，我们采用以下配置开展了实验。

- (1) **线性提示**：即 LLaMA-Adapter 的默认设置（张等人，2024年），在此设置中，提示词向量$\mathbf{P}$会被进行零初始化后直接使用。
- (2) **非线性提示**：采用零初始化机制，对输入的$\mathbf{P}$˜提示词应用非线性的多层感知机，以此生成$\mathbf{P} = \sigma(\mathbf{P})$提示词向量，且这些多层感知机层会在不同层级之间共享。
- (3)**随机初始化提示**：使用采用传统随机初始化方式的输入提示词，也就是方程（4）中的形式，而非采用零初始化机制的形式。
- (4) **微调与低秩分解**：我们将该方法与在……上训练的完全微调版LLaMA模型进行了对比Alpaca数据集（Taori等人，2023年），在该数据集中所有模型参数都会被更新。此外，我们还将其与LoRA技术（Hu等人，2021年）进行了对比测试，后者是一种PEFT方法，会在每一层的权重中插入可训练的低秩分解矩阵；同时也与VeRA技术（Kopiczko等人，2023年）进行了比较，后者同样是PEFT方法，但它使用一对在所有层间共享的随机低秩矩阵，并为每一层单独学习缩放向量。
- (5) **其他提示词微调方法**：包括（Lester等人，2021年）这类方法旨在优化添加到输入文本中的连续型提示词；IA3算法(刘等人，2022年）是一种利用三个可学习向量来重新调整注意力层中键值及位置级前馈激活值的方法。

5.2. 主要结果

I. 零初始化的注意力机制在提示词微调中至关重要，与随机初始化的注意力机制相比，它能提升<span style id='1'>鲁棒性和<span style id='3'>有效性。

我们首先在采用线性提示词设置的前提下，研究零初始化的提示词调优对LLaMA-7B模型和LLaMA-13B模型的影响，并将其性能与传统的随机初始化策略进行对比。

As shown in Table 1, the zero-initialized mechanism enhances stability in LLaMA-13B and significantly boosts the performance of LLaMA-Adapter when using LLaMA-7B, compared to conventional random-initialization strategies. For instance, with LLaMA-7B on Hellaswag and TruthfulQA, the Linear prompt surpasses the Random-Init prompt by 4.17% and 7.71%, respectively. The varying impact of zero-initialization between LLaMA-7B and LLaMA-13B can be attributed to differences in model capacity and expressiveness. LLaMA-13B, with its larger parameter space, naturally generalizes better even with randomly initialized prompts, reducing the relative advantage of zero-initialization. In contrast, LLaMA-7B, with its lower capacity, benefits more from zero-initialization, as it relies heavily on efficient adaptation mechanisms to optimize learning.

总而言之，这些结果与我们第4节中的理论研究结论一致，后者展示了如何通过<code>零初始化</code>来提升参数估计的鲁棒性及样本效率。</code>

attention improves both robustness and sample efficiency in parameter estimation.

II. Non-linear prompts provide the potential to improve the performance of the zero-initialized mechanism and achieve competitive performance with full fine-tuning.

We evaluate the impact of the `non-linear prompt` setting on the LLaMA-Adapter by comparing it with the `linear prompt` setting on LLaMA-7B and LLaMA-13B. Additionally, we benchmark against a fully fine-tuned LLaMA model and a version fine-tuned using LoRA. All models are trained with the Alpaca dataset.

Tables 2 and 3 presents our results with the following observations: first, the `non-linear prompt` consistently matches or outperforms the `linear prompt` across both LLaMA-7B and LLaMA-13B, with performance gains ranging from 1–2%. Notably, on the MMLU and TruthfulQA datasets with LLaMA-13B, the non-linear prompt achieves scores of 51.32 and 38.92, respectively, compared to 49.64 and 34.88 for the linear prompt, demonstrating its effectiveness. Table 3 also shows that the non-linear prompt does not significantly increase training time compared to the linear version.

Secondly, compared to other PEFT methods such as Prompt Tuning, IA3, and VeRA, we see that the `non-linear prompt` significantly outperforms all these PEFT methods in LLaMA-7B version. For example, on the MMLU and TruthfulQA, the `non-linear prompt` surpasses Prompt Tuning by 4.1% and 10.28%, respectively, demonstrating the effectiveness of the new PEFT design.

Finally, when compared to fully fine-tuned and LoRA-based versions of LLaMA-7B, our approach using a non-linear prompt achieves performance nearly on par with full fine-tuning while surpassing LoRA by an average accuracy margin of 0.64%

In summary, these findings validate our theoretical observations and highlight the effectiveness of combining non-linear prompts with zero-initialized attention and improving prompt-tuning performance in LLMs.

III. Sample Efficiency of Zero-Initialized Attention vs. Random-Initialized Attention.

Figures 2 and 3 provide a systematic analysis of the sample efficiency of zero-initialized attention by evaluating its performance under varying data availability. Specifically, we randomly subsample the Alpaca dataset at different fractions {1%, 10%, 30%, 50%, 100%} to simulate low-data scenarios. We then fine-tune the `Non-Linear`, `Linear`, and `Random-Init` prompts on these subsets for both LLaMA-7B and LLaMA-13B and evaluate their performance on the ARC dataset. This experiment allows us to assess how well each initialization strategy adapts to limited data and whether zero-initialized attention provides a consistent advantage in sample efficiency.

We observe that both Non-Linear and Linear prompts significantly enhance sample efficiency in parameter estimation for prefix-tuning in LLMs compared to the Random-Init prompt (which uses conventional attention). In the LLaMA-7B setting (Figure 2), both Non-Linear and Linear prompts outperform Random-Init across all fractions of the Alpaca training set. For example, the Non-Linear prompt exceeds Random-Init by 3.77% when trained on 100% of the dataset, while Linear exceeds it by 2.05%. When trained on 50% of the dataset, Non-Linear and Linear outperform Random-Init by 4.72% and 2.16%, respectively.

In the LLaMA-13B setting (Figure 3), a similar trend is observed, with zero-initialized attention showing consistent advantages over Random-Init, and the Non-Linear prompt slightly outperforming the Linear prompt in most cases. The only exception is at 1% training data, where Linear surpasses Non-Linear by 1.99% in accuracy. In short, these findings corroborate our theoretical results in Section 4, which demonstrate the sample efficiency of the zero-initialized attention mechanism in parameter estimation.

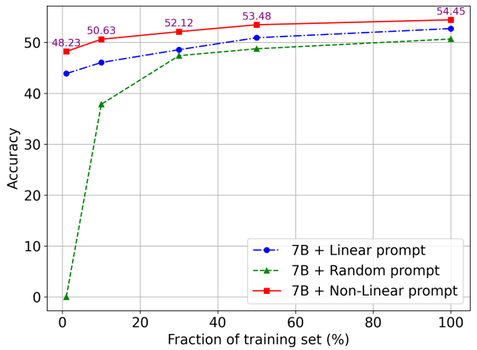


Figure 2. Accuracy of different prompt strategies (Linear, Random, and Non-Linear) with a **LLaMa-7B** model across varying fractions of the training set. The Non-Linear prompt consistently outperforms the other methods, especially in low-data regimes.

6. Conclusion and Limitations

In this paper, we demonstrate that zero-initialization prompt-tuning for adapting LLMs is not just an engineering trick but can be rigorously explained through theoretical properties by drawing connections between attention mechanisms and the mixture-of-experts perspective. Based on these insights, we introduce a novel non-linear prompt-tuning approach that outperforms linear prompts in terms of both performance and robustness while achieving competitive results compared to the full fine-tuning of LLaMA on the

注意力机制能够提升参数估计过程中的鲁棒性，并同时提高样本效率。

II. 非线性提示向量有望提升零初始化机制的性能，使其达到与完整微调相当的出色表现。

我们通过将非线性提示词设置与在LLaMA-7B及LLaMA-13B上使用的线性提示词设置进行对比，来评估该设置对LLaMA-Adapter的影响。此外，我们还将其与经过完整微调的LLaMA模型以及通过LoRA技术进行微调的版本进行了性能基准测试。所有模型均使用Alpaca数据集进行训练。

Tables 2 and 3 presents our results with the following observations: first, the `non-linear prompt` consistently matches or outperforms the `linear prompt` across both LLaMA-7B and LLaMA-13B, with performance gains ranging from 1–2%. Notably, on the MMLU and TruthfulQA datasets with LLaMA-13B, the non-linear prompt achieves scores of 51.32 and 38.92, respectively, compared to 49.64 and 34.88 for the linear prompt, demonstrating its effectiveness. Table 3 also shows that the non-linear prompt does not significantly increase training time compared to the linear version.

其次，与Prompt Tuning、IA3以及VeRA等其他PEFT方法相比，我们发现在该版本的LLaMA-7B模型中，非线性提示词的表现显著优于所有这些PEFT方法。例如，在MMLU和TruthfulQA测试中，非线性提示词的得分分别比Prompt Tuning高出4.1%和10.28%，充分体现了这种新PEFT设计的高效性。

最后，与经过完全微调以及基于LoRA技术的LLaMA-7B版本相比，我们采用非线性提示词的方法所取得的性能几乎可与完全微调相媲美，同时在平均准确率上比LoRA技术高出0.64%。

总而言之，这些研究结果验证了我们的理论观察，并凸显了将非线性提示词与零初始化注意力机制相结合、从而提升大语言模型中提示词调优性能的有效性。

III. 零初始化注意力机制与随机初始化注意力机制的样本效率对比。

图表 2 和3 通过评估在不同数据可用性条件下的表现，对零初始化注意力机制的样本效率进行了系统分析。具体而言，我们以不同的比例{1%、10%、30%、50%、100%} 随机抽取Alpaca数据集的子集，以此模拟数据量较少的场景。随后，我们对LLaMA-7B和LLaMA-13B在这些子集上分别针对非线性、线性策略以及随机初始化提示进行微调，并在ARC数据集上测试它们的性能。该实验有助于我们了解目前尚不清楚各种初始化策略在面对数据量有限的情况时表现如何，也不清楚零初始化注意力机制是否能在样本效率方面带来持续的优势。

我们发现，与采用传统注意力机制的随机初始化提示相比，非线性提示和线性提示都能显著提升大语言模型中前缀微调的参数估计样本效率。在LLaMA-70亿参数量实验设置中（见图2），在Alpaca训练集的各类子集中，非线性提示和线性提示的表现均优于随机初始化提示。例如，在使用全部数据集进行训练时，非线性提示的效率比随机初始化提示高3.77%，而线性提示则高2.05%；在仅使用50%的数据集进行训练时，非线性提示和线性提示的效率分别比随机初始化提示高4.72%和2.16%。

在LLaMA-13B模型设置下（图3），观察到了类似的趋势：零初始化注意力机制始终优于随机初始化提示词，而在大多数情况下，非线性提示词的表现也略好于线性提示词。唯一的例外是在仅使用1%训练数据时，线性策略在准确率上比非线性策略高出1.99%。简而言之，这些研究结果印证了4第节中的理论结论，即零初始化注意力机制在参数估计方面具备更高的样本效率。

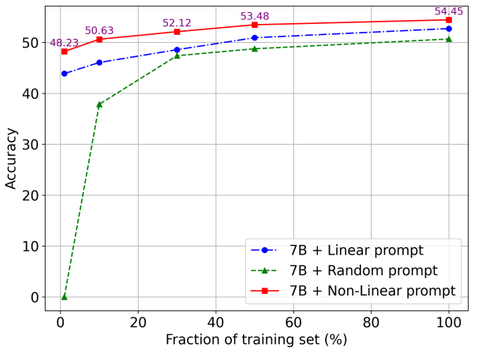


图2。展示在训练集不同比例数据条件下，**LLaMa-7B** 模型采用不同提示词策略（线性策略、随机策略以及非线性策略）时的准确率表现。实验结果表明，非线性提示词的效果始终优于其他方法，尤其在数据量较少的场景中优势更为明显。

6. 结论与局限性

在本文中，我们证明了用于适配大语言模型的零初始化提示词微调并非仅仅是一种工程技巧，而是可以通过将注意力机制与专家混合视角相联系，借助理论特性进行严谨解释的。基于这些见解，我们提出了一种全新的非线性提示词微调方法，该方法在性能与鲁棒性方面均优于线性提示词，同时在针对Alpaca数据集进行完整微调的对比实验中也能取得相当出色的结果。

Table 2. Comparison of `Non-Linear prompt`, `Linear prompt`, and various fine-tuning methods. **Params** denote the total number of parameters updated during the fine-tuning process. **Bold** values indicate better scores between linear and non-linear settings.

Method	Params	ARC			MMLU	Hellaswag	TruthfullQA	Average
		Acc (eas)	*Acc (cha)*	*Acc (aver)*	*Acc*	*Acc*	*Acc*	
LLaMA-7B, Fully Fine-tuning Alpaca	7B	67.47	46.25	56.86	37.25	77.09	42.35	53.39
LLaMA-7B, LoRA Alpaca	4.2M	61.91	42.15	52.03	34.87	77.53	46.14	52.64
LLaMA-7B, Prompt Tuning Alpaca	42K	55.35	37.46	46.41	32.85	75.88	34.76	47.48
LLaMA-7B, IA3 Alpaca	524K	52.06	35.92	43.99	31.65	75.73	32.8	46.04
LLaMA-7B, VeRA Alpaca	541K	49.2	35.49	42.35	30.88	75.59	31.95	45.19
LLaMA-7B + zero-init + linear	1.2M	62.29	43.17	52.73	36.28	**76.79**	**45.53**	52.83
LLaMA-7B + zero-init + non-linear	2.6M	**63.51**	**45.39**	**54.45**	**36.95**	76.67	45.04	**53.28**
LLaMA-13B + zero-init + linear	1.9M	81.78	64.33	73.06	49.64	81.21	34.88	59.70
LLaMA-13B + zero-init + non-linear	3.3M	**82.87**	**66.55**	**74.71**	**51.32**	**81.72**	**38.92**	**61.67**

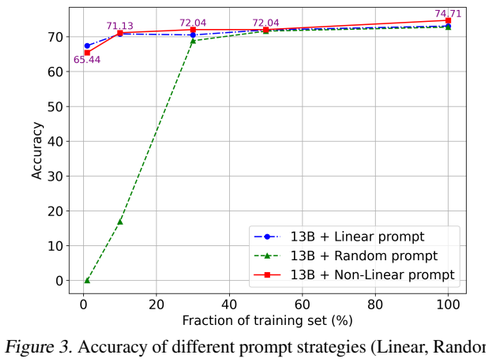


Figure 3. Accuracy of different prompt strategies (Linear, Random, and Non-Linear) using the **LLaMa-13B** model across varying fractions of the training set. The Non-Linear prompt consistently achieves the highest accuracy, particularly in low-resource settings. In contrast, the Random prompt performs significantly worse when only a small portion of the training data is available, but gradually improves as more data is introduced.

Table 3. Efficiency Comparison. The training is conducted on 4 GPUs A100-80GB.

Method	Tuned Params	Storage Space	Training Time
LLaMA-7B + zero-init + linear	1.2M	4.4M	1h54'
LLaMA-7B + zero-init + non-linear	2.6M	9.5M	1h54'
LLaMA-13B + zero-init + linear	1.9M	6.9M	3h17'
LLaMA-13B + zero-init + non-linear	3.3M	12M	3h17'

Alpaca dataset. Our findings are validated across several question-answering tasks, tested on both LLaMA-7B and LLaMA-13B architectures, and align with our theoretical analysis in practical settings. We believe our results will encourage further exploration into initialization techniques and their potential for improving other parameter-efficient fine-tuning methods.

Our study also has some limitations that need further exploration. First, it has only been tested on unimodal LLMs, leaving the performance on multi-modal LLMs unexamined. Multi-modal tasks may require adjustments to the framework in both theoretical perspectives and practical implementations (Li et al., 2023; Nguyen et al., 2024a). Second, the integration of adapters and the optimal number of prompt embeddings across layers have not been systematically explored. Further empirical experiments are needed to refine these components for better efficiency and adaptability in more complex models. Finally, the current study primarily focuses on performance metrics, but the interpretability and stability of non-linear prompts in real-world deployment scenarios require further analysis. Addressing these limitations will contribute to a deeper understanding of the robustness and generalizability of PEFT for LLaMA models.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

Acknowledgment

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表2.本表对比了非线性提示词、线性提示词以及多种微调方法。**参数数量**指的是在微调过程中需要更新的所有参数的总数。**加粗**的数值表示在线性设置与非线性设置下的性能表现更佳。

方法	参数数量	ARC			MMLU	HellaSwag	TruthfullQA	平均值
		准确率（简单场景）	准确率（复杂场景）	准确率（平均场景）	*Acc*	*Acc*	*Acc*	
LLaMA-7B模型，对Alpaca模型进行全微调	7B	67.47	46.25	56.86	37.25	77.09	42.35	53.39
LLaMA-7B模型，基于LoRA技术的Alpaca模型	4.2M	61.91	42.15	52.03	34.87	77.53	46.14	52.64
LLaMA-7B模型结合提示词微调与Alpaca模型	42K	55.35	37.46	46.41	32.85	75.88	34.76	47.48
LLaMA-7B模型结合IA3算法与Alpaca模型	524K	52.06	35.92	43.99	31.65	75.73	32.8	46.04
LLaMA-7B模型结合VeRA技术与Alpaca模型	541K	49.2	35.49	42.35	30.88	75.59	31.95	45.19
LLaMA-7B模型采用零初始化并搭配线性策略	1.2M	62.29	43.17	52.73	36.28	**76.79**	**45.53**	52.83
LLaMA-7B模型采用零初始化并搭配非线性策略	2.6M	**63.51**	**45.39**	**54.45**	**36.95**	76.67	45.04	**53.28**
LLaMA-13B模型采用零初始化并搭配线性策略	1.9M	81.78	64.33	73.06	49.64	81.21	34.88	59.70
LLaMA-13B + 零初始化 + 非线性	3.3M	**82.87**	**66.55**	**74.71**	**51.32**	**81.72**	**38.92**	**61.67**

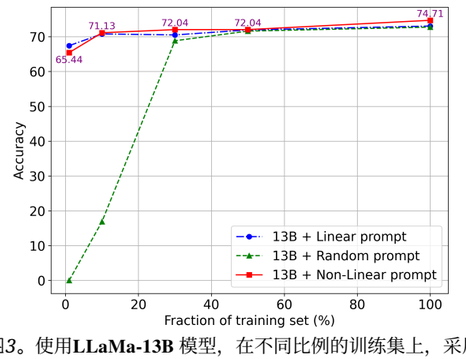


图*3*。使用**LLaMa-13B** 模型，在不同比例的训练集上，采用不同提示词策略（线性策略、随机策略以及非线性策略）时的准确率表现。实验结果显示，非线性提示词始终能获得最高的准确率，尤其是在资源有限的场景下表现更为突出。相比之下，当仅有少量训练数据可用时，随机策略的准确率会显著下降，但随着数据量的增加，其性能也会逐渐提升。

表*3*。效率对比分析。相关训练是在4块A100-80GB型号GPU上完成的。

方法	经过调优参数数量	存储需求空间	训练 Time
LLaMA-7B +零初始化 + 线性策略	1.2M	4.4M	1小时54分钟
LLaMA-7B +零初始化 + 非线性策略	2.6M	9.5M	1小时54分钟
LLaMA-13B +采用零初始化 + 线性策略	1.9M	6.9M	3小时17分钟
LLaMA-13B +采用零初始化 + 非线性策略	3.3M	12M	3小时17分钟

我们的研究结果已在多项问答任务中得到验证，测试对象涵盖了LLaMA-7B和LLaMA-13B两种架构，并且在实际应用场景下也与我们的理论分析相吻合。我们相信，这些成果将会推动人们进一步探索初始化技术，以及其在改进其他参数高效微调方法方面的潜力。

我们的研究也存在一些需要进一步探讨的局限性。首先，它目前仅在单模态大语言模型上进行了测试。

leaving the performance on multi-modal LLMs unexamined. Multi-modal tasks may require adjustments to the framework in both theoretical perspectives and practical implementations (Li et al., 2023; Nguyen et al., 2024a). Second, the integration of adapters and the optimal number of prompt embeddings across layers have not been systematically explored. Further empirical experiments are needed to refine these components for better efficiency and adaptability in more complex models. Finally, the current study primarily focuses on performance metrics, but the interpretability and stability of non-linear prompts in real-world deployment scenarios require further analysis. Addressing these limitations will contribute to a deeper understanding of the robustness and generalizability of PEFT for LLaMA models.

影响声明

本文介绍的研究工作旨在推动机器学习领域的发展。我们的研究可能带来诸多社会层面的影响，但我们认为无需在此特别强调。

致谢

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References

Akbarian, P., Nguyen, H., Han, X., and Ho, N. Quadratic gating functions in mixture of experts: A statistical insight. *arXiv preprint arXiv:2410.11222*, 2024.

Anil, R., Dai, A. M., Firat, O., Johnson, M., Lepikhin, D., Passos, A., Shakeri, S., Taropa, E., Bailey, P., Chen, Z., et al. Palm 2 technical report. *arXiv preprint arXiv:2305.10403*, 2023.

Beeching, E., Fourrier, C., Habib, N., Han, S., Lambert, N., Rajani, N., Sanseviero, O., Tunstall, L., and Wolf, T. Open llm leaderboard. https://huggingface.co/spaces/HuggingFaceH4/open_llm_leaderboard, 2024. Accessed:2024.

Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J. D., Dhariwal, P., Neelakantan, A., Shyam, P., Sastry, G., Askell, A., et al. Language models are few-shot learners. *Advances in neural information processing systems*, 33: 1877–1901, 2020.

Chen, S., Jie, Z., and Ma, L. Llava-mole: Sparse mixture of lora experts for mitigating data conflicts in instruction finetuning mllms. *arXiv preprint arXiv:2401.16160*, 2024.

Chen, T., Chen, X., Du, X., Rashwan, A., Yang, F., Chen, H., Wang, Z., and Li, Y. Adamv-moe: Adaptive multi-task vision mixture-of-experts. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 17346–17357, 2023.

Chen, Z., Deng, Y., Wu, Y., Gu, Q., and Li, Y. Towards understanding the mixture-of-experts layer in deep learning. In Koyejo, S., Mohamed, S., Agarwal, A., Belgrave, D., Cho, K., and Oh, A. (eds.), *Advances in Neural Information Processing Systems*, volume 35, pp. 23049–23062. Curran Associates, Inc., 2022.

Clark, P., Cowhey, I., Etzioni, O., Khot, T., Sabharwal, A., Schoenick, C., and Tafjord, O. Think you have solved question answering? try arc, the ai2 reasoning challenge. *arXiv preprint arXiv:1803.05457*, 2018.

Devlin, J. Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*, 2018.

Fedus, W., Zoph, B., and Shazeer, N. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity. *Journal of Machine Learning Research*, 23(120):1–39, 2022.

Gao, P., Geng, S., Zhang, R., Ma, T., Fang, R., Zhang, Y., Li, H., and Qiao, Y. Clip-adapter: Better vision-language models with feature adapters. *International Journal of Computer Vision*, 132(2):581–595, 2024.

Han, X., Nguyen, H., Harris, C., Ho, N., and Saria, S. Fusemoe: Mixture-of-experts transformers for fleximodal fusion. In *Advances in Neural Information Processing Systems*, 2024.

Hendrycks, D., Burns, C., Basart, S., Zou, A., Mazeika, M., Song, D., and Steinhardt, J. Measuring massive multitask language understanding. *arXiv preprint arXiv:2009.03300*, 2020.

Ho, N., Yang, C.-Y., and Jordan, M. I. Convergence rates for Gaussian mixtures of experts. *Journal of Machine Learning Research*, 23(323):1–81, 2022.

Houlsby, N., Giurgiu, A., Jastrzebski, S., Morrone, B., De Laroussilhe, Q., Gesmundo, A., Attariyan, M., and Gelly, S. Parameter-efficient transfer learning for nlp. In *International conference on machine learning*, pp. 2790–2799. PMLR, 2019.

Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., and Chen, W. Lora: Low-rank adaptation of large language models. *arXiv preprint arXiv:2106.09685*, 2021.

Jacobs, R. A., Jordan, M. I., Nowlan, S. J., and Hinton, G. E. Adaptive mixtures of local experts. *Neural Computation*, 3, 1991.

Jordan, M. I. and Jacobs, R. A. Hierarchical mixtures of experts and the em algorithm. *Neural computation*, 6(2): 181–214, 1994.

Karimi Mahabadi, R., Henderson, J., and Ruder, S. Compacter: Efficient low-rank hypercomplex adapter layers. *Advances in Neural Information Processing Systems*, 34: 1022–1035, 2021.

Kopiczko, D. J., Blankevoort, T., and Asano, Y. M. Vera: Vector-based random matrix adaptation. *arXiv preprint arXiv:2310.11454*, 2023.

Le, M., Nguyen, A., Nguyen, H., Nguyen, T., Pham, T., Van Ngo, L., and Ho, N. Mixture of experts meets prompt-based continual learning. *Advances in Neural Information Processing Systems*, 38, 2024.

Le, M., Nguyen, C., Nguyen, H., Tran, Q., Le, T., and Ho, N. Revisiting prefix-tuning: Statistical benefits of reparameterization among prompts. In *The Thirteenth International Conference on Learning Representations*, 2025.

Lester, B., Al-Rfou, R., and Constant, N. The power of scale for parameter-efficient prompt tuning. *arXiv preprint arXiv:2104.08691*, 2021.

参考文献

Akbarian, P., Nguyen, H., Han, X., 和 Ho, N. 《专家混合模型中的二次门控函数：一种统计学视角》。*arXiv*预印本平台 *arXiv:2410.11222*，2024年。

Anil, R., Dai, A. M., Firat, O., Johnson, M., Lepikhin, D., Passos, A., Shakeri, S., Taropa, E., Bailey, P., Chen, Z. 等人。《Palm 2技术报告》。*arXiv*预印本平台 *arXiv:2305.10403*，2023年。

Beeching, E., Fourrier, C., Habib, N., Han, S., Lambert, N., Rajani, N., Sanseviero, O., Tunstall, L., and Wolf, T. Open llm leaderboard. https://huggingface.co/spaces/HuggingFaceH4/open_llm_leaderboard, 2024. Accessed:2024.

Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J. D., Dhariwal, P., Neelakantan, A., Shyam, P., Sastry, G., Askell, A., et al. Language models are few-shot learners. *Advances in neural information processing systems*, 33: 1877–1901, 2020.

Chen, S., Jie, Z., and Ma, L. Llava-mole: Sparse mixture of lora experts for mitigating data conflicts in instruction finetuning mllms. *arXiv preprint arXiv:2401.16160*, 2024.

Chen, T., Chen, X., Du, X., Rashwan, A., Yang, F., Chen, H., Wang, Z., and Li, Y. Adamv-moe: Adaptive multi-task vision mixture-of-experts. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 17346–17357, 2023.

Chen, Z., Deng, Y., Wu, Y., Gu, Q., and Li, Y. Towards understanding the mixture-of-experts layer in deep learning. In Koyejo, S., Mohamed, S., Agarwal, A., Belgrave, D., Cho, K., and Oh, A. (eds.), *Advances in Neural Information Processing Systems*, volume 35, pp. 23049–23062. Curran Associates, Inc., 2022.

Clark, P., Cowhey, I., Etzioni, O., Khot, T., Sabharwal, A., Schoenick, C., and Tafjord, O. Think you have solved question answering? try arc, the ai2 reasoning challenge. *arXiv preprint arXiv:1803.05457*, 2018.

Devlin, J. Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*, 2018.

Fedus, W., Zoph, B., and Shazeer, N. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity. *Journal of Machine Learning Research*, 23(120):1–39, 2022.

Gao, P., Geng, S., Zhang, R., Ma, T., Fang, R., Zhang, Y., Li, H., and Qiao, Y. Clip-adapter: Better vision-language models with feature adapters. *International Journal of Computer Vision*, 132(2):581–595, 2024.

Han, X., Nguyen, H., Harris, C., Ho, N., and Saria, S. Fusemoe: Mixture-of-experts transformers for fleximodal fusion. In *Advances in Neural Information Processing Systems*, 2024.

Hendrycks, D., Burns, C., Basart, S., Zou, A., Mazeika, M., Song, D., and Steinhardt, J. Measuring massive multitask language understanding. *arXiv preprint arXiv:2009.03300*, 2020.

Ho, N., Yang, C.-Y., and Jordan, M. I. Convergence rates for Gaussian mixtures of experts. *Journal of Machine Learning Research*, 23(323):1–81, 2022.

Houlsby, N., Giurgiu, A., Jastrzebski, S., Morrone, B., De Laroussilhe, Q., Gesmundo, A., Attariyan, M., and Gelly, S. Parameter-efficient transfer learning for nlp. In *International conference on machine learning*, pp. 2790–2799. PMLR, 2019.

Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., and Chen, W. Lora: Low-rank adaptation of large language models. *arXiv preprint arXiv:2106.09685*, 2021.

Jacobs, R. A., Jordan, M. I., Nowlan, S. J., and Hinton, G. E. Adaptive mixtures of local experts. *Neural Computation*, 3, 1991.

Jordan, M. I. and Jacobs, R. A. Hierarchical mixtures of experts and the em algorithm. *Neural computation*, 6(2): 181–214, 1994.

Karimi Mahabadi, R., Henderson, J., and Ruder, S. Compacter: Efficient low-rank hypercomplex adapter layers. *Advances in Neural Information Processing Systems*, 34: 1022–1035, 2021.

Kopiczko, D. J., Blankevoort, T., and Asano, Y. M. Vera: Vector-based random matrix adaptation. *arXiv preprint arXiv:2310.11454*, 2023.

Le, M., Nguyen, A., Nguyen, H., Nguyen, T., Pham, T., Van Ngo, L., and Ho, N. Mixture of experts meets prompt-based continual learning. *Advances in Neural Information Processing Systems*, 38, 2024.

Le, M., Nguyen, C., Nguyen, H., Tran, Q., Le, T., and Ho, N. Revisiting prefix-tuning: Statistical benefits of reparameterization among prompts. In *The Thirteenth International Conference on Learning Representations*, 2025.

Lester, B., Al-Rfou, R., and Constant, N. The power of scale for parameter-efficient prompt tuning. *arXiv preprint arXiv:2104.08691*, 2021.

Li, C., Wong, C., Zhang, S., Usuyama, N., Liu, H., Yang, J., Naumann, T., Poon, H., and Gao, J. Llava-med: Training a large language-and-vision assistant for biomedicine in one day. *Advances in Neural Information Processing Systems*, 36:28541–28564, 2023.

Li, D., Ma, Y., Wang, N., Cheng, Z., Duan, L., Zuo, J., Yang, C., and Tang, M. Mixlora: Enhancing large language models fine-tuning with lora based mixture of experts. *arXiv preprint arXiv:2404.15159*, 2024a.

Li, H., Lin, S., Duan, L., Liang, Y., and Shroff, N. B. Theory on mixture-of-experts in continual learning. *arXiv preprint arXiv:2406.16437*, 2024b.

Li, X., Lian, D., Lu, Z., Bai, J., Chen, Z., and Wang, X. Graphadapter: Tuning vision-language models with dual knowledge graph. *Advances in Neural Information Processing Systems*, 36, 2024c.

Li, X. L. and Liang, P. Prefix-tuning: Optimizing continuous prompts for generation. *arXiv preprint arXiv:2101.00190*, 2021.

Lin, S., Hilton, J., and Evans, O. Truthfulqa: Measuring how models mimic human falsehoods. *arXiv preprint arXiv:2109.07958*, 2021.

Liu, H., Tam, D., Muqeeth, M., Mohta, J., Huang, T., Bansal, M., and Raffel, C. A. Few-shot parameter-efficient fine-tuning is better and cheaper than in-context learning. *Advances in Neural Information Processing Systems*, 35: 1950–1965, 2022.

Liu, X., Ji, K., Fu, Y., Tam, W. L., Du, Z., Yang, Z., and Tang, J. P-tuning v2: Prompt tuning can be comparable to fine-tuning universally across scales and tasks. *arXiv preprint arXiv:2110.07602*, 2021.

Mangrulkar, S., Gugger, S., Debut, L., Belkada, Y., Paul, S., and Bossan, B. Peft: State-of-the-art parameter-efficient fine-tuning methods. *URL: https://github.com/huggingface/peft*, 2022.

Manole, T. and Ho, N. Refined convergence rates for maximum likelihood estimation under finite mixture models. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pp. 14979–15006. PMLR, 17–23 Jul 2022.

Minaee, S., Mikolov, T., Nikzad, N., Chenaghlu, M., Socher, R., Amatriain, X., and Gao, J. Large language models: A survey. *arXiv preprint arXiv:2402.06196*, 2024.

Nguyen, D. M., Diep, N. T., Nguyen, T. Q., Le, H.-B., Nguyen, T., Nguyen, T., Nguyen, T., Ho, N., Xie, P., Wattenhofer, R., et al. Logra-med: Long context multi-graph alignment for medical vision-language model. *arXiv preprint arXiv:2410.02615*, 2024a.

Nguyen, D. M., Le, A. T., Nguyen, T. Q., Diep, N. T., Nguyen, T., Duong-Tran, D., Peters, J., Shen, L., Niepert, M., and Sonntag, D. Dude: Dual distribution-aware context prompt learning for large vision-language model. *Asian Conference on Machine Learning (ACML)*, 2024b.

Nguyen, H., Nguyen, T., and Ho, N. Demystifying softmax gating function in gaussian mixture of experts. *Advances in Neural Information Processing Systems*, 36: 4624–4652, 2023.

Nguyen, H., Akbarian, P., and Ho, N. Is temperature sample efficient for softmax Gaussian mixture of experts? In *Proceedings of the ICML*, 2024c.

Nguyen, H., Akbarian, P., Nguyen, T., and Ho, N. A general theory for softmax gating multinomial logistic mixture of experts. In *Proceedings of the ICML*, 2024d.

Nguyen, H., Han, X., Harris, C. W., Saria, S., and Ho, N. On expert estimation in hierarchical mixture of experts: Beyond softmax gating functions. *arxiv preprint arxiv 2410.02935*, 2024e.

Nguyen, H., Ho, N., and Rinaldo, A. Sigmoid gating is more sample efficient than softmax gating in mixture of experts. In *Advances in Neural Information Processing Systems*, 2024f.

Nguyen, H., Akbarian, P., Pham, T., Nguyen, T., Zhang, S., and Ho, N. Statistical advantages of perturbing cosine router in mixture of experts. In *International Conference on Learning Representations*, 2025.

OpenAI. Chatgpt. https://chat.openai.com, 2025. Accessed: 2025.

OpenAI, J. A., Adler, S., Agarwal, S., Ahmad, L., Akkaya, I., Aleman, F. L., Almeida, D., Altenschmidt, J., Altman, S., Anadkat, S., et al. Gpt-4 technical report, 2024. *URL https://arxiv. org/abs/2303.08774*, 2:6, 2024.

Pardo, L. *Statistical Inference Based on Divergence Measures*. Statistics: A Series of Textbooks and Monographs. CRC Press, 2018. ISBN 9781420034813. URL https://books.google.com/books?id=ziDGGIkhqlMC.

Peng, B., Li, C., He, P., Galley, M., and Gao, J. Instruction tuning with gpt-4. *arXiv preprint arXiv:2304.03277*, 2023.

Pham, Q., Do, G., Nguyen, H., Nguyen, T., Liu, C., Sartipi, M., Nguyen, B. T., Ramasamy, S., Li, X., Hoi, S., and Ho, N. Competesmoe – effective training of

Li, C., Wong, C., Zhang, S., Usuyama, N., Liu, H., Yang, J., Naumann, T., Poon, H., and Gao, J. Llava-med: Training a large language-and-vision assistant for biomedicine in one day. *Advances in Neural Information Processing Systems*, 36:28541–28564, 2023.

Li, D., Ma, Y., Wang, N., Cheng, Z., Duan, L., Zuo, J., Yang, C., and Tang, M. Mixlora: Enhancing large language models fine-tuning with lora based mixture of experts. *arXiv preprint arXiv:2404.15159*, 2024a.

Li, H., Lin, S., Duan, L., Liang, Y., and Shroff, N. B. Theory on mixture-of-experts in continual learning. *arXiv preprint arXiv:2406.16437*, 2024b.

李X、连D、卢Z、白J、陈Z以及王X。Graphadapter：利用双重知识图谱调优视觉语言模型。*Advancesin Neural Information Pro-cessingSystems*，36卷，2024年c期。

李晓磊与梁鹏，《前缀微调：优化用于生成的连续提示词》，*arXiv*预印本平台 *arXiv:2101.00190*，2021年。

Lin, S., Hilton, J., 和 Evans, O. Truthfulqa测试：衡量模型模仿人类虚假行为的能力。*arXiv*预印本平台 *arXiv:2109.07958*，2021年。

刘H.、谭D.、穆基斯M.、莫赫塔J.、黄T.、班萨尔M.以及拉费尔C. A.。少样本参数高效微调比上下文学习更优且成本更低。《神经信息处理系统进展》，35卷：1950–1965页，2022年。

刘X、季K、傅Y、谭W L、杜Z、杨Z与唐J。《P-tuning v2：提示词微调在各类规模与任务中的表现可媲美全量微调》。*arXiv*预印本 *arXiv:2110.07602*，2021年。

Mangrulkar, S., Gugger, S., Debut, L., Belkada, Y., Paul, S. 与 Bossan, B.。Peft：最先进的参数高效微调方法。网址：*https://github. com/huggingface/peft*，2022年。

Manole, T. and Ho, N. Refined convergence rates for maximum likelihood estimation under finite mixture models. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pp. 14979–15006. PMLR, 17–23 Jul 2022.

Minaee, S., Mikolov, T., Nikzad, N., Chenaghlu, M., Socher, R., Amatriain, X., and Gao, J. Large language models: A survey. *arXiv preprint arXiv:2402.06196*, 2024.

Nguyen, D. M., Diep, N. T., Nguyen, T. Q., Le, H.-B., Nguyen, T., Nguyen, T., Nguyen, T., Ho, N., Xie, P., Wattenhofer, R., et al. Logra-med: Long context multi-graph

alignment for medical vision-language model. *arXiv preprint arXiv:2410.02615*, 2024a.

Nguyen, D. M., Le, A. T., Nguyen, T. Q., Diep, N. T., Nguyen, T., Duong-Tran, D., Peters, J., Shen, L., Niepert, M., and Sonntag, D. Dude: Dual distribution-aware context prompt learning for large vision-language model. *Asian Conference on Machine Learning (ACML)*, 2024b.

Nguyen, H., Nguyen, T., and Ho, N. Demystifying softmax gating function in gaussian mixture of experts. *Advances in Neural Information Processing Systems*, 36: 4624–4652, 2023.

Nguyen, H., Akbarian, P., and Ho, N. Is temperature sample efficient for softmax Gaussian mixture of experts? In *Proceedings of the ICML*, 2024c.

Nguyen, H., Akbarian, P., Nguyen, T. 和 Ho, N. 针对多项式逻辑斯蒂特专家混合模型的Softmax门控通用理论。载于*ICML*会议论文集，2024年。

Nguyen, H., Han, X., Harris, C. W., Saria, S. 和 Ho, N.《层次化专家混合模型中的专家估计：超越Softmax门控函数》。*arxiv*预印本 *arxiv 2410.02935*，2024年。

Nguyen, H., Ho, N., 和 Rinaldo, A. 在专家混合模型中，S形函数门控相比Softmax门控更具样本效率。《神经信息处理系统进展》，2024年。

Nguyen, H., Akbarian, P., Pham, T., Nguyen, T., Zhang, S. 以及 Ho, N.《专家混合模型中扰动余弦路由器的统计优势》。载于国际学习表示会议，2025年。

OpenAI. ChatGPT. https://chat.openai.com，2025年。访问时间：2025年。

OpenAI, J. A., Adler, S., Agarwal, S., Ahmad, L., Akkaya, I., Aleman, F. L., Almeida, D., Altenschmidt, J., Altman, S., Anadkat, S. 等人。《Gpt-4技术报告》，2024年。网址：*https://arxiv. org/abs/2303.08774*，2:6，2024年。

Pardo, L. *Statistical Inference Based on Divergence Measures*. Statistics: A Series of Textbooks and Monographs. CRC Press, 2018. ISBN 9781420034813. URL https://books.google.com/books?id=ziDGGIkhqlMC.

Peng, B., Li, C., He, P., Galley, M., and Gao, J. Instruction tuning with gpt-4. *arXiv preprint arXiv:2304.03277*, 2023.

Pham, Q., Do, G., Nguyen, H., Nguyen, T., Liu, C., Sartipi, M., Nguyen, B. T., Ramasamy, S., Li, X., Hoi, S., and Ho, N. Competesmoe – effective training of

sparse mixture of experts via competition. *arXiv preprint arXiv:2402.02526*, 2024.

Radford, A., Wu, J., Child, R., Luan, D., Amodei, D., Sutskever, I., et al. Language models are unsupervised multitask learners. *OpenAI blog*, 1(8):9, 2019.

Raffel, C., Shazeer, N., Roberts, A., Lee, K., Narang, S., Matena, M., Zhou, Y., Li, W., and Liu, P. J. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of machine learning research*, 21 (140):1–67, 2020.

Rücklé, A., Geigle, G., Glockner, M., Beck, T., Pfeiffer, J., Reimers, N., and Gurevych, I. Adapterdrop: On the efficiency of adapters in transformers. *arXiv preprint arXiv:2010.11918*, 2020.

Shazeer, N., Mirhoseini, A., Maziarz, K., Davis, A., Le, Q., Hinton, G., and Dean, J. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. *arXiv preprint arXiv:1701.06538*, 2017.

Shi, Z. and Lipani, A. Dept: Decomposed prompt tuning for parameter-efficient fine-tuning. *arXiv preprint arXiv:2309.05173*, 2023.

Taori, R., Gulrajani, I., Zhang, T., Dubois, Y., Li, X., Guestrin, C., Liang, P., and Hashimoto, T. B. Stanford alpaca: An instruction-following llama model, 2023.

Touvron, H., Lavril, T., Izacard, G., Martinet, X., Lachaux, M.-A., Lacroix, T., Rozière, B., Goyal, N., Hambro, E., Azhar, F., et al. Llama: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971*, 2023a.

Touvron, H., Lavril, T., Izacard, G., Martinet, X., Lachaux, M.-A., Lacroix, T., Rozière, B., Goyal, N., Hambro, E., Azhar, F., et al. Llama: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971*, 2023b.

Touvron, H., Martin, L., Stone, K., Albert, P., Almahairi, A., Babaei, Y., Bashlykov, N., Batra, S., Bhargava, P., Bhosale, S., et al. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*, 2023c.

van de Geer, S. *Empirical processes in M-estimation*. Cambridge University Press, 2000.

Vaswani, A. Attention is all you need. *Advances in Neural Information Processing Systems*, 2017.

Yan, F., Nguyen, H., Le, D., Akbarian, P., and Ho, N. Understanding expert structures on minimax parameter estimation in contaminated mixture of experts. In *Proceedings of The 28th International Conference on Artificial Intelligence and Statistics*, 2025.

Zellers, R., Holtzman, A., Bisk, Y., Farhadi, A., and Choi, Y. Hellaswag: Can a machine really finish your sentence? *arXiv preprint arXiv:1905.07830*, 2019.

Zhang, Q., Chen, M., Bukharin, A., Karampatziakis, N., He, P., Cheng, Y., Chen, W., and Zhao, T. Adalora: Adaptive budget allocation for parameter-efficient fine-tuning. *arXiv preprint arXiv:2303.10512*, 2023.

Zhang, R., Han, J., Liu, C., Zhou, A., Lu, P., Qiao, Y., Li, H., and Gao, P. Llama-adapter: Efficient fine-tuning of large language models with zero-initialized attention. In *The Twelfth International Conference on Learning Representations*, 2024.

Zhou, K., Yang, J., Loy, C. C., and Liu, Z. Learning to prompt for vision-language models. *International Journal of Computer Vision*, 130(9):2337–2348, 2022.

基于竞争机制的稀疏专家混合模型。*arXiv*预印本 *arXiv:2402.02526*，2024年。

Radford, A., Wu, J., Child, R., Luan, D., Amodei, D., Sutskever, I. 等人。语言模型是无需监督的多任务学习器。《*OpenAI*博客》，1(8):9，2019年。

Raffel, C., Shazeer, N., Roberts, A., Lee, K., Narang, S., Matena, M., Zhou, Y., Li, W., 以及 Liu, P. J. 利用统一的文本到文本变换器探索迁移学习的极限。机器学习研究期刊，21 (140)：1–67，2020年。

R¨uckl´e, A., Geigle, G., Glockner, M., Beck, T., Pfeiffer, J., Reimers, N.以及Gurevych, I.。Adapterdrop: 探讨Transformer中适配器的效率问题。*arXiv*预印本平台 *arXiv:2010.11918*，2020年。

Shazeer, N., Mirhoseini, A., Maziarz, K., Davis, A., Le, Q., Hinton, G., and Dean, J. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. *arXiv preprint arXiv:1701.06538*, 2017.

Shi, Z. and Lipani, A. Dept: Decomposed prompt tuning for parameter-efficient fine-tuning. *arXiv preprint arXiv:2309.05173*, 2023.

Taori, R., Gulrajani, I., Zhang, T., Dubois, Y., Li, X., Guestrin, C., Liang, P., and Hashimoto, T. B. Stanford alpaca: An instruction-following llama model, 2023.

Touvron, H., Lavril, T., Izacard, G., Martinet, X., Lachaux, M.-A., Lacroix, T., Rozière, B., Goyal, N., Hambro, E., Azhar, F., et al. Llama: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971*, 2023a.

Touvron, H., Lavril, T., Izacard, G., Martinet, X., Lachaux, M.-A., Lacroix, T., Rozière, B., Goyal, N., Hambro, E., Azhar, F., et al. Llama: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971*, 2023b.

Touvron, H., Martin, L., Stone, K., Albert, P., Almahairi, A., Babaei, Y., Bashlykov, N., Batra, S., Bhargava, P., Bhosale, S., et al. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*, 2023c.

van de Geer, S. *Empirical processes in M-estimation*. Cambridge University Press, 2000.

Vaswani, A. Attention is all you need. *Advances in Neural Information Processing Systems*, 2017.

Yan, F., Nguyen, H., Le, D., Akbarian, P., and Ho, N. Understanding expert structures on minimax parameter estimation in contaminated mixture of experts. In *Proceedings of The 28th International Conference on Artificial Intelligence and Statistics*, 2025.

Zellers, R., Holtzman, A., Bisk, Y., Farhadi, A., and Choi, Y. Hellaswag: Can a machine really finish your sentence? *arXiv preprint arXiv:1905.07830*, 2019.

Zhang, Q., Chen, M., Bukharin, A., Karampatziakis, N., He, P., Cheng, Y., Chen, W., and Zhao, T. Adalora: Adaptive budget allocation for parameter-efficient fine-tuning. *arXiv preprint arXiv:2303.10512*, 2023.

Zhang, R., Han, J., Liu, C., Zhou, A., Lu, P., Qiao, Y., Li, H., and Gao, P. Llama-adapter: Efficient fine-tuning of large language models with zero-initialized attention. In *The Twelfth International Conference on Learning Representations*, 2024.

K·周、J·杨、C·C·洛伊以及Z·刘共同发表了题为《学习为视觉语言模型设计提示词》的文章。国际计算机视觉杂志，第130卷第9期，页码2337–2348，2022年。

Supplement to "On Zero-Initialized Attention: Optimal Prompt and Gating Factor Estimation"

In this supplementary material, we provide detailed proofs of the **main results** in Appendix A. Additional discussion on related works is in Appendix B and additional experimental details are in Appendix C.

A. Proofs

A.1. Proof of Theorem 4.2

Based on the convergence rate of $f_{\widehat{G}_n,\widehat{\alpha}_n}$ to f_{G_*,α_*} in Proposition 4.1, to obtain the conclusion of Theorem 4.2, we only need to demonstrate that

$$\|f_{G,\alpha} - f_{G_*,\alpha_*}\|_{L^2(\mu)} \geq C \cdot (\mathcal{D}(G, G_*) + |\alpha - \alpha_*|)$$

for any $(G, \alpha) \in \mathcal{G}_{L'}(\Theta) \times \Omega$ for some universal constant C. It is equivalent to proving that:

$$\inf_{(G,\alpha)\in\mathcal{G}_{L'}(\Theta)\times\Omega} \|f_{G,\alpha} - f_{G_*,\alpha_*}\|_{L^2(\mu)}/(\mathcal{D}(G, G_*) + |\alpha - \alpha_*|) > 0.$$

To obtain the conclusion for the above inequality, we consider two parts: (i) local part, namely,

$$\lim_{\varepsilon\to 0} \inf_{(G,\alpha)\in\mathcal{G}_{L'}(\Theta)\times\Omega:\mathcal{D}(G,G_*)+|\alpha-\alpha_*|\leq\varepsilon} \|f_{G,\alpha} - f_{G_*,\alpha_*}\|_{L^2(\mu)}/(\mathcal{D}(G, G_*) + |\alpha - \alpha_*|) > 0;$$

(ii) global part, namely, for any $\varepsilon > 0$

$$\inf_{(G,\alpha)\in\mathcal{G}_{L'}(\Theta)\times\Omega:\mathcal{D}(G,G_*)+|\alpha-\alpha_*|>\varepsilon} \|f_{G,\alpha} - f_{G_*,\alpha_*}\|_{L^2(\mu)}/(\mathcal{D}(G, G_*) + |\alpha - \alpha_*|) > 0;$$

Local part: We first start with the local part, which is equivalent to demonstrating that

$$\lim_{\varepsilon\to 0} \inf_{(G,\alpha)\in\mathcal{G}_{L'}(\Theta)\times\Omega:\mathcal{D}(G,G_*)+|\alpha-\alpha_*|\leq\varepsilon} \|f_{G,\alpha} - f_{G_*,\alpha_*}\|_{L^2(\mu)}/(\mathcal{D}(G, G_*) + |\alpha - \alpha_*|) > 0.$$

We prove the above claim by contradiction. Assume by contrary that the above claim does not hold. It indicates that we can find a sequence of mixing measures $G_n := \sum_{j'=1}^{L_n} \exp(\bar{b}_{n,j'})\delta_{\boldsymbol{p}_{n,j'}}$ in $\mathcal{G}_{L'}(\Theta)$ and a sequence of $\alpha_n \in \Omega$ such that when $n \to \infty$, the following limits hold:

$$\begin{cases} \mathcal{D}(G_n, G_*) + |\alpha_n - \alpha_*| \to 0, \\ \|f_{G_n,\alpha_n} - f_{G_*,\alpha_*}\|_{L^2(\mu)}/(\mathcal{D}(G_n, G_*) + |\alpha_n - \alpha_*|) \to 0. \end{cases}$$

The first limit indicates that $\mathcal{D}_n := \mathcal{D}(G_n, G_*) \to 0$ and $\alpha_n \to \alpha_*$ as $n \to \infty$.

For the simplicity of the ensuing presentation, we denote $\mathcal{C}_j^n := \mathcal{C}_j(G_n)$ as a Voronoi cell of G_n induced by the j-th components of G_*. Without loss of generality, we assume that those Voronoi cells do not depend on the sample size, i.e., $\mathcal{C}_j = \mathcal{C}_j^n$, which is possible since our arguments are asymptotic. Therefore, we can rewrite the Voronoi loss $\mathcal{D}_n$ as follows:

$$\mathcal{D}_n := \sum_{j'=1}^{L} \Big| \sum_{i\in\mathcal{C}_{j'}} \exp(\bar{b}_{n,i}) - \exp(\bar{b}_{*,j'}) \Big| + \sum_{j'\in[L]:|\mathcal{C}_{j'}|=1} \sum_{i\in\mathcal{C}_{j'}} \exp(\bar{b}_{n,i})\|\Delta\boldsymbol{p}_{n,ij'}\| + \sum_{j'\in[L]:|\mathcal{C}_{j'}|>1} \sum_{i\in\mathcal{C}_{j'}} \exp(\bar{b}_{n,i})\|\Delta\boldsymbol{p}_{n,ij'}\|^2,$$

where $\Delta\boldsymbol{p}_{n,ij'} = \boldsymbol{p}_{n,i} - \boldsymbol{p}_{*,j'}$ for all $i \in \mathcal{C}_{j'}$.

From the hypothesis, we have $\mathcal{D}_n \to 0$, which implies that $\sum_{i\in\mathcal{C}_j} \exp(\bar{b}_{n,i}) \to \exp(\bar{b}_{*,j})$ and $\boldsymbol{p}_{n,i} \to \boldsymbol{p}_{*,j}$ for any $i \in \mathcal{C}_j, j \in [L]$. To establish the contradiction, our proof consists of three main steps.

《论零初始化注意力机制：最优提示词与门控因子估计》的补充篇

在这份补充材料中，我们提供了附录A中**主要结果**的详细证明。关于相关研究的进一步讨论可见附录B，而更多的实验细节则载于附录C。

A. 证明

A.1. 定理的证明4.2

根据$f_{\widehat{G}_n\text{中的}\widehat{\alpha}_n}$ 到$f_{G_*\text{的}\alpha_*}$ 在命题4.1中的收敛速率，4.2定理的结论即可成立，我们只需证明：

$$\|f_{G,\alpha} - f_{G_*,\alpha_*}\|_{L^2(\mu)} \geq C \cdot (\mathcal{D}(G, G_*) + |\alpha - \alpha_*|)$$

对于任意的 (G, α)，存在某个普适常数 C 使得 $\in \mathcal{G}_{L'}(\Theta) \times \Omega$。这等价于证明：

$$\inf_{(G,\alpha)\in\mathcal{G}_{L'}(\Theta)\times\Omega} \|f_{G,\alpha} - f_{G_*,\alpha_*}\|_{L^2(\mu)}/(\mathcal{D}(G, G_*) + |\alpha - \alpha_*|) > 0.$$

为得出上述不等式的结论，我们将其分为两部分来考虑：(i) 局部部分，即

$$\lim_{\varepsilon\to 0} \inf_{(G,\alpha)\in\mathcal{G}_{L'}(\Theta)\times\Omega:\mathcal{D}(G,G_*)+|\alpha-\alpha_*|\leq\varepsilon} \|f_{G,\alpha} - f_{G_*,\alpha_*}\|_{L^2(\mu)}/(\mathcal{D}(G, G_*) + |\alpha - \alpha_*|) > 0;$$

(ii) 全局部分，即对于任意的 $\varepsilon > 0$

$$\inf_{(G,\alpha)\in\mathcal{G}_{L'}(\Theta)\times\Omega:\mathcal{D}(G,G_*)+|\alpha-\alpha_*|>\varepsilon} \|f_{G,\alpha} - f_{G_*,\alpha_*}\|_{L^2(\mu)}/(\mathcal{D}(G, G_*) + |\alpha - \alpha_*|) > 0;$$

局部部分：我们首先从局部部分开始，这等价于证明该部分。

$$\lim_{\varepsilon\to 0} \inf_{(G,\alpha)\in\mathcal{G}_{L'}(\Theta)\times\Omega:\mathcal{D}(G,G_*)+|\alpha-\alpha_*|\leq\varepsilon} \|f_{G,\alpha} - f_{G_*,\alpha_*}\|_{L^2(\mu)}/(\mathcal{D}(G, G_*) + |\alpha - \alpha_*|) > 0.$$

我们通过反证法来证明上述命题。假设相反，即上述命题不成立。这意味着我们可以找到一组混合度量 G_n : $= \sum_{j'=1}^{L_n} \exp(\bar{b}_{n,j'})\delta_{\boldsymbol{p}_{n,j'}}$，它们属于$\mathcal{G}_{L'}(\Theta)$，同时还能找到一组 $\alpha_n \in \Omega$，使得当$n \to \infty$时，以下极限成立：

$$\begin{cases} \mathcal{D}(G_n, G_*) + |\alpha_n - \alpha_*| \to 0, \\ \|f_{G_n,\alpha_n} - f_{G_*,\alpha_*}\|_{L^2(\mu)}/(\mathcal{D}(G_n, G_*) + |\alpha_n - \alpha_*|) \to 0. \end{cases}$$

第一个极限表明， $\mathcal{D}_n := \mathcal{D}(G_n, G_*) \to 0$ 与 $\alpha_n \to \alpha_*$ 可视为 $n \to \infty$。

为简化后续的阐述，我们将 $\mathcal{C}_j^n := \mathcal{C}_j(G_n)$定义为由 G_*的第 j个分量所诱导的 G_n 的Voronoi单元。在不失一般性的前提下，我们假设这些Voronoi单元不依赖于样本大小，即$\mathcal{C}_j = \mathcal{C}_j^n$，这是可行的，因为我们的论证属于渐近性质。因此，我们可以将Voronoi损失 $\mathcal{D}_n$ 重新表示为如下形式：

$$\mathcal{D}_n := \sum_{j'=1}^{L} \Big| \sum_{i\in\mathcal{C}_{j'}} \exp(\bar{b}_{n,i}) - \exp(\bar{b}_{*,j'}) \Big| + \sum_{j'\in[L]:|\mathcal{C}_{j'}|=1} \sum_{i\in\mathcal{C}_{j'}} \exp(\bar{b}_{n,i})\|\Delta\boldsymbol{p}_{n,ij'}\| + \sum_{j'\in[L]:|\mathcal{C}_{j'}|>1} \sum_{i\in\mathcal{C}_{j'}} \exp(\bar{b}_{n,i})\|\Delta\boldsymbol{p}_{n,ij'}\|^2,$$

其中 $\Delta\boldsymbol{p}_{n,\ ij'} = \boldsymbol{p}_{n,\ i} - \boldsymbol{p}_{*,\ j'}$，适用于所有的 $i \in \mathcal{C}_{j'}$。

根据该假设，我们可以得到 $\mathcal{D}_n \to 0$，这意味着对于任意的$i \in \mathcal{C}_j$值，都有$\exp(\bar{b}_{n,\ i}) \to \exp(\bar{b}_{*,\ j})$以及$\boldsymbol{p}_{n,\ i} \to \boldsymbol{p}_{*,\ j}$。为了证明其中的矛盾，我们的证明过程包含三个主要步骤。

Step 1 - Taylor expansion. To ease the presentation, let us denote

$$f_{G_n}(\boldsymbol{X}) := \sum_{j=1}^{L_n} \frac{\exp((\bar{B}\boldsymbol{p}_{n,j})^\top \boldsymbol{X} + \bar{b}_{n,j})}{\sum_{k=1}^{L_n} \exp((\bar{B}\boldsymbol{p}_{n,k})^\top \boldsymbol{X} + \bar{b}_{n,k})} \cdot \bar{C}\boldsymbol{p}_{n,j},$$

$$f_{G_*}(\boldsymbol{X}) := \sum_{j'=1}^{L} \frac{\exp((\bar{B}\boldsymbol{p}_{*,j'})^\top \boldsymbol{X} + \bar{b}_{*,j'})}{\sum_{k'=1}^{L} \exp((\bar{B}\boldsymbol{p}_{*,k'})^\top \boldsymbol{X} + \bar{b}_{*,k'})} \cdot \bar{C}\boldsymbol{p}_{*,j'}.$$

We now decompose the function $\bar{Q}_n(\boldsymbol{X})$ as follows:

$$\begin{aligned}
\bar{Q}_n(\boldsymbol{X}) :&= \Big[\sum_{k'=1}^{L} \exp((\bar{B}\boldsymbol{p}_{*,k'})^\top \boldsymbol{X} + \bar{b}_{*,k'})\Big] \cdot [f_{G_n,\alpha_n}(\boldsymbol{X}) - f_{G_*,\alpha_*}(\boldsymbol{X})] \\
&= \Big[\sum_{k'=1}^{L} \exp((\bar{B}\boldsymbol{p}_{*,k'})^\top \boldsymbol{X} + \bar{b}_{*,k'})\Big] \cdot \Big(\tanh(\alpha_n) f_{G_n}(\boldsymbol{X}) - \tanh(\alpha_*) f_{G_*}(\boldsymbol{X})\Big) \\
&= \Big[\sum_{k'=1}^{L} \exp((\bar{B}\boldsymbol{p}_{*,k'})^\top \boldsymbol{X} + \bar{b}_{*,k'})\Big] \cdot \tanh(\alpha_n)\Big[f_{G_n}(\boldsymbol{X}) - f_{G_*}(\boldsymbol{X})\Big] \\
&+ \Big[\sum_{k'=1}^{L} \exp((\bar{B}\boldsymbol{p}_{*,k'})^\top \boldsymbol{X} + \bar{b}_{*,k'})\Big] \cdot [\tanh(\alpha_n) - \tanh(\alpha_*)] f_{G_*}(\boldsymbol{X}) \\
&:= \bar{Q}_{n,1}(\boldsymbol{X}) + \bar{Q}_{n,2}(\boldsymbol{X}).
\end{aligned}$$

For that purpose, we will decompose the two terms $\bar{Q}_{n,1}(\boldsymbol{X})$ and $\bar{Q}_{n,2}(\boldsymbol{X})$, respectively.

Decomposition of the function $\bar{Q}_{n,1}(\boldsymbol{X})$. We have

$$\begin{aligned}
\bar{Q}_{n,1}(\boldsymbol{X}) &= \sum_{j=1}^{L} \sum_{i \in \mathcal{C}_j} \tanh(\alpha_n) \exp(\bar{b}_{n,i}) \Big[\exp((\bar{B}\boldsymbol{p}_{n,i})^\top \boldsymbol{X}) \bar{C}\boldsymbol{p}_{n,i} - \exp((\bar{B}\boldsymbol{p}_{*,j})^\top \boldsymbol{X}) \bar{C}\boldsymbol{p}_{*,j}\Big] \\
&- \sum_{j=1}^{L} \sum_{i \in \mathcal{C}_j} \tanh(\alpha_n) \exp(\bar{b}_{n,i}) \Big[\exp((\bar{B}\boldsymbol{p}_{n,i})^\top \boldsymbol{X}) - \exp((\bar{B}\boldsymbol{p}_{*,j})^\top \boldsymbol{X})\Big] f_{G_n}(\boldsymbol{X}) \\
&+ \sum_{j=1}^{L} \tanh(\alpha_n) \Big(\sum_{i \in \mathcal{C}_j} \exp(\bar{b}_{n,i}) - \exp(\bar{b}_{*,j})\Big) \exp((\bar{B}\boldsymbol{p}_{*,j})^\top \boldsymbol{X}) \Big[\bar{C}\boldsymbol{p}_{*,j} - f_{G_n}(\boldsymbol{X})\Big] \\
&:= \bar{A}_n(\boldsymbol{X}) - \bar{B}_n(\boldsymbol{X}) + \bar{C}_n(\boldsymbol{X}).
\end{aligned} \tag{22}$$

We now proceed to decompose the functions $\bar{A}_n(.)$ and $\bar{B}_n(.)$ via Taylor expansion.

Decomposition of the function $\bar{A}_n(\boldsymbol{X})$. We first define the following functions $\bar{U}(\boldsymbol{X}; \boldsymbol{p}) := \exp((\bar{B}\boldsymbol{p})^\top \boldsymbol{X})$ and $\bar{V}(\boldsymbol{p}) = \bar{C}\boldsymbol{p}$. Then, we denote the product of these functions as $\bar{F}(\boldsymbol{X}; \boldsymbol{p}) = \bar{U}(\boldsymbol{X}; \boldsymbol{p})\bar{V}(\boldsymbol{p})$. To decompose $\bar{A}_n(\boldsymbol{X})$, we separately consider Voronoi cells with exactly one element and those with more than one element. It leads to the following decomposition of the function $\bar{A}_n(\boldsymbol{X})$:

$$\begin{aligned}
\bar{A}_n(\boldsymbol{X}) &= \sum_{j:|\mathcal{C}_j|=1} \sum_{i \in \mathcal{C}_j} \tanh(\alpha_n) \exp(\bar{b}_{n,i}) \Big[\bar{F}(\boldsymbol{X}; \boldsymbol{p}_{n,i}) - \bar{F}(\boldsymbol{X}; \boldsymbol{p}_{*,j})\Big] \\
&+ \sum_{j:|\mathcal{C}_j|>1} \sum_{i \in \mathcal{C}_j} \tanh(\alpha_n) \exp(\bar{b}_{n,i}) \Big[\bar{F}(\boldsymbol{X}; \boldsymbol{p}_{n,i}) - \bar{F}(\boldsymbol{X}; \boldsymbol{p}_{*,j})\Big] \\
&:= \bar{A}_{n,1}(\boldsymbol{X}) + \bar{A}_{n,2}(\boldsymbol{X}),
\end{aligned}$$

where we denote $\bar{A}_{n,1}(\boldsymbol{X}) = \sum_{j:|\mathcal{C}_j|=1} \sum_{i \in \mathcal{C}_j} \tanh(\alpha_n) \exp(\bar{b}_{n,i}) \Big[\bar{F}(\boldsymbol{X}; \boldsymbol{p}_{n,i}) - \bar{F}(\boldsymbol{X}; \boldsymbol{p}_{*,j})\Big]$ and $\bar{A}_{n,2}(\boldsymbol{X}) = \sum_{j:|\mathcal{C}_j|>1} \sum_{i \in \mathcal{C}_j} \tanh(\alpha_n) \exp(\bar{b}_{n,i}) \Big[\bar{F}(\boldsymbol{X}; \boldsymbol{p}_{n,i}) - \bar{F}(\boldsymbol{X}; \boldsymbol{p}_{*,j})\Big]$.

第一步——泰勒展开。 为便于阐述，我们先对相关量进行如下表示

$$f_{G_n}(\boldsymbol{X}) := \sum_{j=1}^{L_n} \frac{\exp((\bar{B}\boldsymbol{p}_{n,j})^\top \boldsymbol{X} + \bar{b}_{n,j})}{\sum_{k=1}^{L_n} \exp((\bar{B}\boldsymbol{p}_{n,k})^\top \boldsymbol{X} + \bar{b}_{n,k})} \cdot \bar{C}\boldsymbol{p}_{n,j},$$

$$f_{G_*}(\boldsymbol{X}) := \sum_{j'=1}^{L} \frac{\exp((\bar{B}\boldsymbol{p}_{*,j'})^\top \boldsymbol{X} + \bar{b}_{*,j'})}{\sum_{k'=1}^{L} \exp((\bar{B}\boldsymbol{p}_{*,k'})^\top \boldsymbol{X} + \bar{b}_{*,k'})} \cdot \bar{C}\boldsymbol{p}_{*,j'}.$$

现在我们按如下方式对函数 $Q_n(\boldsymbol{X})$进行分解：

$$\begin{aligned}
\bar{Q}_n(\boldsymbol{X}) :&= \Big[\sum_{k'=1}^{L} \exp((\bar{B}\boldsymbol{p}_{*,k'})^\top \boldsymbol{X} + \bar{b}_{*,k'})\Big] \cdot [f_{G_n,\alpha_n}(\boldsymbol{X}) - f_{G_*,\alpha_*}(\boldsymbol{X})] \\
&= \Big[\sum_{k'=1}^{L} \exp((\bar{B}\boldsymbol{p}_{*,k'})^\top \boldsymbol{X} + \bar{b}_{*,k'})\Big] \cdot \Big(\tanh(\alpha_n) f_{G_n}(\boldsymbol{X}) - \tanh(\alpha_*) f_{G_*}(\boldsymbol{X})\Big) \\
&= \Big[\sum_{k'=1}^{L} \exp((\bar{B}\boldsymbol{p}_{*,k'})^\top \boldsymbol{X} + \bar{b}_{*,k'})\Big] \cdot \tanh(\alpha_n)\Big[f_{G_n}(\boldsymbol{X}) - f_{G_*}(\boldsymbol{X})\Big] \\
&+ \Big[\sum_{k'=1}^{L} \exp((\bar{B}\boldsymbol{p}_{*,k'})^\top \boldsymbol{X} + \bar{b}_{*,k'})\Big] \cdot [\tanh(\alpha_n) - \tanh(\alpha_*)] f_{G_*}(\boldsymbol{X}) \\
&:= \bar{Q}_{n,1}(\boldsymbol{X}) + \bar{Q}_{n,2}(\boldsymbol{X}).
\end{aligned}$$

为此，我们将分别对两项$Q_{n、1}(\boldsymbol{X})$以及$Q_{n、2}(\boldsymbol{X})$进行分解。

函数的分解$Q_{n,\ 1}(\boldsymbol{X})$。我们有

$$\begin{aligned}
\bar{Q}_{n,1}(\boldsymbol{X}) &= \sum_{j=1}^{L} \sum_{i \in \mathcal{C}_j} \tanh(\alpha_n) \exp(\bar{b}_{n,i}) \Big[\exp((\bar{B}\boldsymbol{p}_{n,i})^\top \boldsymbol{X}) \bar{C}\boldsymbol{p}_{n,i} - \exp((\bar{B}\boldsymbol{p}_{*,j})^\top \boldsymbol{X}) \bar{C}\boldsymbol{p}_{*,j}\Big] \\
&- \sum_{j=1}^{L} \sum_{i \in \mathcal{C}_j} \tanh(\alpha_n) \exp(\bar{b}_{n,i}) \Big[\exp((\bar{B}\boldsymbol{p}_{n,i})^\top \boldsymbol{X}) - \exp((\bar{B}\boldsymbol{p}_{*,j})^\top \boldsymbol{X})\Big] f_{G_n}(\boldsymbol{X}) \\
&+ \sum_{j=1}^{L} \tanh(\alpha_n) \Big(\sum_{i \in \mathcal{C}_j} \exp(\bar{b}_{n,i}) - \exp(\bar{b}_{*,j})\Big) \exp((\bar{B}\boldsymbol{p}_{*,j})^\top \boldsymbol{X}) \Big[\bar{C}\boldsymbol{p}_{*,j} - f_{G_n}(\boldsymbol{X})\Big] \\
&:= \bar{A}_n(\boldsymbol{X}) - \bar{B}_n(\boldsymbol{X}) + \bar{C}_n(\boldsymbol{X}).
\end{aligned} \tag{22}$$

接下来，我们将通过泰勒展开法对函数 $A_n(.)$和 $B_n(.)$进行分解。

该函数 $\bar{A}_n(\boldsymbol{X})$的拆分方式。 我们首先定义以下两个函数： $\bar{U}(\boldsymbol{X};\ \boldsymbol{p}) :=$指数函数$((\bar{B}\boldsymbol{p})^\top \boldsymbol{X})$，以及 $\bar{V}(\boldsymbol{p}) = C\boldsymbol{p}$。接着，我们将这两个函数的乘积记为 $F(\boldsymbol{X};\ \boldsymbol{p}) = U(\boldsymbol{X};\ \boldsymbol{p})V(\boldsymbol{p})$。为了对 $A_n(\boldsymbol{X})$进行拆分，我们需要分别考虑仅包含一个元素的Voronoi单元和包含多个元素的Voronoi单元。由此便得到了函数 $A_n(\boldsymbol{X})$的如下拆分形式：

$$\begin{aligned}
\bar{A}_n(\boldsymbol{X}) &= \sum_{j:|\mathcal{C}_j|=1} \sum_{i \in \mathcal{C}_j} \tanh(\alpha_n) \exp(\bar{b}_{n,i}) \Big[\bar{F}(\boldsymbol{X}; \boldsymbol{p}_{n,i}) - \bar{F}(\boldsymbol{X}; \boldsymbol{p}_{*,j})\Big] \\
&+ \sum_{j:|\mathcal{C}_j|>1} \sum_{i \in \mathcal{C}_j} \tanh(\alpha_n) \exp(\bar{b}_{n,i}) \Big[\bar{F}(\boldsymbol{X}; \boldsymbol{p}_{n,i}) - \bar{F}(\boldsymbol{X}; \boldsymbol{p}_{*,j})\Big] \\
&:= \bar{A}_{n,1}(\boldsymbol{X}) + \bar{A}_{n,2}(\boldsymbol{X}),
\end{aligned}$$

其中，我们用 $\bar{A}_{n,1}(\boldsymbol{X}) = \sum_{j:|\mathcal{C}_j|=1} \sum_{i \in \mathcal{C}_j} \tanh(\alpha_n) \exp(\bar{b}_{n,i}) \Big[\bar{F}(\boldsymbol{X}; \boldsymbol{p}_{n,i}) - \bar{F}(\boldsymbol{X}; \boldsymbol{p}_{*,j})\Big]$ 来表示，而用 $\bar{A}_{n,2}(\boldsymbol{X}) = \sum_{j:|\mathcal{C}_j|>1} \sum_{i \in \mathcal{C}_j} \tanh(\alpha_n) \exp(\bar{b}_{n,i}) \Big[\bar{F}(\boldsymbol{X}; \boldsymbol{p}_{n,i}) - \bar{F}(\boldsymbol{X}; \boldsymbol{p}_{*,j})\Big]$来表示。

For the function $\bar{A}_{n,1}(\boldsymbol{X})$, for any indices $i \in \mathcal{C}_j$ and j such that $|\mathcal{C}_j| = 1$, the first-order Taylor expansion entails that

$$\bar{U}(\boldsymbol{X}; \boldsymbol{p}_{n,i}) = \bar{U}(\boldsymbol{X}; \boldsymbol{p}_{*,j}) + \sum_{|\alpha|=1} (\Delta \boldsymbol{p}_{n,ij})^{\alpha} \frac{\partial^{|\alpha|} \bar{U}}{\partial \boldsymbol{p}^{\alpha}} (\boldsymbol{X}; \boldsymbol{p}_{*,j}) + \bar{R}_{ij,1}(\boldsymbol{X}),$$

$$\bar{V}(\boldsymbol{p}_{n,i}) = \bar{V}(\boldsymbol{p}_{*,j}) + \sum_{|\alpha|=1} (\Delta \boldsymbol{p}_{n,ij})^{\alpha} \frac{\partial^{|\alpha|} \bar{V}}{\partial \boldsymbol{p}^{\alpha}} (\boldsymbol{p}_{*,j}) + \bar{R}_{ij,2},$$

where the terms $\bar{R}_{ij,1}(\boldsymbol{X})$ and $\bar{R}_{ij,2}$ are Taylor remainders.

Combining the above results leads to the following formulation of the function $\bar{A}_{n,1}(\boldsymbol{X})$:

$$\begin{aligned}\bar{A}_{n,1}(\boldsymbol{X}) &= \sum_{j:|\mathcal{C}_j|=1} \sum_{i \in \mathcal{C}_j} \frac{\tanh(\alpha_n) \exp(\bar{b}_{n,i})}{\alpha!} \sum_{|\alpha|=1} \Big\{ (\Delta \boldsymbol{p}_{n,ij})^{\alpha} \frac{\partial^{|\alpha|} \bar{U}}{\partial \boldsymbol{p}^{\alpha}} (\boldsymbol{X}; \boldsymbol{p}_{*,j}) \bar{V}(\boldsymbol{p}_{*,j}) \\ &\qquad + (\Delta \boldsymbol{p}_{n,ij})^{\alpha} \frac{\partial^{|\alpha|} \bar{V}}{\partial \boldsymbol{p}^{\alpha}} (\boldsymbol{p}_{*,j}) \bar{U}(\boldsymbol{X}; \boldsymbol{p}_{*,j}) \Big\} + \bar{R}_{n,1}(\boldsymbol{X}) \\ &= \sum_{j:|\mathcal{C}_j|=1} \sum_{|\alpha|=1} \left\{ \bar{M}_{n,j,\alpha} \frac{\partial^{|\alpha|} \bar{U}}{\partial \boldsymbol{p}^{\alpha}} (\boldsymbol{X}; \boldsymbol{p}_{*,j}) \bar{V}(\boldsymbol{p}_{*,j}) + \bar{M}_{n,j,\alpha} \frac{\partial^{|\alpha|} \bar{V}}{\partial \boldsymbol{p}^{\alpha}} (\boldsymbol{p}_{*,j}) \bar{U}(\boldsymbol{X}; \boldsymbol{p}_{*,j}) \right\} + \bar{R}_{n,1}(\boldsymbol{X})\end{aligned}$$

where the function $\bar{R}_{n,1}(\boldsymbol{X})$ is the combination of Taylor remainders and satisfies that $\bar{R}_{n,1}(\boldsymbol{X})/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0$ when $n \to \infty$. Furthermore, the formulations of $\bar{M}_{n,j,\alpha}$ are as follows:

$$\bar{M}_{n,j,\alpha} = \sum_{i \in \mathcal{C}_j} \frac{\tanh(\alpha_n) \exp(\bar{b}_{n,i})}{\alpha!} (\Delta \boldsymbol{p}_{n,ij})^{\alpha},$$

for any $|\alpha| = 1$.

Moving to the function $\bar{A}_{n,2}(\boldsymbol{X})$, the second-order Taylor expansions of the function $\bar{U}(\boldsymbol{X}; \boldsymbol{p}_{n,i})$ around the function $\bar{U}(\boldsymbol{X}; \boldsymbol{p}_{*,j})$ and the function $\bar{V}(\boldsymbol{p}_{n,i})$ around the function $\bar{V}(\boldsymbol{p}_{*,j})$ for any indices $i \in \mathcal{C}_j$ and j such that $|\mathcal{C}_j| > 1$, we obtain the following formulation of the function $\bar{A}_{n,2}(\boldsymbol{X})$:

$$\begin{aligned}\bar{A}_{n,2}(\boldsymbol{X}) &= \sum_{j:|\mathcal{C}_j|>1} \sum_{1 \le |\alpha| \le 2} \left\{ \bar{M}_{n,j,\alpha} \frac{\partial^{|\alpha|} \bar{U}}{\partial \boldsymbol{p}^{\alpha}} (\boldsymbol{X}; \boldsymbol{p}_{*,j}) \bar{V}(\boldsymbol{p}_{*,j}) + \bar{M}_{n,j,\alpha} \frac{\partial^{|\alpha|} \bar{V}}{\partial \boldsymbol{p}^{\alpha}} (\boldsymbol{p}_{*,j}) \bar{U}(\boldsymbol{X}; \boldsymbol{p}_{*,j}) \right\} \\ &\quad + \sum_{j:|\mathcal{C}_j|>1} \sum_{|\alpha|=1,|\beta|=1} \bar{M}_{n,j,\alpha,\beta} \frac{\partial^{|\alpha|} \bar{U}}{\partial \boldsymbol{p}^{\alpha}} (\boldsymbol{X}; \boldsymbol{p}_{*,j}) \frac{\partial^{|\beta|} \bar{V}}{\partial \boldsymbol{p}^{\beta}} (\boldsymbol{p}_{*,j}) + \bar{R}_{n,2}(\boldsymbol{X})\end{aligned}$$

where the function $\bar{R}_{n,2}(\boldsymbol{X})$ is a combination of Taylor remainders and satisfies $\bar{R}_{n,2}(\boldsymbol{X})/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0$ when $n \to \infty$. Furthermore, we define

$$\bar{M}_{n,j,\alpha} = \sum_{i \in \mathcal{C}_j} \frac{\tanh(\alpha_n) \exp(b_{n,i})}{\alpha!} (\Delta \boldsymbol{p}_{n,ij})^{\alpha},$$

for any $|\alpha| = 2$ and

$$\bar{M}_{n,j,\alpha,\beta} = \sum_{i \in \mathcal{C}_j} \frac{\tanh(\alpha_n) \exp(\bar{b}_{n,i})}{\alpha! \beta!} (\Delta \boldsymbol{p}_{n,ij})^{\alpha+\beta},$$

for any $|\alpha| = |\beta| = 1$. From the formulations of the functions $\bar{U}(\boldsymbol{X}; \boldsymbol{p})$ and $\bar{V}(\boldsymbol{p})$, we obtain the following explicit forms of their partial derivatives:

$$\frac{\partial \bar{U}}{\partial \boldsymbol{p}^{(u)}} (\boldsymbol{X}; \boldsymbol{p}) = \exp((\bar{B}\boldsymbol{p})^{\top} \boldsymbol{X}) (\bar{B} 1_u)^{\top} \boldsymbol{X},$$

$$\frac{\partial^2 \bar{U}}{\partial \boldsymbol{p}^{(u)} \partial \boldsymbol{p}^{(v)}} (\boldsymbol{X}; \boldsymbol{p}) = \exp((\bar{B}\boldsymbol{p})^{\top} \boldsymbol{X}) \boldsymbol{X}^{\top} (\bar{B} 1_u) (\bar{B} 1_v)^{\top} \boldsymbol{X},$$

对于该函数 $\bar{A}_{n,1}(\boldsymbol{X})$，对于满足 $|\mathcal{C}_j| = 1$的任意索引 $i \in \mathcal{C}_j$ 和 j，一阶泰勒展开表明

$$\bar{U}(\boldsymbol{X}; \boldsymbol{p}_{n,i}) = \bar{U}(\boldsymbol{X}; \boldsymbol{p}_{*,j}) + \sum_{|\alpha|=1} (\Delta \boldsymbol{p}_{n,ij})^{\alpha} \frac{\partial^{|\alpha|} \bar{U}}{\partial \boldsymbol{p}^{\alpha}} (\boldsymbol{X}; \boldsymbol{p}_{*,j}) + \bar{R}_{ij,1}(\boldsymbol{X}),$$

$$\bar{V}(\boldsymbol{p}_{n,i}) = \bar{V}(\boldsymbol{p}_{*,j}) + \sum_{|\alpha|=1} (\Delta \boldsymbol{p}_{n,ij})^{\alpha} \frac{\partial^{|\alpha|} \bar{V}}{\partial \boldsymbol{p}^{\alpha}} (\boldsymbol{p}_{*,j}) + \bar{R}_{ij,2},$$

其中，项$R_{ij,1}(\boldsymbol{X})$与$R_{ij,2}$ 属于泰勒余项。

将上述结果结合后，可得到该函数的如下表达式$A_{n,\ 1}(\boldsymbol{X})$:

$$\begin{aligned}\bar{A}_{n,1}(\boldsymbol{X}) &= \sum_{j:|\mathcal{C}_j|=1} \sum_{i \in \mathcal{C}_j} \frac{\tanh(\alpha_n) \exp(\bar{b}_{n,i})}{\alpha!} \sum_{|\alpha|=1} \Big\{ (\Delta \boldsymbol{p}_{n,ij})^{\alpha} \frac{\partial^{|\alpha|} \bar{U}}{\partial \boldsymbol{p}^{\alpha}} (\boldsymbol{X}; \boldsymbol{p}_{*,j}) \bar{V}(\boldsymbol{p}_{*,j}) \\ &\qquad + (\Delta \boldsymbol{p}_{n,ij})^{\alpha} \frac{\partial^{|\alpha|} \bar{V}}{\partial \boldsymbol{p}^{\alpha}} (\boldsymbol{p}_{*,j}) \bar{U}(\boldsymbol{X}; \boldsymbol{p}_{*,j}) \Big\} + \bar{R}_{n,1}(\boldsymbol{X}) \\ &= \sum_{j:|\mathcal{C}_j|=1} \sum_{|\alpha|=1} \left\{ \bar{M}_{n,j,\alpha} \frac{\partial^{|\alpha|} \bar{U}}{\partial \boldsymbol{p}^{\alpha}} (\boldsymbol{X}; \boldsymbol{p}_{*,j}) \bar{V}(\boldsymbol{p}_{*,j}) + \bar{M}_{n,j,\alpha} \frac{\partial^{|\alpha|} \bar{V}}{\partial \boldsymbol{p}^{\alpha}} (\boldsymbol{p}_{*,j}) \bar{U}(\boldsymbol{X}; \boldsymbol{p}_{*,j}) \right\} + \bar{R}_{n,1}(\boldsymbol{X})\end{aligned}$$

其中函数 $R_{n,1}(\boldsymbol{X})$为泰勒余项的组合，且满足$R_{n,1}(\boldsymbol{X})/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0$当 $n \to \infty$时。此外，$M_{n,j,\alpha}$ 的表达式如下:

$$\bar{M}_{n,j,\alpha} = \sum_{i \in \mathcal{C}_j} \frac{\tanh(\alpha_n) \exp(\bar{b}_{n,i})}{\alpha!} (\Delta \boldsymbol{p}_{n,ij})^{\alpha},$$

对于任意的 $|\alpha| = 1$。

Moving to the function $\bar{A}_{n,2}(\boldsymbol{X})$, the second-order Taylor expansions of the function $\bar{U}(\boldsymbol{X}; \boldsymbol{p}_{n,i})$ around the function $\bar{U}(\boldsymbol{X}; \boldsymbol{p}_{*,j})$ and the function $\bar{V}(\boldsymbol{p}_{n,i})$ around the function $\bar{V}(\boldsymbol{p}_{*,j})$ for any indices $i \in \mathcal{C}_j$ and j such that $|\mathcal{C}_j| > 1$, we obtain the following formulation of the function $\bar{A}_{n,2}(\boldsymbol{X})$:

$$\begin{aligned}\bar{A}_{n,2}(\boldsymbol{X}) &= \sum_{j:|\mathcal{C}_j|>1} \sum_{1 \le |\alpha| \le 2} \left\{ \bar{M}_{n,j,\alpha} \frac{\partial^{|\alpha|} \bar{U}}{\partial \boldsymbol{p}^{\alpha}} (\boldsymbol{X}; \boldsymbol{p}_{*,j}) \bar{V}(\boldsymbol{p}_{*,j}) + \bar{M}_{n,j,\alpha} \frac{\partial^{|\alpha|} \bar{V}}{\partial \boldsymbol{p}^{\alpha}} (\boldsymbol{p}_{*,j}) \bar{U}(\boldsymbol{X}; \boldsymbol{p}_{*,j}) \right\} \\ &\quad + \sum_{j:|\mathcal{C}_j|>1} \sum_{|\alpha|=1,|\beta|=1} \bar{M}_{n,j,\alpha,\beta} \frac{\partial^{|\alpha|} \bar{U}}{\partial \boldsymbol{p}^{\alpha}} (\boldsymbol{X}; \boldsymbol{p}_{*,j}) \frac{\partial^{|\beta|} \bar{V}}{\partial \boldsymbol{p}^{\beta}} (\boldsymbol{p}_{*,j}) + \bar{R}_{n,2}(\boldsymbol{X})\end{aligned}$$

其中函数 $R_{n,2}(\boldsymbol{X})$是泰勒余项的组合，且满足 $R_{n,2}(\boldsymbol{X})/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0$ 当$n \to \infty$时。此外，我们定义

$$\bar{M}_{n,j,\alpha} = \sum_{i \in \mathcal{C}_j} \frac{\tanh(\alpha_n) \exp(b_{n,i})}{\alpha!} (\Delta \boldsymbol{p}_{n,ij})^{\alpha},$$

对于任意的 $|\alpha| = 2$而言

$$\bar{M}_{n,j,\alpha,\beta} = \sum_{i \in \mathcal{C}_j} \frac{\tanh(\alpha_n) \exp(\bar{b}_{n,i})}{\alpha! \beta!} (\Delta \boldsymbol{p}_{n,ij})^{\alpha+\beta},$$

对于任意的 $|\alpha| = |\beta| = 1$而言。通过 $U(\boldsymbol{X};\ \boldsymbol{p})$与 $V(\boldsymbol{p})$这两个函数的表达式，我们可以得到它们偏导数的如下显式形式:

$$\frac{\partial \bar{U}}{\partial \boldsymbol{p}^{(u)}} (\boldsymbol{X}; \boldsymbol{p}) = \exp((\bar{B}\boldsymbol{p})^{\top} \boldsymbol{X}) (\bar{B} 1_u)^{\top} \boldsymbol{X},$$

$$\frac{\partial^2 \bar{U}}{\partial \boldsymbol{p}^{(u)} \partial \boldsymbol{p}^{(v)}} (\boldsymbol{X}; \boldsymbol{p}) = \exp((\bar{B}\boldsymbol{p})^{\top} \boldsymbol{X}) \boldsymbol{X}^{\top} (\bar{B} 1_u) (\bar{B} 1_v)^{\top} \boldsymbol{X},$$

$$\frac{\partial \bar{V}}{\partial \boldsymbol{p}^{(u)}}(\boldsymbol{p}) = \bar{C}1_u,$$
$$\frac{\partial^2 \bar{V}}{\partial \boldsymbol{p}^{(u)} \partial \boldsymbol{p}^{(v)}}(\boldsymbol{p}) = 0.$$

In these formulations, we use 1_u to denote the vector that its u-th element is 1 and its other elements are 0 for $1 \leq u \leq d$. Plugging these explicit formulations of the derivatives of the functions $\bar{U}(\boldsymbol{X}; \boldsymbol{p})$ and $\bar{V}(\boldsymbol{p})$, we can express the functions $\bar{A}_{n,1}(\boldsymbol{X})$ and $\bar{A}_{n,2}(\boldsymbol{X})$ as follows:

$$\bar{A}_{n,1}(\boldsymbol{X}) = \sum_{j:|\mathcal{C}_j|=1} \exp((\bar{B}\boldsymbol{p}_{*,j})^\top \boldsymbol{X}) \big[\mathcal{L}_{1,n}(\boldsymbol{p}_{*,j}) + \mathcal{L}_{2,n}(\boldsymbol{p}_{*,j})^\top \bar{B}^\top \boldsymbol{X}\big] + \bar{R}_{n,1}(\boldsymbol{X}),$$
$$\bar{A}_{n,2}(\boldsymbol{X}) = \sum_{j:|\mathcal{C}_j|>1} \exp((\bar{B}\boldsymbol{p}_{*,j})^\top \boldsymbol{X}) \big[\bar{\mathcal{L}}_{1,n}(\boldsymbol{p}_{*,j}) + \bar{\mathcal{L}}_{2,n}(\boldsymbol{p}_{*,j})^\top \bar{B}^\top \boldsymbol{X} + (\bar{B}^\top \boldsymbol{X})^\top \bar{\mathcal{L}}_{3,n}(\boldsymbol{p}_{*,j}) \bar{B}^\top \boldsymbol{X}\big] + \bar{R}_{n,2}(\boldsymbol{X}),$$

where the functions $\mathcal{L}_{1,n}(\boldsymbol{p}), \mathcal{L}_{2,n}(\boldsymbol{p}), \bar{\mathcal{L}}_{1,n}(\boldsymbol{p}), \bar{\mathcal{L}}_{2,n}(\boldsymbol{p})$, and $\bar{\mathcal{L}}_{3,n}(\boldsymbol{p})$ are defined as follows:

$$\mathcal{L}_{1,n}(\boldsymbol{p}) = \sum_{u=1}^{d} \bar{M}_{n,j,1_u} \bar{C} 1_u,$$
$$\mathcal{L}_{2,n}(\boldsymbol{p}) = \sum_{u=1}^{d} \bar{M}_{n,j,1_u} 1_u \bar{C}\boldsymbol{p},$$
$$\bar{\mathcal{L}}_{1,n}(\boldsymbol{p}) = \sum_{u=1}^{d} \bar{M}_{n,j,1_u} C 1_u,$$
$$\bar{\mathcal{L}}_{2,n}(\boldsymbol{p}) = \sum_{u=1}^{d} \bar{M}_{n,j,1_u} 1_u \bar{C}\boldsymbol{p} + \sum_{1\leq u,v\leq d} \bar{M}_{n,j,1_v,1_u} \bar{C} 1_u 1_v$$
$$\bar{\mathcal{L}}_{3,n}(\boldsymbol{p}) = \sum_{1\leq u,v\leq d} \bar{M}_{n,j,1_{uv}} 1_u 1_v^\top \bar{C}\boldsymbol{p}.$$

In these formulations, we denote 1_{uv} as the matrix that its (u, v)-th element is 1 and its other elements are 0 for any $1 \leq u, v \leq d$.

Decomposition of the function $\bar{B}_n(\boldsymbol{X})$. Similar to the decomposition of the function $\bar{A}_n(\boldsymbol{X})$, we can decompose the function $\bar{B}_n(\boldsymbol{X})$ as follows:

$$\begin{aligned}\bar{B}_n(\boldsymbol{X}) &= \sum_{j:|\mathcal{C}_j|=1} \sum_{i\in\mathcal{C}_j} \tanh(\alpha_n) \exp(\bar{b}_{n,i}) \Big[\bar{U}(\boldsymbol{X}; \boldsymbol{p}_{n,i}) - \bar{U}(\boldsymbol{X}; \boldsymbol{p}_{*,j})\Big] f_{G_n}(\boldsymbol{X}) \\ &\quad + \sum_{j:|\mathcal{C}_j|>1} \sum_{i\in\mathcal{C}_j} \tanh(\alpha_n) \exp(\bar{b}_{n,i}) \Big[\bar{U}(\boldsymbol{X}; \boldsymbol{p}_{n,i}) - \bar{U}(\boldsymbol{X}; \boldsymbol{p}_{*,j})\Big] f_{G_n}(\boldsymbol{X}) \\ &:= \bar{B}_{n,1}(\boldsymbol{X}) + \bar{B}_{n,2}(\boldsymbol{X})\end{aligned}$$

where we denote $\bar{B}_{n,1}(\boldsymbol{X}) = \sum_{j:|\mathcal{C}_j|=1} \sum_{i\in\mathcal{C}_j} \tanh(\alpha_n) \exp(\bar{b}_{n,i}) \Big[\bar{U}(\boldsymbol{X}; \boldsymbol{p}_{n,i}) - \bar{U}(\boldsymbol{X}; \boldsymbol{p}_{*,j})\Big] f_{G_n}(\boldsymbol{X})$ and $\bar{B}_{n,2}(\boldsymbol{X}) = \sum_{j:|\mathcal{C}_j|>1} \sum_{i\in\mathcal{C}_j} \tanh(\alpha_n) \exp(\bar{b}_{n,i}) \Big[\bar{U}(\boldsymbol{X}; \boldsymbol{p}_{n,i}) - \bar{U}(\boldsymbol{X}; \boldsymbol{p}_{*,j})\Big] f_{G_n}(\boldsymbol{X})$.

Similar to the Taylor expansions for the functions $\bar{A}_{n,1}(\boldsymbol{X})$ and $\bar{A}_{n,2}(\boldsymbol{X})$, by using the first-order Taylor expansion to $\bar{B}_{n,1}(\boldsymbol{X})$ and the second-order Taylor expansion to $\bar{B}_{n,2}(\boldsymbol{X})$, we obtain that

$$\bar{B}_{n,1}(\boldsymbol{X}) = \sum_{j:|\mathcal{C}_j|=1} \sum_{|\alpha|=1} \bar{M}_{n,j,\alpha} \frac{\partial^{|\alpha|} \bar{U}}{\partial \boldsymbol{p}^\alpha}(\boldsymbol{X}; \boldsymbol{p}_{*,j}) f_{G_n}(\boldsymbol{X}) + \bar{R}_{n,3}(\boldsymbol{X})$$

$$\frac{\partial \bar{V}}{\partial \boldsymbol{p}^{(u)}}(\boldsymbol{p}) = \bar{C}1_u,$$
$$\frac{\partial^2 \bar{V}}{\partial \boldsymbol{p}^{(u)} \partial \boldsymbol{p}^{(v)}}(\boldsymbol{p}) = 0.$$

在这些表达式中，我们使用 1_u 来表示一个向量——对于 $1 \leq u \leq d$.而言，该向量的第 u个元素为1，其余元素均为0。将 $U(\boldsymbol{X};\ \boldsymbol{p})$与 $V(\boldsymbol{p})$这两个函数导数的显式表达式代入后，我们便可以如下表示$A_{n,1}(\boldsymbol{X})$以及$A_{n,2}(\boldsymbol{X})$这些函数:

$$\bar{A}_{n,1}(\boldsymbol{X}) = \sum_{j:|\mathcal{C}_j|=1} \exp((\bar{B}\boldsymbol{p}_{*,j})^\top \boldsymbol{X}) \big[\mathcal{L}_{1,n}(\boldsymbol{p}_{*,j}) + \mathcal{L}_{2,n}(\boldsymbol{p}_{*,j})^\top \bar{B}^\top \boldsymbol{X}\big] + \bar{R}_{n,1}(\boldsymbol{X}),$$
$$\bar{A}_{n,2}(\boldsymbol{X}) = \sum_{j:|\mathcal{C}_j|>1} \exp((\bar{B}\boldsymbol{p}_{*,j})^\top \boldsymbol{X}) \big[\bar{\mathcal{L}}_{1,n}(\boldsymbol{p}_{*,j}) + \bar{\mathcal{L}}_{2,n}(\boldsymbol{p}_{*,j})^\top \bar{B}^\top \boldsymbol{X} + (\bar{B}^\top \boldsymbol{X})^\top \bar{\mathcal{L}}_{3,n}(\boldsymbol{p}_{*,j}) \bar{B}^\top \boldsymbol{X}\big] + \bar{R}_{n,2}(\boldsymbol{X}),$$

其中，函数 $\mathcal{L}_{1,n}(\boldsymbol{p})$ $\mathcal{L}_{2,n}(\boldsymbol{p})$ $\mathcal{L}_{1,n}(\boldsymbol{p})$$\mathcal{L}_{2,n}(\boldsymbol{p})$以及$\mathcal{L}_{3,n}(\boldsymbol{p})$的定义如下:

$$\mathcal{L}_{1,n}(\boldsymbol{p}) = \sum_{u=1}^{d} \bar{M}_{n,j,1_u} \bar{C} 1_u,$$
$$\mathcal{L}_{2,n}(\boldsymbol{p}) = \sum_{u=1}^{d} \bar{M}_{n,j,1_u} 1_u \bar{C}\boldsymbol{p},$$
$$\bar{\mathcal{L}}_{1,n}(\boldsymbol{p}) = \sum_{u=1}^{d} \bar{M}_{n,j,1_u} C 1_u,$$
$$\bar{\mathcal{L}}_{2,n}(\boldsymbol{p}) = \sum_{u=1}^{d} \bar{M}_{n,j,1_u} 1_u \bar{C}\boldsymbol{p} + \sum_{1\leq u,v\leq d} \bar{M}_{n,j,1_v,1_u} \bar{C} 1_u 1_v$$
$$\bar{\mathcal{L}}_{3,n}(\boldsymbol{p}) = \sum_{1\leq u,v\leq d} \bar{M}_{n,j,1_{uv}} 1_u 1_v^\top \bar{C}\boldsymbol{p}.$$

在这些公式中，我们将 1_{uv} 定义为这样一种矩阵：对于任意的$1 \leq u$、 $v \leq d$，其$(u$、 $v)$-th元素为1，其余元素均为0。

函数 $B_n(\boldsymbol{X})$ 的分解。 与函数 $A_n(\boldsymbol{X})$ 的分解方式类似，我们可以将函数 $B_n(\boldsymbol{X})$按如下方式分解:

$$\begin{aligned}\bar{B}_n(\boldsymbol{X}) &= \sum_{j:|\mathcal{C}_j|=1} \sum_{i\in\mathcal{C}_j} \tanh(\alpha_n) \exp(\bar{b}_{n,i}) \Big[\bar{U}(\boldsymbol{X}; \boldsymbol{p}_{n,i}) - \bar{U}(\boldsymbol{X}; \boldsymbol{p}_{*,j})\Big] f_{G_n}(\boldsymbol{X}) \\ &\quad + \sum_{j:|\mathcal{C}_j|>1} \sum_{i\in\mathcal{C}_j} \tanh(\alpha_n) \exp(\bar{b}_{n,i}) \Big[\bar{U}(\boldsymbol{X}; \boldsymbol{p}_{n,i}) - \bar{U}(\boldsymbol{X}; \boldsymbol{p}_{*,j})\Big] f_{G_n}(\boldsymbol{X}) \\ &:= \bar{B}_{n,1}(\boldsymbol{X}) + \bar{B}_{n,2}(\boldsymbol{X})\end{aligned}$$

其中，我们用$\bar{B}_{n,1}(\boldsymbol{X}) = \sum_{j:|\mathcal{C}_j|=1} \sum_{i\in\mathcal{C}_j} \tanh(\alpha_n) \exp(\bar{b}_{n,i}) \Big[\bar{U}(\boldsymbol{X}; \boldsymbol{p}_{n,i}) - \bar{U}(\boldsymbol{X}; \boldsymbol{p}_{*,j})\Big] f_{G_n}(\boldsymbol{X})$来表示，而用$\bar{B}_{n,2}(\boldsymbol{X}) = \sum_{j:|\mathcal{C}_j|>1} \sum_{i\in\mathcal{C}_j} \tanh(\alpha_n) \exp(\bar{b}_{n,i}) \Big[\bar{U}(\boldsymbol{X}; \boldsymbol{p}_{n,i}) - \bar{U}(\boldsymbol{X}; \boldsymbol{p}_{*,j})\Big] f_{G_n}(\boldsymbol{X})$来表示。

与函数 $A_{n,1}(\boldsymbol{X})$以及$A_{n,2}(\boldsymbol{X})$的泰勒级数展开类似，通过使用一阶泰勒级数对$B_{n,1}(\boldsymbol{X})$进行展开，再利用二阶泰勒级数对$B_{n,2}(\boldsymbol{X})$进行展开，我们便可以得出结果。

$$\bar{B}_{n,1}(\boldsymbol{X}) = \sum_{j:|\mathcal{C}_j|=1} \sum_{|\alpha|=1} \bar{M}_{n,j,\alpha} \frac{\partial^{|\alpha|} \bar{U}}{\partial \boldsymbol{p}^\alpha}(\boldsymbol{X}; \boldsymbol{p}_{*,j}) f_{G_n}(\boldsymbol{X}) + \bar{R}_{n,3}(\boldsymbol{X})$$

$$\bar{B}_{n,2}(\boldsymbol{X}) = \sum_{j:|\mathcal{C}_j|=1} \sum_{1\leq|\alpha|\leq 2} \bar{M}_{n,j,\alpha} \frac{\partial^{|\alpha|}\bar{U}}{\partial \boldsymbol{p}^{\alpha}}(\boldsymbol{X};\boldsymbol{p}_{*,j}) f_{G_n}(\boldsymbol{X}) + \bar{R}_{n,4}(\boldsymbol{X})$$

where the functions $\bar{R}_{n,3}(\boldsymbol{X}), \bar{R}_{n,4}(\boldsymbol{X})$ are Taylor remainders. Furthermore, they satisfy that $\bar{R}_{n,3}(\boldsymbol{X})/(\mathcal{D}_n+|\alpha_n-\alpha_*|) \to 0$ and $\bar{R}_{n,4}(\boldsymbol{X})/(\mathcal{D}_n+|\alpha_n-\alpha_*|) \to 0$ when $n \to \infty$. Given the explicit formulations of the derivatives of the functions $\bar{U}(\boldsymbol{X};\boldsymbol{p})$ and $\bar{V}(\boldsymbol{p})$, the functions $\bar{B}_{n,1}(\boldsymbol{X})$ and $\bar{B}_{n,2}(\boldsymbol{X})$ can be then rewritten as follows:

$$\bar{B}_{n,1}(\boldsymbol{X}) = \sum_{j:|\mathcal{C}_j|=1} \exp((\bar{B}\boldsymbol{p}_{*,j})^\top \boldsymbol{X}) \mathcal{N}_{1,n}(\boldsymbol{p}_{*,j})^\top \boldsymbol{X} f_{G_n}(\boldsymbol{X}) + \bar{R}_{n,3}(\boldsymbol{X}),$$

$$\begin{aligned}\bar{B}_{n,2}(\boldsymbol{X}) = \sum_{j:|\mathcal{C}_j|>1} \exp((\bar{B}\boldsymbol{p}_{*,j})^\top \boldsymbol{X}) \big[\bar{\mathcal{N}}_{1,n}(\boldsymbol{p}_{*,j})^\top \bar{B}^\top \boldsymbol{X} \\ + (\bar{B}^\top \boldsymbol{X})^\top \bar{\mathcal{N}}_{2,n}(\boldsymbol{p}_{*,j})(\bar{B}^\top \boldsymbol{X})\big] f_{G_n}(\boldsymbol{X}) + \bar{R}_{n,4}(\boldsymbol{X}).\end{aligned}$$

Here, the functions $\mathcal{N}_{1,n}(\boldsymbol{X})$, $\bar{\mathcal{N}}_{1,n}(\boldsymbol{X})$, and $\bar{\mathcal{N}}_{2,n}(\boldsymbol{X})$ have the following formulations:

$$\begin{aligned}\mathcal{N}_{1,n}(\boldsymbol{p}) &= \sum_{u=1}^{d} \bar{M}_{n,j,1_u} 1_u, \\ \bar{\mathcal{N}}_{1,n}(\boldsymbol{p}) &= \sum_{u=1}^{d} \bar{M}_{n,j,1_u} 1_u, \\ \bar{\mathcal{N}}_{2,n}(\boldsymbol{p}) &= \sum_{1\leq u,v\leq d} \bar{M}_{n,j,1_{uv}} 1_u 1_v^\top.\end{aligned}$$

Collecting all of the above results with the decomposition of the functions $\bar{A}_n(\boldsymbol{X})$ and $\bar{B}_n(\boldsymbol{X})$, we can represent the function $\bar{Q}_{n,1}(\boldsymbol{X})$ in equation (22) as follows:

$$\begin{aligned}\bar{Q}_{n,1}(\boldsymbol{X}) = &\sum_{j:|\mathcal{C}_j|=1} \exp((\bar{B}\boldsymbol{p}_{*,j})^\top \boldsymbol{X}) \big[\bar{\mathcal{L}}'_{1,n}(\boldsymbol{p}_{*,j}) + \bar{\mathcal{L}}_{2,n}(\boldsymbol{p}_{*,j})^\top \bar{B}^\top \boldsymbol{X}\big) \\ &+ \sum_{j:|\mathcal{C}_j|>1} \exp((\bar{B}\boldsymbol{p}_{*,j})^\top \boldsymbol{X}) \big[\bar{\mathcal{L}}'_{1,n}(\boldsymbol{p}_{*,j}) + \bar{\mathcal{L}}_{2,n}(\boldsymbol{p}_{*,j})^\top \bar{B}^\top \boldsymbol{X} + (\bar{B}^\top \boldsymbol{X})^\top \bar{\mathcal{L}}_{3,n}(\boldsymbol{p}_{*,j}) \bar{B}^\top \boldsymbol{X}\big] \\ &- \sum_{j:|\mathcal{C}_j|=1} \exp((\bar{B}\boldsymbol{p}_{*,j})^\top \boldsymbol{X}) \big[\bar{M}_{n,j,0_d} + \mathcal{N}_{1,n}(\boldsymbol{p}_{*,j})^\top \bar{B}^\top \boldsymbol{X}\big] f_{G_n}(\boldsymbol{X}) \\ &- \sum_{j:|\mathcal{C}_j|>1} \exp((\bar{B}\boldsymbol{p}_{*,j})^\top \boldsymbol{X}) \big[\bar{M}_{n,j,0_d} + \bar{\mathcal{N}}_{1,n}(\boldsymbol{p}_{*,j})^\top \bar{B}^\top \boldsymbol{X} + (\bar{B}^\top \boldsymbol{X})^\top \bar{\mathcal{N}}_{2,n}(\boldsymbol{p}_{*,j}) \bar{B}^\top \boldsymbol{X}\big] f_{G_n}(\boldsymbol{X}) \\ &+ \bar{R}_{n,1}(\boldsymbol{X}) + \bar{R}_{n,2}(\boldsymbol{X}) - \bar{R}_{n,3}(\boldsymbol{X}) - \bar{R}_{n,4}(\boldsymbol{X}) \qquad (23)\end{aligned}$$

where we define $\bar{M}_{n,j,0_d} = \tanh(\alpha_n)\Big(\sum_{i\in\mathcal{C}_j} \exp(\bar{b}_{n,i}) - \exp(\bar{b}_{*,j})\Big)$ for any index $1 \leq j \leq L$, $\mathcal{L}'_{1,n}(\boldsymbol{p}_{*,j}) = \mathcal{L}_{1,n}(\boldsymbol{p}_{*,j}) + \bar{M}_{n,j,0_d}\bar{C}\boldsymbol{p}_{*,j}$, and $\bar{\mathcal{L}}'_{1,n}(\boldsymbol{p}_{*,j}) = \bar{\mathcal{L}}_{1,n}(\boldsymbol{p}_{*,j}) + \bar{M}_{n,j,0_d}\bar{C}\boldsymbol{p}_{*,j}$.

Decomposition of the function $\bar{Q}_{n,2}(\boldsymbol{X})$. An application of the first-order Taylor expansion leads to the following expression for the function $\bar{Q}_{n,2}(\boldsymbol{X})$:

$$\begin{aligned}\bar{Q}_{n,2}(\boldsymbol{X}) &= [\tanh(\alpha_n) - \tanh(\alpha_*)] \cdot \Big[\sum_{k'=1}^{L} \exp((\bar{B}\boldsymbol{p}_{*,k'})^\top \boldsymbol{X} + \bar{b}_{*,k'})\Big] \cdot f_{G_*}(\boldsymbol{X}) \\ &= (\alpha_n - \alpha_*)[1 - \tanh^2(\alpha_*)] \cdot \Big[\sum_{k'=1}^{L} \exp((\bar{B}\boldsymbol{p}_{*,k'})^\top \boldsymbol{X} + \bar{b}_{*,k'})\Big] \cdot f_{G_*}(\boldsymbol{X}) + \bar{R}_5(\boldsymbol{X}), \qquad (24)\end{aligned}$$

where the function $\bar{R}_{n,5}(\boldsymbol{X})$ is Taylor remainder and satisfies that $\bar{R}_{n,5}(\boldsymbol{X})/(\mathcal{D}_n+|\alpha_n-\alpha_*|) \to 0$ as $n \to \infty$.

$$\bar{B}_{n,2}(\boldsymbol{X}) = \sum_{j:|\mathcal{C}_j|=1} \sum_{1\leq|\alpha|\leq 2} \bar{M}_{n,j,\alpha} \frac{\partial^{|\alpha|}\bar{U}}{\partial \boldsymbol{p}^{\alpha}}(\boldsymbol{X};\boldsymbol{p}_{*,j}) f_{G_n}(\boldsymbol{X}) + \bar{R}_{n,4}(\boldsymbol{X})$$

where the functions $\bar{R}_{n,3}(\boldsymbol{X})$ $\bar{R}_{n,4}(\boldsymbol{X})$ are Taylor remainders. Furthermore, they satisfy that $\bar{R}_{n,3}(\boldsymbol{X})/(\mathcal{D}_n+|\alpha_n-\alpha_*|) \to 0$ and $\bar{R}_{n,4}(\boldsymbol{X})/(\mathcal{D}_n+|\alpha_n-\alpha_*|) \to 0$ when $n \to \infty$. Given the explicit formulations of the derivatives of the functions $\bar{U}(\boldsymbol{X};\boldsymbol{p})$ and $\bar{V}(\boldsymbol{p})$, the functions $\bar{B}_{n,1}(\boldsymbol{X})$ and $\bar{B}_{n,2}(\boldsymbol{X})$ can be then rewritten as follows:

$$\bar{B}_{n,1}(\boldsymbol{X}) = \sum_{j:|\mathcal{C}_j|=1} \exp((\bar{B}\boldsymbol{p}_{*,j})^\top \boldsymbol{X}) \mathcal{N}_{1,n}(\boldsymbol{p}_{*,j})^\top \boldsymbol{X} f_{G_n}(\boldsymbol{X}) + \bar{R}_{n,3}(\boldsymbol{X}),$$

$$\begin{aligned}\bar{B}_{n,2}(\boldsymbol{X}) = \sum_{j:|\mathcal{C}_j|>1} \exp((\bar{B}\boldsymbol{p}_{*,j})^\top \boldsymbol{X}) \big[\bar{\mathcal{N}}_{1,n}(\boldsymbol{p}_{*,j})^\top \bar{B}^\top \boldsymbol{X} \\ + (\bar{B}^\top \boldsymbol{X})^\top \bar{\mathcal{N}}_{2,n}(\boldsymbol{p}_{*,j})(\bar{B}^\top \boldsymbol{X})\big] f_{G_n}(\boldsymbol{X}) + \bar{R}_{n,4}(\boldsymbol{X}).\end{aligned}$$

此处，函数$\mathcal{N}_{1,n}(\boldsymbol{X})$、$\bar{\mathcal{N}}_{1,n}(\boldsymbol{X})$以及$\bar{\mathcal{N}}_{2,n}(\boldsymbol{X})$的表达式如下：

$$\begin{aligned}\mathcal{N}_{1,n}(\boldsymbol{p}) &= \sum_{u=1}^{d} \bar{M}_{n,j,1_u} 1_u, \\ \bar{\mathcal{N}}_{1,n}(\boldsymbol{p}) &= \sum_{u=1}^{d} \bar{M}_{n,j,1_u} 1_u, \\ \bar{\mathcal{N}}_{2,n}(\boldsymbol{p}) &= \sum_{1\leq u,v\leq d} \bar{M}_{n,j,1_{uv}} 1_u 1_v^\top.\end{aligned}$$

将上述所有结果与函数的分解相结合 $\bar{A}_n(\boldsymbol{X})$ 以及 $\bar{B}_n(\boldsymbol{X})$，我们就可以如下表示该函数$\bar{Q}_{n,1}(\boldsymbol{X})$ 不等式（22）：

$$\begin{aligned}\bar{Q}_{n,1}(\boldsymbol{X}) = &\sum_{j:|\mathcal{C}_j|=1} \exp((\bar{B}\boldsymbol{p}_{*,j})^\top \boldsymbol{X}) \big[\bar{\mathcal{L}}'_{1,n}(\boldsymbol{p}_{*,j}) + \bar{\mathcal{L}}_{2,n}(\boldsymbol{p}_{*,j})^\top \bar{B}^\top \boldsymbol{X}\big) \\ &+ \sum_{j:|\mathcal{C}_j|>1} \exp((\bar{B}\boldsymbol{p}_{*,j})^\top \boldsymbol{X}) \big[\bar{\mathcal{L}}'_{1,n}(\boldsymbol{p}_{*,j}) + \bar{\mathcal{L}}_{2,n}(\boldsymbol{p}_{*,j})^\top \bar{B}^\top \boldsymbol{X} + (\bar{B}^\top \boldsymbol{X})^\top \bar{\mathcal{L}}_{3,n}(\boldsymbol{p}_{*,j}) \bar{B}^\top \boldsymbol{X}\big] \\ &- \sum_{j:|\mathcal{C}_j|=1} \exp((\bar{B}\boldsymbol{p}_{*,j})^\top \boldsymbol{X}) \big[\bar{M}_{n,j,0_d} + \mathcal{N}_{1,n}(\boldsymbol{p}_{*,j})^\top \bar{B}^\top \boldsymbol{X}\big] f_{G_n}(\boldsymbol{X}) \\ &- \sum_{j:|\mathcal{C}_j|>1} \exp((\bar{B}\boldsymbol{p}_{*,j})^\top \boldsymbol{X}) \big[\bar{M}_{n,j,0_d} + \bar{\mathcal{N}}_{1,n}(\boldsymbol{p}_{*,j})^\top \bar{B}^\top \boldsymbol{X} + (\bar{B}^\top \boldsymbol{X})^\top \bar{\mathcal{N}}_{2,n}(\boldsymbol{p}_{*,j}) \bar{B}^\top \boldsymbol{X}\big] f_{G_n}(\boldsymbol{X}) \\ &+ \bar{R}_{n,1}(\boldsymbol{X}) + \bar{R}_{n,2}(\boldsymbol{X}) - \bar{R}_{n,3}(\boldsymbol{X}) - \bar{R}_{n,4}(\boldsymbol{X}) \qquad (23)\end{aligned}$$

在此我们定义 $\bar{M}_{n,j,0_d}$ = 为$\tanh(\alpha_n)\Big(\sum_{i\in\mathcal{C}_j} \exp(\bar{b}_{n,i}),$ − 为$\exp(\bar{b}_{*,j})\Big)$，其中 $1 \leq j \leq L$, $\mathcal{L}'_{1,n}$分别对应$(\boldsymbol{p}_{*,j})$、 = $\mathcal{L}_{1,n}(\boldsymbol{p}_{*,j})$+ $M_{n,j,0_d}C\boldsymbol{p}_{*,j}$, 而$\mathcal{L}'_{1,n}$则对应$(\boldsymbol{p}_{*,j})$= $\mathcal{L}_{1,n}(\boldsymbol{p}_{*,j})$+ $M_{n,j,0_d}C\boldsymbol{p}_{*,j_\circ}$

该函数的可分解形式为$Q_{n,2}(\boldsymbol{X})$。通过一阶泰勒展开，可得到该函数的如下表达式：

$$\begin{aligned}\bar{Q}_{n,2}(\boldsymbol{X}) &= [\tanh(\alpha_n) - \tanh(\alpha_*)] \cdot \Big[\sum_{k'=1}^{L} \exp((\bar{B}\boldsymbol{p}_{*,k'})^\top \boldsymbol{X} + \bar{b}_{*,k'})\Big] \cdot f_{G_*}(\boldsymbol{X}) \\ &= (\alpha_n - \alpha_*)[1 - \tanh^2(\alpha_*)] \cdot \Big[\sum_{k'=1}^{L} \exp((\bar{B}\boldsymbol{p}_{*,k'})^\top \boldsymbol{X} + \bar{b}_{*,k'})\Big] \cdot f_{G_*}(\boldsymbol{X}) + \bar{R}_5(\boldsymbol{X}), \qquad (24)\end{aligned}$$

其中函数 $\bar{R}_{n,5}(\boldsymbol{X})$为泰勒余项，且满足 $\bar{R}_{n,5}(\boldsymbol{X})/(\mathcal{D}_n+|\alpha_n-\alpha_*|) \to 0$ 等于 $n \to \infty_\circ$

Step 2 - Non-vanishing coefficients. The results of equations (23) and (24) indicate that $[\bar{Q}_n(\boldsymbol{X}) - \sum_{i=1}^{5} \bar{R}_{n,i}(\boldsymbol{X})]/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$ can be represented as a combination of the linearly independent functions $\exp((\bar{B}\boldsymbol{p}_{*,j})^\top \boldsymbol{X})$, $(\bar{B}^\top \boldsymbol{X})^{(u)} \exp((\bar{B}\boldsymbol{p}_{*,j})^\top \boldsymbol{X})$, $(\bar{B}^\top \boldsymbol{X})^{(u)}(\bar{B}^\top \boldsymbol{X})^{(v)} \exp((\bar{B}\boldsymbol{p}_{*,j})^\top \boldsymbol{X})$, $\exp((\bar{B}\boldsymbol{p}_{*,j})^\top \boldsymbol{X}) f_{G_n}(\boldsymbol{X})$, $(\bar{B}^\top \boldsymbol{X})^{(u)} \exp((\bar{B}\boldsymbol{p}_{*,j})^\top \boldsymbol{X}) f_{G_n}(\boldsymbol{X})$, $(\bar{B}^\top \boldsymbol{X})^{(u)}(\bar{B}^\top \boldsymbol{X})^{(v)} \exp((\bar{B}\boldsymbol{p}_{*,j})^\top \boldsymbol{X}) f_{G_n}(\boldsymbol{X})$, and $[1 - \tanh^2(\alpha_*)] \cdot \left[\sum_{k'=1}^{L} \exp((\bar{B}\boldsymbol{p}_{*,k'})^\top \boldsymbol{X} + \bar{b}_{*,k'})\right] \cdot f_{G_*}(\boldsymbol{X})$ for any $1 \le j \le L$ and $1 \le u, v \le d$.

Our claim is that at least one of the coefficients of these linearly independent terms in the formulation of $[\bar{Q}_n(\boldsymbol{X}) - \sum_{i=1}^{5} \bar{R}_{n,i}(\boldsymbol{X})]/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$ does not go to 0 as $n \to \infty$. Assume by contrary that this claim does not hold, which means that all the coefficients of these linearly independent terms go to 0 as $n \to \infty$. Therefore, as $n \to \infty$ we obtain that

$$\mathcal{L}_{1,n}(\boldsymbol{p}_{*,j})/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0,\ \mathcal{L}_{2,n}(\boldsymbol{p}_{*,j})^{(u)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0,\ \bar{\mathcal{L}}_{1,n}(\boldsymbol{p}_{*,j})/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0,$$
$$\bar{\mathcal{L}}_{2,n}(\boldsymbol{p}_{*,j})^{(u)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0,\ \bar{\mathcal{L}}_{2,n}(\boldsymbol{p}_{*,j})^{(u)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0,\ \bar{\mathcal{L}}_{2,n}(\boldsymbol{p}_{*,j})^{(u)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0,$$
$$\bar{\mathcal{L}}_{3,n}(\boldsymbol{p}_{*,j})^{(uv)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0,\ \mathcal{N}_{1,n}(\boldsymbol{p}_{*,j})/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0,\ \bar{\mathcal{N}}_{1,n}((\boldsymbol{p}_{*,j})^{(u)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0,$$
$$\bar{\mathcal{N}}_{2,n}(\boldsymbol{p}_{*,j})^{(uv)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0,\ \bar{M}_{n,j,0_d}/\mathcal{D}_n \to 0,\ (\alpha_n - \alpha_*)/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0$$

for any $1 \le u, v \le d$ and $1 \le j \le L$.

As $(\alpha_n - \alpha_*)/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0$, we deduce that

$$\frac{|\alpha_n - \alpha_*|}{(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to 0. \tag{25}$$

Note that since $\alpha_n \to \alpha_* \neq 0$ as $n \to \infty$, we have $1/\tanh(\alpha_n) \not\to \infty$. Then, as $\bar{M}_{n,j,0_d}/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0$, it implies that

$$\frac{|\sum_{i \in \mathcal{C}_j} \exp(\bar{b}_{n,i}) - \exp(\bar{b}_{*,j})|}{(\mathcal{D}_n + |\alpha_n - \alpha_*|)} = \frac{1}{\tanh(\alpha_n)} \cdot \frac{|\bar{M}_{n,j,0_d}|}{(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to 0,$$

for any $1 \le j \le L$. By varying the index j from 1 to L in these limits and summing them up, we obtain that

$$\frac{\sum_{j=1}^{L} |\sum_{i \in \mathcal{C}_j} \exp(\bar{b}_{n,i}) - \exp(\bar{b}_{*,j})|}{(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to 0. \tag{26}$$

Now, we consider indices $j \in [L]$ such that $|\mathcal{C}_j| = 1$, i.e., the corresponding Voronoi cell has only one element. From the hypothesis, we have $\mathcal{L}_{2,n}(\boldsymbol{p}_{*,j})^{(u)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0$, which leads to $\bar{M}_{n,j,1_u}/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0$. Hence, we find that

$$\frac{\sum_{i \in \mathcal{C}_j} \exp(\bar{b}_{n,i}) \|\Delta \boldsymbol{p}_{n,ij}\|}{(\mathcal{D}_n + |\alpha_n - \alpha_*|)} = \frac{\sum_{u=1}^{d} |\bar{M}_{n,j,1_u}|}{\tanh(\alpha_n)(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to 0.$$

That limit directly implies the following result:

$$\frac{\sum_{j:|\mathcal{C}_j|=1} \sum_{i \in \mathcal{C}_j} \exp(\bar{b}_{n,i}) \|\Delta \boldsymbol{p}_{n,ij}\|}{(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to 0. \tag{27}$$

We now move to the Voronoi cells having more than one element, namely, we consider indices $j \in [L]$ satisfying $|\mathcal{C}_j| > 1$. The limit $\bar{\mathcal{L}}_{3,n}(\boldsymbol{p}_{*,j})^{(uu)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0$ induces that

$$\frac{\sum_{i \in \mathcal{C}_j} \exp(\bar{b}_{n,i}) \|\Delta \boldsymbol{p}_{n,ij}\|^2}{(\mathcal{D}_n + |\alpha_n - \alpha_*|)} = \frac{\sum_{u=1}^{d} \bar{\mathcal{L}}_{3,n}(\boldsymbol{p}_{*,j})^{(uu)}}{\tanh(\alpha_n)(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to 0.$$

By varying the indices j in these limits over all the Voronoi cells $\mathcal{C}_j$ having more than one element, we find that

$$\frac{\sum_{j:|\mathcal{C}_j|>1} \sum_{i \in \mathcal{C}_j} \exp(\bar{b}_{n,i}) \|\Delta \boldsymbol{p}_{n,ij}\|^2}{(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to 0. \tag{28}$$

Step 2 - Non-vanishing coefficients. The results of equations (23) and (24) indicate that $[\bar{Q}_n(\boldsymbol{X}) - \sum_{i=1}^{5} \bar{R}_{n,i}(\boldsymbol{X})]/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$ can be represented as a combination of the linearly independent functions $\exp((\bar{B}\boldsymbol{p}_{*,j})^\top \boldsymbol{X})$, $(\bar{B}^\top \boldsymbol{X})^{(u)} \exp((\bar{B}\boldsymbol{p}_{*,j})^\top \boldsymbol{X})$, $(\bar{B}^\top \boldsymbol{X})^{(u)}(\bar{B}^\top \boldsymbol{X})^{(v)} \exp((\bar{B}\boldsymbol{p}_{*,j})^\top \boldsymbol{X})$, $\exp((\bar{B}\boldsymbol{p}_{*,j})^\top \boldsymbol{X}) f_{G_n}(\boldsymbol{X})$, $(\bar{B}^\top \boldsymbol{X})^{(u)} \exp((\bar{B}\boldsymbol{p}_{*,j})^\top \boldsymbol{X}) f_{G_n}(\boldsymbol{X})$, $(\bar{B}^\top \boldsymbol{X})^{(u)}(\bar{B}^\top \boldsymbol{X})^{(v)} \exp((\bar{B}\boldsymbol{p}_{*,j})^\top \boldsymbol{X}) f_{G_n}(\boldsymbol{X})$, and $[1 - \tanh^2(\alpha_*)] \cdot \left[\sum_{k'=1}^{L} \exp((\bar{B}\boldsymbol{p}_{*,k'})^\top \boldsymbol{X} + \bar{b}_{*,k'})\right] \cdot f_{G_*}(\boldsymbol{X})$ for any $1 \le j \le L$ and $1 \le u, v \le d$.

¯我们的结论是，在 $[Q_n(\boldsymbol{X}) - \sum_{i=1}^{5} \bar{R}_{n,\ i}(\boldsymbol{X})]/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$的表述中，这些线性无关项的系数至少有一个不会像 $n \to \infty$那样趋于0。假设相反的情况成立，即该结论不成立，这意味着这些线性无关项的所有系数都会像 $n \to \infty$那样趋于0。因此，正如 $n \to \infty$ 所示，我们得出了这样的结果。

$$\mathcal{L}_{1,n}(\boldsymbol{p}_{*,j})/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0,\ \mathcal{L}_{2,n}(\boldsymbol{p}_{*,j})^{(u)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0,\ \bar{\mathcal{L}}_{1,n}(\boldsymbol{p}_{*,j})/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0,$$
$$\bar{\mathcal{L}}_{2,n}(\boldsymbol{p}_{*,j})^{(u)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0,\ \bar{\mathcal{L}}_{2,n}(\boldsymbol{p}_{*,j})^{(u)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0,\ \bar{\mathcal{L}}_{2,n}(\boldsymbol{p}_{*,j})^{(u)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0,$$
$$\bar{\mathcal{L}}_{3,n}(\boldsymbol{p}_{*,j})^{(uv)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0,\ \mathcal{N}_{1,n}(\boldsymbol{p}_{*,j})/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0,\ \bar{\mathcal{N}}_{1,n}((\boldsymbol{p}_{*,j})^{(u)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0,$$
$$\bar{\mathcal{N}}_{2,n}(\boldsymbol{p}_{*,j})^{(uv)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0,\ \bar{M}_{n,j,0_d}/\mathcal{D}_n \to 0,\ (\alpha_n - \alpha_*)/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0$$

对于任意的 $1 \le u,\ v \le d$ 以及 $1 \le j \le L$。

由于 $(\alpha_n - \alpha_*)/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0$，我们由此推断出

$$\frac{|\alpha_n - \alpha_*|}{(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to 0. \tag{25}$$

¯请注意，由于 $\alpha_n \to \alpha_* \neq 0$ 相当于 $n \to \infty$，因此我们有 $1/\tanh(\alpha_n) \not\to \infty$。进而，因为$M_{n,j,0_d}/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0$，这就意味着

$$\frac{|\sum_{i \in \mathcal{C}_j} \exp(\bar{b}_{n,i}) - \exp(\bar{b}_{*,j})|}{(\mathcal{D}_n + |\alpha_n - \alpha_*|)} = \frac{1}{\tanh(\alpha_n)} \cdot \frac{|\bar{M}_{n,j,0_d}|}{(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to 0,$$

对于任意的 $1 \le j \le L$. 通过在这些范围内将指数 j 从1变化到 L 并求和，我们可以得到

$$\frac{\sum_{j=1}^{L} |\sum_{i \in \mathcal{C}_j} \exp(\bar{b}_{n,i}) - \exp(\bar{b}_{*,j})|}{(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to 0. \tag{26}$$

现在，我们考虑满足 $|\mathcal{C}_j| = 1$条件的索引 $j \in [L]$，即对应的Voronoi单元仅包含一个元素。根据假设，我们有$\mathcal{L}_{2,\ n}(\boldsymbol{p}_{*,j})^{(u)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0$，进而得到$\bar{M}_{n,j,1_u}/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0$。因此，我们得出结论：

$$\frac{\sum_{i \in \mathcal{C}_j} \exp(\bar{b}_{n,i}) \|\Delta \boldsymbol{p}_{n,ij}\|}{(\mathcal{D}_n + |\alpha_n - \alpha_*|)} = \frac{\sum_{u=1}^{d} |\bar{M}_{n,j,1_u}|}{\tanh(\alpha_n)(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to 0.$$

该极限可直接推导出以下结果：

$$\frac{\sum_{j:|\mathcal{C}_j|=1} \sum_{i \in \mathcal{C}_j} \exp(\bar{b}_{n,i}) \|\Delta \boldsymbol{p}_{n,ij}\|}{(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to 0. \tag{27}$$

接下来我们研究包含多个元素的Voronoi单元，即考虑满足 $|\mathcal{C}_j| > 1$条件的索引 $j \in [L]$。极限$\bar{\mathcal{L}}_{3,\ n}(\boldsymbol{p}_{*,\ j})^{(uu)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0$ 会促使出现如下情况。

$$\frac{\sum_{i \in \mathcal{C}_j} \exp(\bar{b}_{n,i}) \|\Delta \boldsymbol{p}_{n,ij}\|^2}{(\mathcal{D}_n + |\alpha_n - \alpha_*|)} = \frac{\sum_{u=1}^{d} \bar{\mathcal{L}}_{3,n}(\boldsymbol{p}_{*,j})^{(uu)}}{\tanh(\alpha_n)(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to 0.$$

通过改变in的值。在所有包含多个元素的Voronoi单元中，将 j 在这些范围内的随机数进行抽样后，我们发现

$$\frac{\sum_{j:|\mathcal{C}_j|>1} \sum_{i \in \mathcal{C}_j} \exp(\bar{b}_{n,i}) \|\Delta \boldsymbol{p}_{n,ij}\|^2}{(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to 0. \tag{28}$$

Combining the results from equations (25), (26), (27), and (28) leads to

$$1 = \frac{\mathcal{D}_n + |\alpha_n - \alpha_*|}{\mathcal{D}_n + |\alpha_n - \alpha_*|} \to 0$$

as $n \to \infty$, which cannot hold. As a consequence, at least one of the coefficients of the terms in the formulations of $[Q_n(\boldsymbol{X}) - \sum_{i=1}^{5} \bar{R}_{n,i}(\boldsymbol{X})]/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$ does not go to 0 as $n \to \infty$.

Step 3 - Application of the Fatou's lemma. We denote m_n as the maximum of the absolute values of $\mathcal{L}'_{1,n}(\boldsymbol{p}_{*,j})/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$, $\mathcal{L}_{2,n}(\boldsymbol{p}_{*,j})^{(u)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$, $\bar{\mathcal{L}}'_{1,n}(\boldsymbol{p}_{*,j})/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$, $\bar{\mathcal{L}}_{2,n}(\boldsymbol{p}_{*,j})^{(u)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$, $\bar{\mathcal{L}}_{3,n}(\boldsymbol{p}_{*,j})^{(uv)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$, $\mathcal{N}_{1,n}(\boldsymbol{p}_{*,j})/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$, $\bar{\mathcal{N}}_{1,n}((\boldsymbol{p}_{*,j})^{(u)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$, $\bar{\mathcal{N}}_{2,n}(\boldsymbol{p}_{*,j})^{(uv)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$, $\bar{M}_{n,j,0_d}/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$, and $(\alpha_n - \alpha_*)/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$ for all $1 \leq u, v \leq d$. From the result of Step 2 in the proof, we have $1/m_n \not\to \infty$ as $n \to \infty$.

Recall that $\|f_{G_n,\alpha_n} - f_{G_*,\alpha_*}\|_{L^2(\mu)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0$ as $n \to \infty$, which indicates that $\|f_{G_n,\alpha_n} - f_{G_*,\alpha_*}\|_{L^2(\mu)}/(m_n(\mathcal{D}_n + |\alpha_n - \alpha_*|)) \to 0$. Furthermore, since the $L^2(\mu)$ norm is equivalent to the $L^1(\mu)$ norm, we have $\|f_{G_n,\alpha_n} - f_{G_*,\alpha_*}\|_{L^1(\mu)}/(m_n(\mathcal{D}_n + |\alpha_n - \alpha_*|)) \to 0$. An application of Fatou's lemma leads to

$$0 = \lim_{n\to\infty} \frac{\|f_{G_n,\alpha_n} - f_{G_*,\alpha_*}\|_{L^1(\mu)}}{m_n(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \geq \int \liminf_{n\to\infty} \frac{\|f_{G_n,\alpha_n}(\boldsymbol{X}) - f_{G_*,\alpha_*}(\boldsymbol{X})\|_1}{m_n(\mathcal{D}_n + |\alpha_n - \alpha_*|)} d\mu(\boldsymbol{X}) \geq 0.$$

It indicates that $\liminf_{n\to\infty} \frac{\|f_{G_n}(\boldsymbol{X}) - f_{G_*}(\boldsymbol{X})\|_1}{m_n(\mathcal{D}_n + |\alpha_n - \alpha_*|)} = 0$ for almost surely $\boldsymbol{X}$. As $n \to \infty$, we denote

$$\begin{aligned}
&\frac{\mathcal{L}'_{1,n}(\boldsymbol{p}_{*,j})}{m_n(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to \widehat{\alpha}_j, \quad \frac{\mathcal{L}_{2,n}(\boldsymbol{p}_{*,j})}{m_n(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to \beta_j, \\
&\frac{\bar{\mathcal{L}}'_{1,n}(\boldsymbol{p}_{*,j})}{m_n(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to \bar{\alpha}_j, \quad \frac{\bar{\mathcal{L}}_{2,n}(\boldsymbol{p}_{*,j})}{m_n(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to \bar{\beta}_j, \quad \frac{\bar{\mathcal{L}}_{3,n}(\boldsymbol{p}_{*,j})}{m_n(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to \bar{\gamma}_j, \\
&\frac{\bar{M}_{n,j,0_d}}{m_n(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to \tilde{\alpha}_j, \quad \frac{\mathcal{N}_{1,n}(\boldsymbol{p}_{*,j})}{m_n(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to \tilde{\beta}_j, \\
&\frac{\bar{\mathcal{N}}_{1,n}(\boldsymbol{p}_{*,j})}{m_n(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to \widehat{\beta}_j, \quad \frac{\bar{\mathcal{N}}_{2,n}(\boldsymbol{p}_{*,j})}{m_n(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to \widehat{\gamma}_j, \quad \frac{\alpha_n - \alpha_*}{m_n(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to \tau
\end{aligned}$$

for any $1 \leq j \leq L$. Here, from the definition of m_n, at least one coefficient among $\{\widehat{\alpha}_j, \beta_j, \tilde{\alpha}_j, \tilde{\beta}_j\}_{j:|\mathcal{C}_j|=1}$, $\{\bar{\alpha}_j, \bar{\beta}_j, \bar{\gamma}_j, \tilde{\alpha}_j, \widehat{\beta}_j, \widehat{\gamma}_j\}_{j:|\mathcal{C}_j|>1}$, and τ is different from 0. Then, the equation

$$\liminf_{n\to\infty} \frac{\|Q_n(\boldsymbol{X})\|_1}{m_n(\mathcal{D}_n + |\alpha_n - \alpha_*|)} = \Big[\sum_{k'=1}^{L} \exp((\bar{B}\boldsymbol{p}_{*,k'})^\top \boldsymbol{X} + \bar{b}_{*,k'})\Big] \cdot \liminf_{n\to\infty} \frac{\|f_{G_n,\alpha_n}(\boldsymbol{X}) - f_{G_*,\alpha_*}(\boldsymbol{X})\|_1}{m_n(\mathcal{D}_n + |\alpha_n - \alpha_*|)} = 0$$

leads to

$$\begin{aligned}
&\sum_{j:|\mathcal{C}_j|=1} \exp((\bar{B}\boldsymbol{p}_{*,j})^\top \boldsymbol{X})(\alpha_j + \beta_j^\top(\bar{B}^\top \boldsymbol{X})) \\
&+ \sum_{j:|\mathcal{C}_j|>1} \exp((\bar{B}\boldsymbol{p}_{*,j})^\top \boldsymbol{X})\big[\bar{\alpha}_j + \bar{\beta}_j^\top(B^\top \boldsymbol{X}) + (\bar{B}^\top \boldsymbol{X})^\top \bar{\gamma}_j(\bar{B}^\top \boldsymbol{X})\big] \\
&- \sum_{j:|\mathcal{C}_j|=1} \exp((\bar{B}\boldsymbol{p}_{*,j})^\top \boldsymbol{X})(\tilde{\alpha}_j + \tilde{\beta}_j^\top(\bar{B}^\top \boldsymbol{X}))f_{G_*}(\boldsymbol{X}) \\
&- \sum_{j:|\mathcal{C}_j|>1} \exp((\bar{B}\boldsymbol{p}_{*,j})^\top \boldsymbol{X})\big[\tilde{\alpha}_j + \widehat{\beta}_j^\top(\bar{B}^\top \boldsymbol{X}) + (\bar{B}^\top \boldsymbol{X})^\top \widehat{\gamma}_j \bar{B}^\top \boldsymbol{X}\big] f_{G_*}(\boldsymbol{X}) \\
&+ \tau[1 - \tanh^2(\alpha_*)] \cdot \Big[\sum_{k'=1}^{L} \exp((\bar{B}\boldsymbol{p}_{*,k'})^\top \boldsymbol{X} + \bar{b}_{*,k'})\Big] \cdot f_{G_*}(\boldsymbol{X}) = \boldsymbol{0}_d
\end{aligned}$$

for almost surely $\boldsymbol{X}$. That equation only holds if and only if all the coefficients $\{\widehat{\alpha}_j, \beta_j, \tilde{\alpha}_j, \tilde{\beta}_j\}_{j:|\mathcal{C}_j|=1}$,

结合方程（25）、（26）、（27）以及（28）的结果，可得到

$$1 = \frac{\mathcal{D}_n + |\alpha_n - \alpha_*|}{\mathcal{D}_n + |\alpha_n - \alpha_*|} \to 0$$

由于 $n \to \infty$无法成立。因此，在$[Q_n(\boldsymbol{X})- \sum_{i=1}^{5} \bar{R}_{n,\ i}(\boldsymbol{X})]/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$的表达式中，各项的系数至少有一个不会像 $n \to \infty$那样趋于0。

Step 3 - Application of the Fatou's lemma. We denote m_n as the maximum of the absolute values of $\mathcal{L}'_{1,n}(\boldsymbol{p}_{*,j})/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$, $\mathcal{L}_{2,n}(\boldsymbol{p}_{*,j})^{(u)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$, $\bar{\mathcal{L}}'_{1,n}(\boldsymbol{p}_{*,j})/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$, $\bar{\mathcal{L}}_{2,n}(\boldsymbol{p}_{*,j})^{(u)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$, $\bar{\mathcal{L}}_{3,n}(\boldsymbol{p}_{*,j})^{(uv)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$, $\mathcal{N}_{1,n}(\boldsymbol{p}_{*,j})/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$, $\bar{\mathcal{N}}_{1,n}((\boldsymbol{p}_{*,j})^{(u)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$, $\bar{\mathcal{N}}_{2,n}(\boldsymbol{p}_{*,j})^{(uv)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$, $\bar{M}_{n,j,0_d}/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$, and $(\alpha_n - \alpha_*)/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$ for all $1 \leq u, v \leq d$. From the result of Step 2 in the proof, we have $1/m_n \not\to \infty$ as $n \to \infty$.

请记住，$\|f_{G_n,\ \alpha_n} - f_{G_*,\ \alpha_*}\|_{L^2(}$ $_{f_{G_*}\ \alpha_*\|_{L^2}\ \mu}$ / $_{m_n}$ $_{\mathcal{D}_n + |\alpha_n - \alpha_*|}$ $\to$ 0 可表示为 $n \to \infty$，这说明$\|f_{G_n,\ ()}(())$。此外，由于()范数与()范数等价，因此我们有$\|f_{G_n,\ \alpha_n} - f_{G_*,\ \alpha_*}\|_{L^1(\mu)}$ /$(m_n(\mathcal{D}_n + |\alpha_n - \alpha_*|)) \to 0$。应用Fatou引理可得出

$$0 = \lim_{n\to\infty} \frac{\|f_{G_n,\alpha_n} - f_{G_*,\alpha_*}\|_{L^1(\mu)}}{m_n(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \geq \int \liminf_{n\to\infty} \frac{\|f_{G_n,\alpha_n}(\boldsymbol{X}) - f_{G_*,\alpha_*}(\boldsymbol{X})\|_1}{m_n(\mathcal{D}_n + |\alpha_n - \alpha_*|)} d\mu(\boldsymbol{X}) \geq 0.$$

这表明对于几乎必然成立的 $\boldsymbol{X}$，有$\liminf_{n\to\infty} \frac{\|f_{G_n}(\quad \boldsymbol{X} \ - f_{G_*}\ \boldsymbol{X}\ \|_1)\ (\ )\ (\ ) = 0}{m_n\ \mathcal{D}_n + |\alpha_n - \alpha_*|}$。我们将 $n \to \infty$记为

$$\begin{aligned}
&\frac{\mathcal{L}'_{1,n}(\boldsymbol{p}_{*,j})}{m_n(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to \widehat{\alpha}_j, \quad \frac{\mathcal{L}_{2,n}(\boldsymbol{p}_{*,j})}{m_n(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to \beta_j, \\
&\frac{\bar{\mathcal{L}}'_{1,n}(\boldsymbol{p}_{*,j})}{m_n(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to \bar{\alpha}_j, \quad \frac{\bar{\mathcal{L}}_{2,n}(\boldsymbol{p}_{*,j})}{m_n(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to \bar{\beta}_j, \quad \frac{\bar{\mathcal{L}}_{3,n}(\boldsymbol{p}_{*,j})}{m_n(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to \bar{\gamma}_j, \\
&\frac{\bar{M}_{n,j,0_d}}{m_n(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to \tilde{\alpha}_j, \quad \frac{\mathcal{N}_{1,n}(\boldsymbol{p}_{*,j})}{m_n(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to \tilde{\beta}_j, \\
&\frac{\bar{\mathcal{N}}_{1,n}(\boldsymbol{p}_{*,j})}{m_n(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to \widehat{\beta}_j, \quad \frac{\bar{\mathcal{N}}_{2,n}(\boldsymbol{p}_{*,j})}{m_n(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to \widehat{\gamma}_j, \quad \frac{\alpha_n - \alpha_*}{m_n(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to \tau
\end{aligned}$$

˜对于任意的 $1 \leq j \leq L$而言。根据 m_n的定义， $\{\widehat{\alpha}_j, \beta_j, \tilde{\alpha}_j, \beta_j\}_{j:|\mathcal{C}_j|=1}$、 - ⌒、 $\{\bar{\alpha}_j, \beta_j, \bar{\gamma}_j, \tilde{\alpha}_j, \beta_j, \widehat{\gamma}_j\}_{j:|\mathcal{C}_j|>1}$以及 τ这些系数中至少有一个不等于0。进而，该方程

$$\liminf_{n\to\infty} \frac{\|Q_n(\boldsymbol{X})\|_1}{m_n(\mathcal{D}_n + |\alpha_n - \alpha_*|)} = \Big[\sum_{k'=1}^{L} \exp((\bar{B}\boldsymbol{p}_{*,k'})^\top \boldsymbol{X} + \bar{b}_{*,k'})\Big] \cdot \liminf_{n\to\infty} \frac{\|f_{G_n,\alpha_n}(\boldsymbol{X}) - f_{G_*,\alpha_*}(\boldsymbol{X})\|_1}{m_n(\mathcal{D}_n + |\alpha_n - \alpha_*|)} = 0$$

会导出相应结果。

$$\begin{aligned}
&\sum_{j:|\mathcal{C}_j|=1} \exp((\bar{B}\boldsymbol{p}_{*,j})^\top \boldsymbol{X})(\alpha_j + \beta_j^\top(\bar{B}^\top \boldsymbol{X})) \\
&+ \sum_{j:|\mathcal{C}_j|>1} \exp((\bar{B}\boldsymbol{p}_{*,j})^\top \boldsymbol{X})\big[\bar{\alpha}_j + \bar{\beta}_j^\top(B^\top \boldsymbol{X}) + (\bar{B}^\top \boldsymbol{X})^\top \bar{\gamma}_j(\bar{B}^\top \boldsymbol{X})\big] \\
&- \sum_{j:|\mathcal{C}_j|=1} \exp((\bar{B}\boldsymbol{p}_{*,j})^\top \boldsymbol{X})(\tilde{\alpha}_j + \tilde{\beta}_j^\top(\bar{B}^\top \boldsymbol{X}))f_{G_*}(\boldsymbol{X}) \\
&- \sum_{j:|\mathcal{C}_j|>1} \exp((\bar{B}\boldsymbol{p}_{*,j})^\top \boldsymbol{X})\big[\tilde{\alpha}_j + \widehat{\beta}_j^\top(\bar{B}^\top \boldsymbol{X}) + (\bar{B}^\top \boldsymbol{X})^\top \widehat{\gamma}_j \bar{B}^\top \boldsymbol{X}\big] f_{G_*}(\boldsymbol{X}) \\
&+ \tau[1 - \tanh^2(\alpha_*)] \cdot \Big[\sum_{k'=1}^{L} \exp((\bar{B}\boldsymbol{p}_{*,k'})^\top \boldsymbol{X} + \bar{b}_{*,k'})\Big] \cdot f_{G_*}(\boldsymbol{X}) = \boldsymbol{0}_d
\end{aligned}$$

˜几乎可以肯定地，结果为 $\boldsymbol{X}$。只有当所有系数 $\{\widehat{\alpha}_j, \beta_j, \tilde{\alpha}_j, \beta_j\}_{j:|\mathcal{C}_j|=1}$都满足特定条件时，该等式才成立。

$\{\bar{\alpha}_j, \bar{\beta}_j, \bar{\gamma}_j, \tilde{\alpha}_j, \widehat{\beta}_j, \widehat{\gamma}_j\}_{j:|\mathcal{C}_j|>1}$, and τ are 0, which is a contradiction.

It indicates that we indeed have the conclusion of the local part, namely,

$$\lim_{\varepsilon\to 0} \inf_{G\in\mathcal{G}_{L'}(\Omega):(\mathcal{D}(G,G_*)+|\alpha-\alpha_*|)\leq\varepsilon} \|f_{G,\alpha} - f_{G_*,\alpha_*}\|_{L^2(\mu)}/(\mathcal{D}(G,G_*)+|\alpha-\alpha_*|) > 0.$$

Global part: The result of the local part implies that we can find a positive constant ε' such that

$$\inf_{G\in\mathcal{G}_{L'}(\Omega):(\mathcal{D}(G,G_*)+|\alpha-\alpha_*|)\leq\varepsilon'} \|f_{G,\alpha} - f_{G_*,\alpha_*}\|_{L^2(\mu)}/(\mathcal{D}(G,G_*)+|\alpha-\alpha_*|) > 0.$$

Therefore, to obtain the conclusion of the theorem it is sufficient to prove that

$$\inf_{G\in\mathcal{G}_{L'}(\Omega):(\mathcal{D}(G,G_*)+|\alpha-\alpha_*|)>\varepsilon'} \|f_{G,\alpha} - f_{G_*,\alpha_*}\|_{L^2(\mu)}/(\mathcal{D}(G,G_*)+|\alpha-\alpha_*|) > 0.$$

Assume by contrary that the above claim does not hold. Then there exists a sequence of measures $G'_n := \sum_{j'=1}^{L'} \exp(\bar{b}_{n,j'})\delta_{\boldsymbol{p}_{n,j'}}$ in $\mathcal{G}_{L'}(\Theta)$ and $\alpha'_n \in \Omega$ such that we have

$$\begin{cases} \mathcal{D}(G'_n, G_*) + |\alpha'_n - \alpha_*| > \varepsilon' \\ \|f_{G'_n,\alpha'_n} - f_{G_*,\alpha_*}\|_{L^2(\mu)}/(\mathcal{D}(G'_n, G_*) + |\alpha'_n - \alpha_*|) \to 0. \end{cases}$$

These limits indicate that $\|f_{G'_n,\alpha'_n} - f_{G_*,\alpha_*}\|_{L^2(\mu)} \to 0$ as $n \to \infty$.
Recall that the sets Θ and Ω are compact. Therefore, there exists a mixing measure G' in $\mathcal{G}_{L'}(\Omega)$ such that one of (G'_n, α'_n)'s subsequences converges to (G', α'). Since $\mathcal{D}(G'_n, G_*) + |\alpha'_n - \alpha_*| > \varepsilon'$, we deduce that $\mathcal{D}(G', G_*) + |\alpha' - \alpha_*| > \varepsilon'$.
An application of the Fatou's lemma leads to

$$0 = \lim_{n\to\infty} \|f_{G'_n,\alpha'_n} - f_{G_*,\alpha_*}\|_{L^2(\mu)} \geq \int \liminf_{n\to\infty} \left\|f_{G'_n,\alpha'_n}(\boldsymbol{X}) - f_{G_*,\alpha_*}(\boldsymbol{X})\right\|_2^2 d\mu(\boldsymbol{X}).$$

Hence, we have $f_{G',\alpha'}(\boldsymbol{X}) = f_{G_*,\alpha_*}(\boldsymbol{X})$ for $\mu-$almost surely $\boldsymbol{X}$. From the identifiability property (cf. the end of this proof), we deduce that $(G', \alpha') \equiv (G_*, \alpha_*)$. It follows that $\mathcal{D}(G', G_*) + |\alpha' - \alpha_*| = 0$. It contradicts to the hypothesis that $\mathcal{D}(G', G_*) + |\alpha' - \alpha_*| > \varepsilon' > 0$.
As a consequence, the proof of the global part is completed. We obtain the conclusion of the theorem.

Proof for the identifiability property. We now demonstrate that if $f_{\bar{G},\bar{\alpha}}(\boldsymbol{X}) = f_{\bar{G}_*,\bar{\alpha}_*}(\boldsymbol{X})$ for almost every $\boldsymbol{X}$, then we obtain that $(\bar{G}, \bar{\alpha}) \equiv (\bar{G}_*, \bar{\alpha}_*)$.

To ease the presentation we denote the following notations:

$$\begin{aligned}
\mathrm{softmax}^{\text{Pretrain}}_{\bar{G}}(u) &= \frac{\exp(u)}{\sum_{k=1}^{N}\exp(\boldsymbol{X}^\top \bar{A}_k^0 \boldsymbol{X} + \bar{a}_k^0)}, \\
\mathrm{softmax}^{\text{Prompt}}_{\bar{G}}(u') &= \frac{\exp(u)}{\sum_{j'=1}^{L'}\exp((\bar{B}\boldsymbol{p}_{j'})^\top \boldsymbol{X} + \bar{b}_{j'})}, \\
\mathrm{softmax}^{\text{Pretrain}}_{\bar{G}_*}(u_*) &= \frac{\exp(u_*)}{\sum_{k=1}^{N}\exp(\boldsymbol{X}^\top \bar{A}_k^0 \boldsymbol{X} + \bar{a}_k^0)}, \\
\mathrm{softmax}^{\text{Prompt}}_{\bar{G}_*}(u'_*) &= \frac{\exp(u_*)}{\sum_{j'=1}^{L}\exp((\bar{B}\boldsymbol{p}_{*,j'})^\top \boldsymbol{X} + \bar{b}_{*,j'})}.
\end{aligned}$$

Here, u, u', u_*, and u'_* in these equations satisfy:

$$\begin{aligned}
u &\in \{\boldsymbol{X}^\top \bar{A}_j^0 \boldsymbol{X} + \bar{a}_j^0 : j \in [N]\},\ u' \in \{(\bar{B}\boldsymbol{p}_{j'})^\top \boldsymbol{X} + \bar{b}_{j'} : j' \in [L']\} \\
u_* &\in \{\boldsymbol{X}^\top \bar{A}_j^0 \boldsymbol{X} + \bar{a}_j^0 : j \in [N]\},\ u'_* \in \{(\bar{B}\boldsymbol{p}_{*,j'})^\top \boldsymbol{X} + \bar{b}_{*,j'} : j' \in [L]\}.
\end{aligned}$$

The equation $f_{\bar{G},\bar{\alpha}}(\boldsymbol{X}) = f_{\bar{G}_*,\bar{\alpha}_*}(\boldsymbol{X})$ for almost every $\boldsymbol{X}$ indicates that

$$\sum_{j=1}^{N} \mathrm{softmax}^{\text{Pretrain}}_{\bar{G}}(\boldsymbol{X}^\top \bar{A}_j^0 \boldsymbol{X} + \bar{a}_j^0))h(\boldsymbol{X}, \bar{\eta}_j^0) + \tanh(\bar{\alpha}) \sum_{j'=1}^{L'} \mathrm{softmax}^{\text{Prompt}}_{\bar{G}}((\bar{B}\boldsymbol{p}_{j'})^\top \boldsymbol{X} + \bar{b}_{j'})\bar{C}\boldsymbol{p}_{j'}$$

– $\sim\{\bar{\alpha}_j, \bar{\beta}_j, \bar{\gamma}_j, \tilde{\alpha}_j, \bar{\beta}_j, \widehat{\gamma}_j\}_{j:|\mathcal{C}_j|>1}$，同时 τ的值为0，这与前提相矛盾。

这表明我们确实得到了局部部分的结论，即：

$$\lim_{\varepsilon\to 0} \inf_{G\in\mathcal{G}_{L'}(\Omega):(\mathcal{D}(G,G_*)+|\alpha-\alpha_*|)\leq\varepsilon} \|f_{G,\alpha} - f_{G_*,\alpha_*}\|_{L^2(\mu)}/(\mathcal{D}(G,G_*)+|\alpha-\alpha_*|) > 0.$$

全局部分： 局部部分的结果表明，我们可以找到一个正常数 ε'，使得

$$\inf_{G\in\mathcal{G}_{L'}(\Omega):(\mathcal{D}(G,G_*)+|\alpha-\alpha_*|)\leq\varepsilon'} \|f_{G,\alpha} - f_{G_*,\alpha_*}\|_{L^2(\mu)}/(\mathcal{D}(G,G_*)+|\alpha-\alpha_*|) > 0.$$

因此，要得出该定理的结论，只需证明如下内容即可。

$$\inf_{G\in\mathcal{G}_{L'}(\Omega):(\mathcal{D}(G,G_*)+|\alpha-\alpha_*|)>\varepsilon'} \|f_{G,\alpha} - f_{G_*,\alpha_*}\|_{L^2(\mu)}/(\mathcal{D}(G,G_*)+|\alpha-\alpha_*|) > 0.$$

假设上述结论不成立，即其并非真命题。那么在 $\mathcal{G}_{L'}(\Theta)$ 与 $\alpha'_n \in \Omega$ 中存在一组测度序列 $G'_n := \sum_{j'=1}^{L'}$指数函数$(\bar{b}_{n,\ j'})\delta_{\boldsymbol{p}_{n,\ j'}}$，使得我们拥有

$$\begin{cases} \mathcal{D}(G'_n, G_*) + |\alpha'_n - \alpha_*| > \varepsilon' \\ \|f_{G'_n,\alpha'_n} - f_{G_*,\alpha_*}\|_{L^2(\mu)}/(\mathcal{D}(G'_n, G_*) + |\alpha'_n - \alpha_*|) \to 0. \end{cases}$$

这些界限表明$\|f_{G'_{n,}\ \alpha'_n} - f_{G_{*,}\ \alpha_*}\|_{L^2\ (\mu)} \to 0$ 即 $n \to \infty$。
请记住，集合 Θ 和Ω都是紧致的。因此，在 $\mathcal{G}_{L'}(\Omega)$中存在一个混合度量 G'，使得$(G'_{n,}\alpha'_n)$的某个子序列会收敛到$(G',\ \alpha')$。由于 $\mathcal{D}(G'_{n,}G_*)\ +|\alpha'_n - \alpha_*| > \varepsilon'$，我们可以推断出 $\mathcal{D}(G', G_*)\ +|\alpha' - \alpha_*| > \varepsilon'$。应用法图引理即可得到

$$0 = \lim_{n\to\infty} \|f_{G'_n,\alpha'_n} - f_{G_*,\alpha_*}\|_{L^2(\mu)} \geq \int \liminf_{n\to\infty} \left\|f_{G'_n,\alpha'_n}(\boldsymbol{X}) - f_{G_*,\alpha_*}(\boldsymbol{X})\right\|_2^2 d\mu(\boldsymbol{X}).$$

因此，对于 μ–而言，几乎可以肯定地有$\alpha'(\boldsymbol{X}) = f_{G_*,\alpha_*}(\boldsymbol{X})$。根据可识别性性质（参见本证明的结尾），我们可以推断出$(G', \alpha')\ \equiv (G_*, \alpha_*)$。由此可知 $\mathcal{D}(G', G_*)\ +|\alpha' - \alpha_*| = 0$。这一结果与假设$\mathcal{D}(G', G_*)\ +|\alpha' - \alpha_*| > \varepsilon' > 0$相矛盾。

作为共同作者 在nsequence中，全局部分的证明已经完成。我们由此得到了该定理的结论。

关于可识别性性质的证明。我们现在要证明：如果对于几乎所有的 $\boldsymbol{X}$，都有$f_{\bar{G},\ \bar{\alpha}}(\boldsymbol{X}) = f_{\bar{G}_*,\ \bar{\alpha}_*}(\boldsymbol{X})$成立，那么我们能够得出$(G,\ \bar{\alpha})\ \equiv (G_*,\ \bar{\alpha}_*)$这一结论。

在此，这些方程中的{v1}、{v2}、{v3}以及{v4}均满足如下条件：

$$\begin{aligned}
\mathrm{softmax}^{\text{Pretrain}}_{\bar{G}}(u) &= \frac{\exp(u)}{\sum_{k=1}^{N}\exp(\boldsymbol{X}^\top \bar{A}_k^0 \boldsymbol{X} + \bar{a}_k^0)}, \\
\mathrm{softmax}^{\text{Prompt}}_{\bar{G}}(u') &= \frac{\exp(u)}{\sum_{j'=1}^{L'}\exp((\bar{B}\boldsymbol{p}_{j'})^\top \boldsymbol{X} + \bar{b}_{j'})}, \\
\mathrm{softmax}^{\text{Pretrain}}_{\bar{G}_*}(u_*) &= \frac{\exp(u_*)}{\sum_{k=1}^{N}\exp(\boldsymbol{X}^\top \bar{A}_k^0 \boldsymbol{X} + \bar{a}_k^0)}, \\
\mathrm{softmax}^{\text{Prompt}}_{\bar{G}_*}(u'_*) &= \frac{\exp(u_*)}{\sum_{j'=1}^{L}\exp((\bar{B}\boldsymbol{p}_{*,j'})^\top \boldsymbol{X} + \bar{b}_{*,j'})}.
\end{aligned}$$

在此，这些方程中的 u、 u'、 u_*和 u'_* 满足：

$$\begin{aligned}
u &\in \{\boldsymbol{X}^\top \bar{A}_j^0 \boldsymbol{X} + \bar{a}_j^0 : j \in [N]\},\ u' \in \{(\bar{B}\boldsymbol{p}_{j'})^\top \boldsymbol{X} + \bar{b}_{j'} : j' \in [L']\} \\
u_* &\in \{\boldsymbol{X}^\top \bar{A}_j^0 \boldsymbol{X} + \bar{a}_j^0 : j \in [N]\},\ u'_* \in \{(\bar{B}\boldsymbol{p}_{*,j'})^\top \boldsymbol{X} + \bar{b}_{*,j'} : j' \in [L]\}.
\end{aligned}$$

对于几乎所有的 $\boldsymbol{X}$而言，方程$f_{\bar{G},\ \bar{\alpha}}(\boldsymbol{X}) = f_{\bar{G}_{*,}\ \bar{\alpha}_*}(\boldsymbol{X})$都表明

$$\sum_{j=1}^{N} \mathrm{softmax}^{\text{Pretrain}}_{\bar{G}}(\boldsymbol{X}^\top \bar{A}_j^0 \boldsymbol{X} + \bar{a}_j^0))h(\boldsymbol{X}, \bar{\eta}_j^0) + \tanh(\bar{\alpha}) \sum_{j'=1}^{L'} \mathrm{softmax}^{\text{Prompt}}_{\bar{G}}((\bar{B}\boldsymbol{p}_{j'})^\top \boldsymbol{X} + \bar{b}_{j'})\bar{C}\boldsymbol{p}_{j'}$$

$$= \sum_{j=1}^{N} \text{softmax}^{\text{Pretrain}}_{\bar{G}_*}(\boldsymbol{X}^\top \bar{A}_j^0 \boldsymbol{X} + \bar{a}_j^0))h(\boldsymbol{X}, \bar{\eta}_j^0) + \tanh(\bar{\alpha}_*) \sum_{j'=1}^{L} \text{softmax}^{\text{Prompt}}_{\bar{G}_*}((\bar{B}\boldsymbol{p}_{*,j'})^\top \boldsymbol{X} + \bar{b}_{*,j'})\bar{C}\boldsymbol{p}_{*,j'}. \quad (29)$$

The above equation only holds when $L = L'$. Furthermore, we also have that

$$\{\text{softmax}^{\text{Prompt}}_{\bar{G}}((\bar{B}\boldsymbol{p}_{j'})^\top \boldsymbol{X} + \bar{b}_{j'}) : j' \in [L']\} = \{\text{softmax}^{\text{Prompt}}_{\bar{G}_*}((\bar{B}\boldsymbol{p}_{*,j'})^\top \boldsymbol{X} + \bar{b}_{*,j'}) : j' \in [L]\},$$

for almost every $\boldsymbol{X}$. By relabelling the indices, we can assume that

$$\text{softmax}^{\text{Prompt}}_{\bar{G}}((\bar{B}\boldsymbol{p}_{j'})^\top \boldsymbol{X} + \bar{b}_{j'}) = \text{softmax}^{\text{Prompt}}_{\bar{G}_*}((\bar{B}\boldsymbol{p}_{*,j'})^\top \boldsymbol{X} + \bar{b}_{*,j'}),$$

for almost every $\boldsymbol{X}$ and any $j' \in [L]$. From the translation invariant property of the softmax function, the above equations only hold when $\bar{b}_{j'} = \bar{b}_{*,j'} + \bar{r}$ for some $\bar{r} \in \mathbb{R}$ and any $j' \in [L]$. Given these results, equation (29) leads to

$$\tanh(\bar{\alpha}) \sum_{j=1}^{L} \exp(\bar{b}_j) \exp((\bar{B}\boldsymbol{p}_j)^\top \boldsymbol{X})\bar{C}\boldsymbol{p}_j = \tanh(\bar{\alpha}_*) \sum_{j=1}^{L} \exp(\bar{b}_{*,j}) \exp((\bar{B}\boldsymbol{p}_{*,j})^\top \boldsymbol{X})\bar{C}\boldsymbol{p}_{*,j}, \quad (30)$$

for almost surely $\boldsymbol{X}$.

Now, we partition the set $\{1, 2, \ldots, L\}$ into m subsets $\bar{K}_1, \bar{K}_2, \ldots, \bar{K}_m$ where $m \leq L$, such that $\exp(\bar{b}_j) = \exp(\bar{b}_{*,j'})$ for any $j, j' \in \bar{K}_i$ and $i \in [m]$. It is clear that $\exp(\bar{b}_j) \neq \exp(\bar{b}_{*,j'})$ when j, j' belong to different subsets $\bar{K}_i$. Collecting these results, equation (30) can be rewritten as follows:

$$\tanh(\bar{\alpha}) \sum_{i=1}^{m} \sum_{j \in \bar{K}_i} \exp(\bar{b}_j) \exp((\bar{B}\boldsymbol{p}_j)^\top \boldsymbol{X})\bar{C}\boldsymbol{p}_j = \tanh(\bar{\alpha}_*) \sum_{i=1}^{m} \sum_{j \in \bar{K}_i} \exp(\bar{b}_{*,j}) \exp((\bar{B}\boldsymbol{p}_{*,j})^\top \boldsymbol{X})\bar{C}\boldsymbol{p}_{*,j},$$

for almost surely $\boldsymbol{X}$. Hence, we achieve that

$$\{((\bar{B}\boldsymbol{p}_j)^\top, \boldsymbol{p}_j) : j \in \bar{K}_i\} = \{((\bar{B}\boldsymbol{p}_{*,j})^\top, \boldsymbol{p}_{*,j}) : j \in \bar{K}_i\} \text{ and } \tanh(\bar{\alpha}) = \tanh(\bar{\alpha}_*).$$

It naturally leads to

$$\{\boldsymbol{p}_j : j \in \bar{K}_i\} = \{\boldsymbol{p}_{*,j} : j \in \bar{K}_i\}.$$

Without loss of generality, $\boldsymbol{p}_j = \boldsymbol{p}_{*,j}$ for all $j \in \bar{K}_i$. As a consequence, we obtain that $\bar{\alpha} = \bar{\alpha}_*$ and

$$\sum_{i=1}^{m} \sum_{j \in \bar{K}_i} \exp(\bar{b}_j)\delta_{\boldsymbol{p}_j} = \sum_{i=1}^{m} \sum_{j \in \bar{K}_i} \exp(\bar{b}_{*,j})\delta_{\boldsymbol{p}_{*,j}}.$$

It is equivalent to $(\bar{G}, \bar{\alpha}) \equiv (\bar{G}_*, \bar{\alpha}_*)$. We achieve the conclusion of the identifiability claim.

A.2. Proof of Theorem 4.4

Similar to the proof of Theorem 4.2 in Appendix A.1, we only need to demonstrate that

$$\|g_{G,\alpha} - g_{G_*,\alpha_*}\|_{L^2(\mu)} \geq C \cdot (\mathcal{D}(G, G_*) + |\alpha - \alpha_*|)$$

for any $(G, \alpha) \in \mathcal{G}_{L'}(\Theta) \times \Omega$ for some universal constant C. It is equivalent to proving that:

$$\inf_{(G,\alpha) \in \mathcal{G}_{L'}(\Theta) \times \Omega} \|g_{G,\alpha} - g_{G_*,\alpha_*}\|_{L^2(\mu)} / (\mathcal{D}(G, G_*) + |\alpha - \alpha_*|) > 0.$$

To obtain the conclusion for the above inequality, we consider two parts: (i) local part, namely,

$$\lim_{\varepsilon \to 0} \inf_{(G,\alpha) \in \mathcal{G}_{L'}(\Theta) \times \Omega : \mathcal{D}(G,G_*) + |\alpha - \alpha_*| \leq \varepsilon} \|g_{G,\alpha} - g_{G_*,\alpha_*}\|_{L^2(\mu)} / (\mathcal{D}(G, G_*) + |\alpha - \alpha_*|) > 0;$$

$$= \sum_{j=1}^{N} \text{softmax}^{\text{Pretrain}}_{\bar{G}_*}(\boldsymbol{X}^\top \bar{A}_j^0 \boldsymbol{X} + \bar{a}_j^0))h(\boldsymbol{X}, \bar{\eta}_j^0) + \tanh(\bar{\alpha}_*) \sum_{j'=1}^{L} \text{softmax}^{\text{Prompt}}_{\bar{G}_*}((\bar{B}\boldsymbol{p}_{*,j'})^\top \boldsymbol{X} + \bar{b}_{*,j'})\bar{C}\boldsymbol{p}_{*,j'}. \quad (29)$$

上述方程仅在 $L = L'$时成立。此外，我们还有如下结论

$$\{\text{softmax}^{\text{Prompt}}_{\bar{G}}((\bar{B}\boldsymbol{p}_{j'})^\top \boldsymbol{X} + \bar{b}_{j'}) : j' \in [L']\} = \{\text{softmax}^{\text{Prompt}}_{\bar{G}_*}((\bar{B}\boldsymbol{p}_{*,j'})^\top \boldsymbol{X} + \bar{b}_{*,j'}) : j' \in [L]\},$$

对于几乎所有的 $\boldsymbol{X}$而言。通过重新标记索引，我们可以假设

$$\text{softmax}^{\text{Prompt}}_{\bar{G}}((\bar{B}\boldsymbol{p}_{j'})^\top \boldsymbol{X} + \bar{b}_{j'}) = \text{softmax}^{\text{Prompt}}_{\bar{G}_*}((\bar{B}\boldsymbol{p}_{*,j'})^\top \boldsymbol{X} + \bar{b}_{*,j'}),$$

对于几乎所有的 $\boldsymbol{X}$ 以及任意的 $j' \in [L]$而言。根据softmax函数的平移不变性，上述方程式 仅在 $b_{j'} = b_*$ 时成立，$j' + \bar{r}$ 对于某些 $\bar{r} \in \mathbb{R}$ 和任意的 $j' \in [L]$来说也是如此。基于这些结果，方程（29）可推导出

$$\tanh(\bar{\alpha}) \sum_{j=1}^{L} \exp(\bar{b}_j) \exp((\bar{B}\boldsymbol{p}_j)^\top \boldsymbol{X})\bar{C}\boldsymbol{p}_j = \tanh(\bar{\alpha}_*) \sum_{j=1}^{L} \exp(\bar{b}_{*,j}) \exp((\bar{B}\boldsymbol{p}_{*,j})^\top \boldsymbol{X})\bar{C}\boldsymbol{p}_{*,j}, \quad (30)$$

对于几乎必然成立的情况 $\boldsymbol{X}$。

现在，我们将集合 $\{1, 2, \ldots, L\}$ 划分为 m 个子集 K_1、K_2、……、K_m，其中 $m \leq L$满足对于任意的 $j j' \in K_i$ $i \in [m]$，都有$\exp(b_j) = \exp(b_{*,j'})$。显然，当 $j j'$ 属于不同的子集 K_i时，有$\exp(b_j) \neq \exp(b_{*,j'})$。综合这些结果，方程(30)可以改写为如下形式：

$$\tanh(\bar{\alpha}) \sum_{i=1}^{m} \sum_{j \in \bar{K}_i} \exp(\bar{b}_j) \exp((\bar{B}\boldsymbol{p}_j)^\top \boldsymbol{X})\bar{C}\boldsymbol{p}_j = \tanh(\bar{\alpha}_*) \sum_{i=1}^{m} \sum_{j \in \bar{K}_i} \exp(\bar{b}_{*,j}) \exp((\bar{B}\boldsymbol{p}_{*,j})^\top \boldsymbol{X})\bar{C}\boldsymbol{p}_{*,j},$$

对于几乎必然成立的情况 $\boldsymbol{X}$。因此，我们实现了这一目标。

$$\{((\bar{B}\boldsymbol{p}_j)^\top, \boldsymbol{p}_j) : j \in \bar{K}_i\} = \{((\bar{B}\boldsymbol{p}_{*,j})^\top, \boldsymbol{p}_{*,j}) : j \in \bar{K}_i\} \text{ and } \tanh(\bar{\alpha}) = \tanh(\bar{\alpha}_*).$$

这自然而然地会引出

$$\{\boldsymbol{p}_j : j \in \bar{K}_i\} = \{\boldsymbol{p}_{*,j} : j \in \bar{K}_i\}.$$

With 为一般性考虑，$\boldsymbol{p}_j = \boldsymbol{p}_{*,j}$ 对所有的 $j \in K_i$均成立。因此，我们可以得出 $\bar{\alpha} = \bar{\alpha}_*$ 且

$$\sum_{i=1}^{m} \sum_{j \in \bar{K}_i} \exp(\bar{b}_j)\delta_{\boldsymbol{p}_j} = \sum_{i=1}^{m} \sum_{j \in \bar{K}_i} \exp(\bar{b}_{*,j})\delta_{\boldsymbol{p}_{*,j}}.$$

它等价于 $(G, \bar{\alpha}) \equiv (G_*, \bar{\alpha}_*)$。由此我们得出了可识别性命题的结论。

A.2. 定理的证明4.4

与附录 A.1中定理 4.2的证明类似，我们只需证明

$$\|g_{G,\alpha} - g_{G_*,\alpha_*}\|_{L^2(\mu)} \geq C \cdot (\mathcal{D}(G, G_*) + |\alpha - \alpha_*|)$$

对于任意的 (G, α)，存在某个普适常数 C 使得 $\in \mathcal{G}_{L'}(\Theta) \times \Omega$，这等价于证明：

$$\inf_{(G,\alpha) \in \mathcal{G}_{L'}(\Theta) \times \Omega} \|g_{G,\alpha} - g_{G_*,\alpha_*}\|_{L^2(\mu)} / (\mathcal{D}(G, G_*) + |\alpha - \alpha_*|) > 0.$$

为得出上述不等式的结论，我们需要从两个部分入手：(i) 局部部分，即

$$\lim_{\varepsilon \to 0} \inf_{(G,\alpha) \in \mathcal{G}_{L'}(\Theta) \times \Omega : \mathcal{D}(G,G_*) + |\alpha - \alpha_*| \leq \varepsilon} \|g_{G,\alpha} - g_{G_*,\alpha_*}\|_{L^2(\mu)} / (\mathcal{D}(G, G_*) + |\alpha - \alpha_*|) > 0;$$

(ii) global part, namely, for any $\varepsilon > 0$

$$\inf_{(G,\alpha)\in\mathcal{G}_{L'}(\Theta)\times\Omega:\mathcal{D}(G,G_*)+|\alpha-\alpha_*|>\varepsilon} \|g_{G,\alpha} - g_{G_*,\alpha_*}\|_{L^2(\mu)}/(\mathcal{D}(G,G_*) + |\alpha-\alpha_*|) > 0;$$

Since the global part can be argued in a similar fashion to Appendix A.1, we will focus only on proving the local part in this appendix. Additionally, we will impose the following essential yet mild assumptions in the activation function σ to facilitate our arguments:

Assumptions. We assume that the activation function σ meet the following assumptions:

(A.1) (Uniform Lipschitz) Let $F(\boldsymbol{X};\boldsymbol{p}) := \exp((B\sigma(\boldsymbol{p}))^\top \boldsymbol{X})C\sigma(\boldsymbol{p})$. Then, for any $r \in \{1,2\}$, we have

$$\sum_{|\alpha|=r}\left|\Big(\frac{\partial^{|\alpha|}F}{\partial \boldsymbol{p}^\alpha}(\boldsymbol{X};\boldsymbol{p}) - \frac{\partial^{|\alpha|}F}{\partial \boldsymbol{p}^\alpha}(\boldsymbol{X};\boldsymbol{p}')\Big)\gamma^\alpha\right| \leq c\|\boldsymbol{p}-\boldsymbol{p}'\|^\zeta\|\gamma\|^r,$$

for any vector $\gamma \in \mathbb{R}^d$ and for some positive constants ζ and c which are independent of $\boldsymbol{X}$ and $\boldsymbol{p}, \boldsymbol{p}'$. Here, $\alpha \in \mathbb{N}^d$.

(A.2) (Injective) If there exist parameters $\boldsymbol{p}$ and $\boldsymbol{p}'$ such that $\sigma(\boldsymbol{p}) = \sigma(\boldsymbol{p}')$, then we obtain that $\boldsymbol{p} = \boldsymbol{p}'$.

Local part: We first start with the local part, which is equivalent to demonstrating that

$$\lim_{\varepsilon\to 0}\inf_{(G,\alpha)\in\mathcal{G}_{L'}(\Theta)\times\Omega:\mathcal{D}(G,G_*)+|\alpha-\alpha_*|\leq\varepsilon} \|g_{G,\alpha} - g_{G_*,\alpha_*}\|_{L^2(\mu)}/(\mathcal{D}(G,G_*) + |\alpha-\alpha_*|) > 0.$$

We prove the above claim by contradiction. Assume by contrary that the above claim does not hold. It indicates that we can find a sequence of mixing measures $G_n := \sum_{j'=1}^{L_n}\exp(\bar{b}_{n,j'})\delta_{\boldsymbol{p}_{n,j'}}$ in $\mathcal{G}_{L'}(\Theta)$ and a sequence of $\alpha_n \in \Omega$ such that when $n \to \infty$, the following limits hold:

$$\begin{cases} \mathcal{D}(G_n,G_*) + |\alpha_n - \alpha_*| \to 0, \\ \|g_{G_n,\alpha_n} - g_{G_*,\alpha_*}\|_{L^2(\mu)}/(\mathcal{D}(G_n,G_*) + |\alpha_n-\alpha_*|) \to 0. \end{cases}$$

The first limit indicates that $\mathcal{D}_n := \mathcal{D}(G_n,G_*) \to 0$ and $\alpha_n \to \alpha_*$ as $n \to \infty$.

For the simplicity of the ensuing presentation, we denote $\mathcal{C}_j^n := \mathcal{C}_j(G_n)$ as a Voronoi cell of G_n induced by the j-th components of G_*. Without loss of generality, we assume that those Voronoi cells do not depend on the sample size, i.e., $\mathcal{C}_j = \mathcal{C}_j^n$, which is possible since our arguments are asymptotic. Therefore, we can rewrite the Voronoi loss $\mathcal{D}_n$ as follows:

$$\mathcal{D}_n := \sum_{j'=1}^{L}\Big|\sum_{i\in\mathcal{C}_{j'}}\exp(\bar{b}_{n,i}) - \exp(\bar{b}_{*,j'})\Big| + \sum_{j'\in[L]:|\mathcal{C}_{j'}|=1}\sum_{i\in\mathcal{C}_{j'}}\exp(\bar{b}_{n,i})\|\Delta\boldsymbol{p}_{n,ij'}\| + \sum_{j'\in[L]:|\mathcal{C}_{j'}|>1}\sum_{i\in\mathcal{C}_{j'}}\exp(\bar{b}_{n,i})\|\Delta\boldsymbol{p}_{n,ij'}\|^2,$$

where $\Delta\boldsymbol{p}_{n,ij'} = \boldsymbol{p}_{n,i} - \boldsymbol{p}_{*,j'}$ for all $i \in \mathcal{C}_{j'}$.

From the hypothesis, we have $\mathcal{D}_n \to 0$, which implies that $\sum_{i\in\mathcal{C}_j}\exp(\bar{b}_{n,i}) \to \exp(\bar{b}_{*,j})$ and $\boldsymbol{p}_{n,i} \to \boldsymbol{p}_{*,j}$ for any $i \in \mathcal{C}_j, j \in [L]$. Similar to the proof of Theorem 4.2, to establish the contradiction our proof consists of three main steps.

Step 1 - Taylor expansion. To ease the presentation, let us denote

$$g_{G_n}(\boldsymbol{X}) := \sum_{j=1}^{L_n}\frac{\exp((\bar{B}\sigma(\boldsymbol{p}_{n,j}))^\top\boldsymbol{X} + \bar{b}_{n,j})}{\sum_{k=1}^{L_n}\exp((\bar{B}\sigma(\boldsymbol{p}_{n,k}))^\top\boldsymbol{X} + \bar{b}_{n,k})}\cdot\bar{C}\sigma(\boldsymbol{p}_{n,j}),$$

$$g_{G_*}(\boldsymbol{X}) := \sum_{j'=1}^{L}\frac{\exp((\bar{B}\sigma(\boldsymbol{p}_{*,j'}))^\top\boldsymbol{X} + \bar{b}_{*,j'})}{\sum_{k'=1}^{L}\exp((\bar{B}\sigma(\boldsymbol{p}_{*,k'}))^\top\boldsymbol{X} + \bar{b}_{*,k'})}\cdot\bar{C}\sigma(\boldsymbol{p}_{*,j'}).$$

(ii) 全局部分，即对于任意的 $\varepsilon > 0$

$$\inf_{(G,\alpha)\in\mathcal{G}_{L'}(\Theta)\times\Omega:\mathcal{D}(G,G_*)+|\alpha-\alpha_*|>\varepsilon} \|g_{G,\alpha} - g_{G_*,\alpha_*}\|_{L^2(\mu)}/(\mathcal{D}(G,G_*) + |\alpha-\alpha_*|) > 0;$$

由于全局部分可以采用与附录 A.1中类似的方法进行论证，因此在本附录中我们仅专注于证明局部部分。此外，为了便于推导，我们将在激活函数 σ 中设定以下既必要又较为宽松的假设：

假设条件。我们假设激活函数 σ 满足以下假设：

(A.1)（一致利普希茨）设 $F(\boldsymbol{X};\boldsymbol{p}) := \exp((B\sigma(\boldsymbol{p}))^\top\boldsymbol{X})C\sigma(\boldsymbol{p})$。那么，对于任意的 $r \in \{1,2\}$，我们有

$$\sum_{|\alpha|=r}\left|\Big(\frac{\partial^{|\alpha|}F}{\partial \boldsymbol{p}^\alpha}(\boldsymbol{X};\boldsymbol{p}) - \frac{\partial^{|\alpha|}F}{\partial \boldsymbol{p}^\alpha}(\boldsymbol{X};\boldsymbol{p}')\Big)\gamma^\alpha\right| \leq c\|\boldsymbol{p}-\boldsymbol{p}'\|^\zeta\|\gamma\|^r,$$

对于任意向量 $\gamma \in \mathbb{R}^d$，以及某些与 $\boldsymbol{X}$、$\boldsymbol{p}$、$\boldsymbol{p}'$无关的正常数 ζ 和 c，此处为 $\alpha \in \mathbb{N}^d$。

(A.2)（单射性）若存在参数 $\boldsymbol{p}$ 和 $\boldsymbol{p}'$，使得 $\sigma(\boldsymbol{p}) = \sigma(\boldsymbol{p}')$，则可得出 $\boldsymbol{p} = \boldsymbol{p}'$。

局部部分： 我们首先从局部部分开始，这等价于证明

$$\lim_{\varepsilon\to 0}\inf_{(G,\alpha)\in\mathcal{G}_{L'}(\Theta)\times\Omega:\mathcal{D}(G,G_*)+|\alpha-\alpha_*|\leq\varepsilon} \|g_{G,\alpha} - g_{G_*,\alpha_*}\|_{L^2(\mu)}/(\mathcal{D}(G,G_*) + |\alpha-\alpha_*|) > 0.$$

我们通过反证法来证明上述命题。假设相反，即上述命题不成立。这意味着我们可以找到一列混合度量 $G_n := \sum_{j'=1}^{L_n}\exp(\bar{b}_{n,j'})\delta_{\boldsymbol{p}_{n,j'}}$，它们属于 $\mathcal{G}_{L'}(\Theta)$，同时还可以找到一列 $\alpha_n \in \Omega$，使得当$n \to \infty$时，以下极限成立：

$$\begin{cases} \mathcal{D}(G_n,G_*) + |\alpha_n - \alpha_*| \to 0, \\ \|g_{G_n,\alpha_n} - g_{G_*,\alpha_*}\|_{L^2(\mu)}/(\mathcal{D}(G_n,G_*) + |\alpha_n-\alpha_*|) \to 0. \end{cases}$$

第一个极限表明 $\mathcal{D}_n := \mathcal{D}(G_n,G_*) \to 0$ 以及 $\alpha_n \to \alpha_*$ 可表示为 $n \to \infty$。

为简化后续的阐述，我们将 $\mathcal{C}_j^n := \mathcal{C}_j(G_n)$表示为由 G_*的第 j个分量所诱导的 G_n 的Voronoi单元。在不失一般性的前提下，我们假设这些Voronoi单元与样本大小无关，即$\mathcal{C}_j = \mathcal{C}_j^n$，这是可行的，因为我们的论证属于渐近性质。因此，我们可以将Voronoi损失 $\mathcal{D}_n$ 重新表述为如下形式：

$$\mathcal{D}_n := \sum_{j'=1}^{L}\Big|\sum_{i\in\mathcal{C}_{j'}}\exp(\bar{b}_{n,i}) - \exp(\bar{b}_{*,j'})\Big| + \sum_{j'\in[L]:|\mathcal{C}_{j'}|=1}\sum_{i\in\mathcal{C}_{j'}}\exp(\bar{b}_{n,i})\|\Delta\boldsymbol{p}_{n,ij'}\| + \sum_{j'\in[L]:|\mathcal{C}_{j'}|>1}\sum_{i\in\mathcal{C}_{j'}}\exp(\bar{b}_{n,i})\|\Delta\boldsymbol{p}_{n,ij'}\|^2,$$

其中$\Delta\boldsymbol{p}_{n,ij'} = \boldsymbol{p}_{n,i} - \boldsymbol{p}_{*,j'}$对所有的 $i \in \mathcal{C}_j$均成立。

根据假设，我们有 $\mathcal{D}_n \to 0$，这表明 $\sum_{i\in\mathcal{C}_j}$指数函数($b_{n,i}$) $\to$ 等于指数函数($b_{*,j}$)，同时 $\boldsymbol{p}_{n,i} \to \boldsymbol{p}_{*,j}$ 对任意$i \in \mathcal{C}_j$、 $j \in [L]$都成立。与定理4.2的证明类似，为得出矛盾结论，我们的证明包含三个主要步骤。

步骤1——泰勒级数展开。为便于阐述，我们将其记为

$$g_{G_n}(\boldsymbol{X}) := \sum_{j=1}^{L_n}\frac{\exp((\bar{B}\sigma(\boldsymbol{p}_{n,j}))^\top\boldsymbol{X} + \bar{b}_{n,j})}{\sum_{k=1}^{L_n}\exp((\bar{B}\sigma(\boldsymbol{p}_{n,k}))^\top\boldsymbol{X} + \bar{b}_{n,k})}\cdot\bar{C}\sigma(\boldsymbol{p}_{n,j}),$$

$$g_{G_*}(\boldsymbol{X}) := \sum_{j'=1}^{L}\frac{\exp((\bar{B}\sigma(\boldsymbol{p}_{*,j'}))^\top\boldsymbol{X} + \bar{b}_{*,j'})}{\sum_{k'=1}^{L}\exp((\bar{B}\sigma(\boldsymbol{p}_{*,k'}))^\top\boldsymbol{X} + \bar{b}_{*,k'})}\cdot\bar{C}\sigma(\boldsymbol{p}_{*,j'}).$$

We now perform the following decomposition:

$$\begin{aligned}\widetilde{Q}_n(\boldsymbol{X}) :&= \Big[\sum_{k'=1}^{L}\exp((\bar{B}\sigma(\boldsymbol{p}_{*,k'}))^\top\boldsymbol{X}+\bar{b}_{*,k'})\Big]\cdot[g_{G_n,\alpha_n}(\boldsymbol{X})-g_{G_*,\alpha_*}(\boldsymbol{X})]\\ &= \Big[\sum_{k'=1}^{L}\exp((\bar{B}\sigma(\boldsymbol{p}_{*,k'}))^\top\boldsymbol{X}+\bar{b}_{*,k'})\Big]\cdot\Big(\tanh(\alpha_n)g_{G_n}(\boldsymbol{X})-\tanh(\alpha_*)g_{G_*}(\boldsymbol{X})\Big)\\ &= \Big[\sum_{k'=1}^{L}\exp((\bar{B}\sigma(\boldsymbol{p}_{*,k'}))^\top\boldsymbol{X}+\bar{b}_{*,k'})\Big]\cdot\tanh(\alpha_n)\Big[g_{G_n}(\boldsymbol{X})-g_{G_*}(\boldsymbol{X})\Big]\\ &+ \Big[\sum_{k'=1}^{L}\exp((\bar{B}\sigma(\boldsymbol{p}_{*,k'}))^\top\boldsymbol{X}+\bar{b}_{*,k'})\Big]\cdot[\tanh(\alpha_n)-\tanh(\alpha_*)]g_{G_*}(\boldsymbol{X})\\ &:= \widetilde{Q}_{n,1}(\boldsymbol{X})+\widetilde{Q}_{n,2}(\boldsymbol{X}).\end{aligned}$$

For that purpose, we will decompose the two terms $\widetilde{Q}_{n,1}(\boldsymbol{X})$ and $\widetilde{Q}_{n,2}(\boldsymbol{X})$, respectively.

Decomposition of the function $\widetilde{Q}_{n,1}(\boldsymbol{X})$. We have

$$\begin{aligned}\widetilde{Q}_{n,1}(\boldsymbol{X}) &= \sum_{j=1}^{L}\sum_{i\in\mathcal{C}_j}\tanh(\alpha_n)\exp(\bar{b}_{n,i})\Big[\exp((\bar{B}\sigma(\boldsymbol{p}_{n,i}))^\top\boldsymbol{X})\bar{C}\sigma(\boldsymbol{p}_{n,i})-\exp((\bar{B}\sigma(\boldsymbol{p}_{*,j}))^\top\boldsymbol{X})\bar{C}\sigma(\boldsymbol{p}_{*,j})\Big]\\ &- \sum_{j=1}^{L}\sum_{i\in\mathcal{C}_j}\tanh(\alpha_n)\exp(\bar{b}_{n,i})\Big[\exp((\bar{B}\sigma(\boldsymbol{p}_{n,i}))^\top\boldsymbol{X})-\exp((\bar{B}\sigma(\boldsymbol{p}_{*,j}))^\top\boldsymbol{X})\Big]g_{G_n}(\boldsymbol{X})\\ &+ \sum_{j=1}^{L}\tanh(\alpha_n)\Big(\sum_{i\in\mathcal{C}_j}\exp(\bar{b}_{n,i})-\exp(\bar{b}_{*,j})\Big)\exp((\bar{B}\sigma(\boldsymbol{p}_{*,j}))^\top\boldsymbol{X})\Big[\bar{C}\sigma(\boldsymbol{p}_{*,j})-g_{G_n}(\boldsymbol{X})\Big]\\ &:= \widetilde{A}_n(\boldsymbol{X})-\widetilde{B}_n(\boldsymbol{X})+\widetilde{C}_n(\boldsymbol{X}).\end{aligned}\tag{31}$$

Decomposition of the function $\widetilde{A}_n(\boldsymbol{X})$. To ease the presentation, we define the following functions $\widetilde{U}(\boldsymbol{X};\boldsymbol{p}) := \exp((\bar{B}\sigma(\boldsymbol{p}))^\top\boldsymbol{X})$ and $\widetilde{V}(\boldsymbol{p})=\bar{C}\sigma(\boldsymbol{p})$. Then, we denote the product of these functions as $\widetilde{F}(\boldsymbol{X};\boldsymbol{p})=\widetilde{U}(\boldsymbol{X};\boldsymbol{p})\widetilde{V}(\boldsymbol{p})$. To decompose $\widetilde{A}_n(\boldsymbol{X})$, we separately consider Voronoi cells with exactly one element and those with more than one element. It leads to the following decomposition of the function $\widetilde{A}_n(\boldsymbol{X})$:

$$\begin{aligned}\widetilde{A}_n(\boldsymbol{X}) &= \sum_{j:|\mathcal{C}_j|=1}\sum_{i\in\mathcal{C}_j}\tanh(\alpha_n)\exp(\bar{b}_{n,i})\Big[\widetilde{F}(\boldsymbol{X};\boldsymbol{p}_{n,i})-\widetilde{F}(\boldsymbol{X};\boldsymbol{p}_{*,j})\Big]\\ &+ \sum_{j:|\mathcal{C}_j|>1}\sum_{i\in\mathcal{C}_j}\tanh(\alpha_n)\exp(\bar{b}_{n,i})\Big[\widetilde{F}(\boldsymbol{X};\boldsymbol{p}_{n,i})-\widetilde{F}(\boldsymbol{X};\boldsymbol{p}_{*,j})\Big]\\ &:= \widetilde{A}_{n,1}(\boldsymbol{X})+\widetilde{A}_{n,2}(\boldsymbol{X}),\end{aligned}$$

where we denote $\widetilde{A}_{n,1}(\boldsymbol{X}) = \sum_{j:|\mathcal{C}_j|=1}\sum_{i\in\mathcal{C}_j}\tanh(\alpha_n)\exp(\bar{b}_{n,i})\Big[\widetilde{F}(\boldsymbol{X};\boldsymbol{p}_{n,i})-\widetilde{F}(\boldsymbol{X};\boldsymbol{p}_{*,j})\Big]$ and $\widetilde{A}_{n,2}(\boldsymbol{X}) = \sum_{j:|\mathcal{C}_j|>1}\sum_{i\in\mathcal{C}_j}\tanh(\alpha_n)\exp(\bar{b}_{n,i})\Big[\widetilde{F}(\boldsymbol{X};\boldsymbol{p}_{n,i})-\widetilde{F}(\boldsymbol{X};\boldsymbol{p}_{*,j})\Big]$.

For the function $\widetilde{A}_{n,1}(\boldsymbol{X})$, for any indices $i\in\mathcal{C}_j$ and j such that $|\mathcal{C}_j|=1$, the first-order Taylor expansion entails that

$$\widetilde{U}(\boldsymbol{X};\boldsymbol{p}_{n,i})=\widetilde{U}(\boldsymbol{X};\boldsymbol{p}_{*,j})+\sum_{|\alpha|=1}(\Delta\boldsymbol{p}_{n,ij})^\alpha\frac{\partial^{|\alpha|}\widetilde{U}}{\partial\boldsymbol{p}^\alpha}(\boldsymbol{X};\boldsymbol{p}_{*,j})+\widetilde{R}_{ij,1}(\boldsymbol{X}),$$

$$\widetilde{V}(\boldsymbol{p}_{n,i})=\widetilde{V}(\boldsymbol{p}_{*,j})+\sum_{|\alpha|=1}(\Delta\boldsymbol{p}_{n,ij})^\alpha\frac{\partial^{|\alpha|}\widetilde{V}}{\partial\boldsymbol{p}^\alpha}(\boldsymbol{p}_{*,j})+\widetilde{R}_{ij,2},$$

现在我们来进行如下分解：

$$\begin{aligned}\widetilde{Q}_n(\boldsymbol{X}) :&= \Big[\sum_{k'=1}^{L}\exp((\bar{B}\sigma(\boldsymbol{p}_{*,k'}))^\top\boldsymbol{X}+\bar{b}_{*,k'})\Big]\cdot[g_{G_n,\alpha_n}(\boldsymbol{X})-g_{G_*,\alpha_*}(\boldsymbol{X})]\\ &= \Big[\sum_{k'=1}^{L}\exp((\bar{B}\sigma(\boldsymbol{p}_{*,k'}))^\top\boldsymbol{X}+\bar{b}_{*,k'})\Big]\cdot\Big(\tanh(\alpha_n)g_{G_n}(\boldsymbol{X})-\tanh(\alpha_*)g_{G_*}(\boldsymbol{X})\Big)\\ &= \Big[\sum_{k'=1}^{L}\exp((\bar{B}\sigma(\boldsymbol{p}_{*,k'}))^\top\boldsymbol{X}+\bar{b}_{*,k'})\Big]\cdot\tanh(\alpha_n)\Big[g_{G_n}(\boldsymbol{X})-g_{G_*}(\boldsymbol{X})\Big]\\ &+ \Big[\sum_{k'=1}^{L}\exp((\bar{B}\sigma(\boldsymbol{p}_{*,k'}))^\top\boldsymbol{X}+\bar{b}_{*,k'})\Big]\cdot[\tanh(\alpha_n)-\tanh(\alpha_*)]g_{G_*}(\boldsymbol{X})\\ &:= \widetilde{Q}_{n,1}(\boldsymbol{X})+\widetilde{Q}_{n,2}(\boldsymbol{X}).\end{aligned}$$

~为此，我们将分别对两项$Q_{n,1}(\boldsymbol{X})$和$Q_{n,2}(\boldsymbol{X})$进行分解。

~**函数的分解**$Q_{n,1}(\boldsymbol{X})$。我们有

$$\begin{aligned}\widetilde{Q}_{n,1}(\boldsymbol{X}) &= \sum_{j=1}^{L}\sum_{i\in\mathcal{C}_j}\tanh(\alpha_n)\exp(\bar{b}_{n,i})\Big[\exp((\bar{B}\sigma(\boldsymbol{p}_{n,i}))^\top\boldsymbol{X})\bar{C}\sigma(\boldsymbol{p}_{n,i})-\exp((\bar{B}\sigma(\boldsymbol{p}_{*,j}))^\top\boldsymbol{X})\bar{C}\sigma(\boldsymbol{p}_{*,j})\Big]\\ &- \sum_{j=1}^{L}\sum_{i\in\mathcal{C}_j}\tanh(\alpha_n)\exp(\bar{b}_{n,i})\Big[\exp((\bar{B}\sigma(\boldsymbol{p}_{n,i}))^\top\boldsymbol{X})-\exp((\bar{B}\sigma(\boldsymbol{p}_{*,j}))^\top\boldsymbol{X})\Big]g_{G_n}(\boldsymbol{X})\\ &+ \sum_{j=1}^{L}\tanh(\alpha_n)\Big(\sum_{i\in\mathcal{C}_j}\exp(\bar{b}_{n,i})-\exp(\bar{b}_{*,j})\Big)\exp((\bar{B}\sigma(\boldsymbol{p}_{*,j}))^\top\boldsymbol{X})\Big[\bar{C}\sigma(\boldsymbol{p}_{*,j})-g_{G_n}(\boldsymbol{X})\Big]\\ &:= \widetilde{A}_n(\boldsymbol{X})-\widetilde{B}_n(\boldsymbol{X})+\widetilde{C}_n(\boldsymbol{X}).\end{aligned}\tag{31}$$

~**该函数** $A_n(\boldsymbol{X})$**的拆分形式**。为便于表述，我们定义如下函数：$U(\boldsymbol{X};\boldsymbol{p})$:=exp(($\bar{B}\sigma(\boldsymbol{p}))^\top\boldsymbol{X}$) 以及 $\widetilde{V}(\boldsymbol{p})=\bar{C}\sigma(\boldsymbol{p})$。随后，我们将这些函数的乘积记为 $\widetilde{F}(\boldsymbol{X};\boldsymbol{p})=\widetilde{U}(\boldsymbol{X};\boldsymbol{p})\widetilde{V}(\boldsymbol{p})$。为~拆分 $A_n(\boldsymbol{X})$，我们需要分别考虑仅包含一个元素的 Voronoi 单元和包含多个元素的 Voronoi单元。由此可得函数 $A_n(\boldsymbol{X})$的如下拆分形式：

$$\begin{aligned}\widetilde{A}_n(\boldsymbol{X}) &= \sum_{j:|\mathcal{C}_j|=1}\sum_{i\in\mathcal{C}_j}\tanh(\alpha_n)\exp(\bar{b}_{n,i})\Big[\widetilde{F}(\boldsymbol{X};\boldsymbol{p}_{n,i})-\widetilde{F}(\boldsymbol{X};\boldsymbol{p}_{*,j})\Big]\\ &+ \sum_{j:|\mathcal{C}_j|>1}\sum_{i\in\mathcal{C}_j}\tanh(\alpha_n)\exp(\bar{b}_{n,i})\Big[\widetilde{F}(\boldsymbol{X};\boldsymbol{p}_{n,i})-\widetilde{F}(\boldsymbol{X};\boldsymbol{p}_{*,j})\Big]\\ &:= \widetilde{A}_{n,1}(\boldsymbol{X})+\widetilde{A}_{n,2}(\boldsymbol{X}),\end{aligned}$$

其中，$\widetilde{A}_{n,1}(\boldsymbol{X})$ 表示 $=\sum_{j:|\mathcal{C}_j|=1}\sum_{i\in\mathcal{C}_j}\tanh(\alpha_n)\exp(\bar{b}_{n,i})\Big[\widetilde{F}(\boldsymbol{X};\boldsymbol{p}_{n,i})-\widetilde{F}(\boldsymbol{X};\boldsymbol{p}_{*,j})\Big]$，而 $\widetilde{A}_{n,2}(\boldsymbol{X})$ 表示 $=\sum_{j:|\mathcal{C}_j|>1}\sum_{i\in\mathcal{C}_j}\tanh(\alpha_n)\exp(\bar{b}_{n,i})\Big[\widetilde{F}(\boldsymbol{X};\boldsymbol{p}_{n,i})-\widetilde{F}(\boldsymbol{X};\boldsymbol{p}_{*,j})\Big]$。

~对于函数 $A_{n,1}(\boldsymbol{X})$而言，只要存在满足 $|\mathcal{C}_j|=1$条件的任意索引 $i\in\mathcal{C}_j$ 和 j，其一阶泰勒展开便可以用来表示该函数。

$$\widetilde{U}(\boldsymbol{X};\boldsymbol{p}_{n,i})=\widetilde{U}(\boldsymbol{X};\boldsymbol{p}_{*,j})+\sum_{|\alpha|=1}(\Delta\boldsymbol{p}_{n,ij})^\alpha\frac{\partial^{|\alpha|}\widetilde{U}}{\partial\boldsymbol{p}^\alpha}(\boldsymbol{X};\boldsymbol{p}_{*,j})+\widetilde{R}_{ij,1}(\boldsymbol{X}),$$

$$\widetilde{V}(\boldsymbol{p}_{n,i})=\widetilde{V}(\boldsymbol{p}_{*,j})+\sum_{|\alpha|=1}(\Delta\boldsymbol{p}_{n,ij})^\alpha\frac{\partial^{|\alpha|}\widetilde{V}}{\partial\boldsymbol{p}^\alpha}(\boldsymbol{p}_{*,j})+\widetilde{R}_{ij,2},$$

where the terms $\widetilde{R}_{ij,1}(\boldsymbol{X})$ and $\widetilde{R}_{ij,2}$ are Taylor remainders.

Combining the above results leads to the following formulation of the function $\widetilde{A}_{n,1}(\boldsymbol{X})$:

$$\begin{aligned}\widetilde{A}_{n,1}(\boldsymbol{X}) &= \sum_{j:|\mathcal{C}_j|=1}\sum_{i\in\mathcal{C}_j}\frac{\tanh(\alpha_n)\exp(\bar{b}_{n,i})}{\alpha!}\sum_{|\alpha|=1}\Big\{(\Delta\boldsymbol{p}_{n,ij})^{\alpha}\frac{\partial^{|\alpha|}\widetilde{U}}{\partial\boldsymbol{p}^{\alpha}}(\boldsymbol{X};\boldsymbol{p}_{*,j})\widetilde{V}(\boldsymbol{p}_{*,j}) \\ &\qquad + (\Delta\boldsymbol{p}_{n,ij})^{\alpha}\frac{\partial^{|\alpha|}\widetilde{V}}{\partial\boldsymbol{p}^{\alpha}}(\boldsymbol{p}_{*,j})\widetilde{U}(\boldsymbol{X};\boldsymbol{p}_{*,j})\Big\} + \widetilde{R}_{n,1}(\boldsymbol{X}) \\ &= \sum_{j:|\mathcal{C}_j|=1}\sum_{|\alpha|=1}\Big\{\bar{M}_{n,j,\alpha}\frac{\partial^{|\alpha|}\widetilde{U}}{\partial\boldsymbol{p}^{\alpha}}(\boldsymbol{X};\boldsymbol{p}_{*,j})\widetilde{V}(\boldsymbol{p}_{*,j}) + \bar{M}_{n,j,\alpha}\frac{\partial^{|\alpha|}\widetilde{V}}{\partial\boldsymbol{p}^{\alpha}}(\boldsymbol{p}_{*,j})\widetilde{U}(\boldsymbol{X};\boldsymbol{p}_{*,j})\Big\} + \widetilde{R}_{n,1}(\boldsymbol{X})\end{aligned}$$

where the function $\widetilde{R}_{n,1}(\boldsymbol{X})$ is the combination of Taylor remainders and satisfies that $\widetilde{R}_{n,1}(\boldsymbol{X})/(\mathcal{D}_n+|\alpha_n-\alpha_*|)\to 0$ when $n\to\infty$. Furthermore, the formulations of $\bar{M}_{n,j,\alpha}$ are as follows:

$$\bar{M}_{n,j,\alpha} = \sum_{i\in\mathcal{C}_j}\frac{\tanh(\alpha_n)\exp(\bar{b}_{n,i})}{\alpha!}(\Delta\boldsymbol{p}_{n,ij})^{\alpha},$$

for any $|\alpha|=1$.

Moving to the function $\widetilde{A}_{n,2}(\boldsymbol{X})$, the second-order Taylor expansions of the function $\widetilde{U}(\boldsymbol{X};\boldsymbol{p}_{n,i})$ around the function $\widetilde{U}(\boldsymbol{X};\boldsymbol{p}_{*,j})$ and the function $\widetilde{V}(\boldsymbol{p}_{n,i})$ around the function $\widetilde{V}(\boldsymbol{p}_{*,j})$ for any indices $i\in\mathcal{C}_j$ and j such that $|\mathcal{C}_j|>1$, we obtain the following formulation of the function $\widetilde{A}_{n,2}(\boldsymbol{X})$:

$$\begin{aligned}\widetilde{A}_{n,2}(\boldsymbol{X}) &= \sum_{j:|\mathcal{C}_j|>1}\sum_{1\le|\alpha|\le 2}\Big\{\bar{M}_{n,j,\alpha}\frac{\partial^{|\alpha|}\widetilde{U}}{\partial\boldsymbol{p}^{\alpha}}(\boldsymbol{X};\boldsymbol{p}_{*,j})\widetilde{V}(\boldsymbol{p}_{*,j}) + \bar{M}_{n,j,\alpha}\frac{\partial^{|\alpha|}\widetilde{V}}{\partial\boldsymbol{p}^{\alpha}}(\boldsymbol{p}_{*,j})\widetilde{U}(\boldsymbol{X};\boldsymbol{p}_{*,j})\Big\} \\ &\quad + \sum_{j:|\mathcal{C}_j|>1}\sum_{|\alpha|=1,|\beta|=1}\bar{M}_{n,j,\alpha,\beta}\frac{\partial^{|\alpha|}\widetilde{U}}{\partial\boldsymbol{p}^{\alpha}}(\boldsymbol{X};\boldsymbol{p}_{*,j})\frac{\partial^{|\beta|}\widetilde{V}}{\partial\boldsymbol{p}^{\beta}}(\boldsymbol{p}_{*,j}) + \widetilde{R}_{n,2}(\boldsymbol{X})\end{aligned}$$

where the function $\widetilde{R}_{n,2}(\boldsymbol{X})$ is a combination of Taylor remainders and satisfies $\widetilde{R}_{n,2}(\boldsymbol{X})/(\mathcal{D}_n+|\alpha_n-\alpha_*|)\to 0$ when $n\to\infty$. Furthermore, we define

$$\bar{M}_{n,j,\alpha} = \sum_{i\in\mathcal{C}_j}\frac{\tanh(\alpha_n)\exp(b_{n,i})}{\alpha!}(\Delta\boldsymbol{p}_{n,ij})^{\alpha},$$

for any $|\alpha|=2$ and

$$\bar{M}_{n,j,\alpha,\beta} = \sum_{i\in\mathcal{C}_j}\frac{\tanh(\alpha_n)\exp(\bar{b}_{n,i})}{\alpha!\beta!}(\Delta\boldsymbol{p}_{n,ij})^{\alpha+\beta},$$

for any $|\alpha|=|\beta|=1$. From the formulations of the functions $\widetilde{U}(\boldsymbol{X};\boldsymbol{p})$ and $\widetilde{V}(\boldsymbol{p})$, we obtain the following explicit forms of their partial derivatives:

$$\begin{aligned}\frac{\partial\widetilde{U}}{\partial\boldsymbol{p}^{(u)}}(\boldsymbol{X};\boldsymbol{p}) &= \exp((\bar{B}\sigma(\boldsymbol{p}))^{\top}\boldsymbol{X})\Big(\bar{B}\frac{\partial\sigma}{\partial\boldsymbol{p}^{(u)}}(\boldsymbol{p})\Big)^{\top}\boldsymbol{X}, \\ \frac{\partial^2\widetilde{U}}{\partial\boldsymbol{p}^{(u)}\partial\boldsymbol{p}^{(v)}}(\boldsymbol{X};\boldsymbol{p}) &= \exp((\bar{B}\sigma(\boldsymbol{p}))^{\top}\boldsymbol{X})\Big\{\Big(\bar{B}\frac{\partial^2\sigma}{\partial\boldsymbol{p}^{(u)}\partial\boldsymbol{p}^{(v)}}(\boldsymbol{p})\Big)^{\top}\boldsymbol{X} + \boldsymbol{X}^{\top}\Big(\bar{B}\frac{\partial\sigma}{\partial\boldsymbol{p}^{(u)}}(\boldsymbol{p})\Big)\Big(\bar{B}\frac{\partial\sigma}{\partial\boldsymbol{p}^{(v)}}(\boldsymbol{p})\Big)^{\top}\boldsymbol{X}\Big\}, \\ \frac{\partial\widetilde{V}}{\partial\boldsymbol{p}^{(u)}}(\boldsymbol{p}) &= \bar{C}\frac{\partial\sigma}{\partial\boldsymbol{p}^{(u)}}(\boldsymbol{p}), \\ \frac{\partial^2\widetilde{V}}{\partial\boldsymbol{p}^{(u)}\partial\boldsymbol{p}^{(v)}}(\boldsymbol{p}) &= \bar{C}\frac{\partial^2\sigma}{\partial\boldsymbol{p}^{(u)}\partial\boldsymbol{p}^{(v)}}(\boldsymbol{p}).\end{aligned}$$

Plugging these explicit formulations of the derivatives of the functions $\widetilde{U}(\boldsymbol{X};\boldsymbol{p})$ and $\widetilde{V}(\boldsymbol{p})$, we can express the functions

~其中$R_{ij,\ 1}(\boldsymbol{X})$以及$R_{ij,\ 2}$这些项属于泰勒余项。

~将上述结果结合后，可得到该函数的如下表达式$A_{n,1}(\boldsymbol{X})$:

$$\begin{aligned}\widetilde{A}_{n,1}(\boldsymbol{X}) &= \sum_{j:|\mathcal{C}_j|=1}\sum_{i\in\mathcal{C}_j}\frac{\tanh(\alpha_n)\exp(\bar{b}_{n,i})}{\alpha!}\sum_{|\alpha|=1}\Big\{(\Delta\boldsymbol{p}_{n,ij})^{\alpha}\frac{\partial^{|\alpha|}\widetilde{U}}{\partial\boldsymbol{p}^{\alpha}}(\boldsymbol{X};\boldsymbol{p}_{*,j})\widetilde{V}(\boldsymbol{p}_{*,j}) \\ &\qquad + (\Delta\boldsymbol{p}_{n,ij})^{\alpha}\frac{\partial^{|\alpha|}\widetilde{V}}{\partial\boldsymbol{p}^{\alpha}}(\boldsymbol{p}_{*,j})\widetilde{U}(\boldsymbol{X};\boldsymbol{p}_{*,j})\Big\} + \widetilde{R}_{n,1}(\boldsymbol{X}) \\ &= \sum_{j:|\mathcal{C}_j|=1}\sum_{|\alpha|=1}\Big\{\bar{M}_{n,j,\alpha}\frac{\partial^{|\alpha|}\widetilde{U}}{\partial\boldsymbol{p}^{\alpha}}(\boldsymbol{X};\boldsymbol{p}_{*,j})\widetilde{V}(\boldsymbol{p}_{*,j}) + \bar{M}_{n,j,\alpha}\frac{\partial^{|\alpha|}\widetilde{V}}{\partial\boldsymbol{p}^{\alpha}}(\boldsymbol{p}_{*,j})\widetilde{U}(\boldsymbol{X};\boldsymbol{p}_{*,j})\Big\} + \widetilde{R}_{n,1}(\boldsymbol{X})\end{aligned}$$

~其中函数$R_{n,\ 1}(\boldsymbol{X})$为泰勒余项的组合，并满足当$n\to\infty$时$R_{n,\ 1}(\boldsymbol{X})/(\mathcal{D}_n+|\alpha_n-\alpha_*|)\ \to 0$。此外，$M_{n、\ j、\ \alpha}$的表述如下:

$$\bar{M}_{n,j,\alpha} = \sum_{i\in\mathcal{C}_j}\frac{\tanh(\alpha_n)\exp(\bar{b}_{n,i})}{\alpha!}(\Delta\boldsymbol{p}_{n,ij})^{\alpha},$$

对于任意的$|\alpha|=1$。

Moving to the function $\widetilde{A}_{n,2}(\boldsymbol{X})$, the second-order Taylor expansions of the function $\widetilde{U}(\boldsymbol{X};\boldsymbol{p}_{n,i})$ around the function $\widetilde{U}(\boldsymbol{X};\boldsymbol{p}_{*,j})$ and the function $\widetilde{V}(\boldsymbol{p}_{n,i})$ around the function $\widetilde{V}(\boldsymbol{p}_{*,j})$ for any indices $i\in\mathcal{C}_j$ and j such that $|\mathcal{C}_j|>1$, we obtain the following formulation of the function $\widetilde{A}_{n,2}(\boldsymbol{X})$:

$$\begin{aligned}\widetilde{A}_{n,2}(\boldsymbol{X}) &= \sum_{j:|\mathcal{C}_j|>1}\sum_{1\le|\alpha|\le 2}\Big\{\bar{M}_{n,j,\alpha}\frac{\partial^{|\alpha|}\widetilde{U}}{\partial\boldsymbol{p}^{\alpha}}(\boldsymbol{X};\boldsymbol{p}_{*,j})\widetilde{V}(\boldsymbol{p}_{*,j}) + \bar{M}_{n,j,\alpha}\frac{\partial^{|\alpha|}\widetilde{V}}{\partial\boldsymbol{p}^{\alpha}}(\boldsymbol{p}_{*,j})\widetilde{U}(\boldsymbol{X};\boldsymbol{p}_{*,j})\Big\} \\ &\quad + \sum_{j:|\mathcal{C}_j|>1}\sum_{|\alpha|=1,|\beta|=1}\bar{M}_{n,j,\alpha,\beta}\frac{\partial^{|\alpha|}\widetilde{U}}{\partial\boldsymbol{p}^{\alpha}}(\boldsymbol{X};\boldsymbol{p}_{*,j})\frac{\partial^{|\beta|}\widetilde{V}}{\partial\boldsymbol{p}^{\beta}}(\boldsymbol{p}_{*,j}) + \widetilde{R}_{n,2}(\boldsymbol{X})\end{aligned}$$

~其中函数$R_{n,\ 2}(\boldsymbol{X})$是由泰勒余项组合而成的，且满足$R_{n,\ 2}(\boldsymbol{X})/(\mathcal{D}_n+|\alpha_n-\alpha_*|)\to 0$当$n\to\infty$时成立。此外，我们进一步定义了……

$$\bar{M}_{n,j,\alpha} = \sum_{i\in\mathcal{C}_j}\frac{\tanh(\alpha_n)\exp(b_{n,i})}{\alpha!}(\Delta\boldsymbol{p}_{n,ij})^{\alpha},$$

对于任意的$|\alpha|=2$以及…
…

$$\bar{M}_{n,j,\alpha,\beta} = \sum_{i\in\mathcal{C}_j}\frac{\tanh(\alpha_n)\exp(\bar{b}_{n,i})}{\alpha!\beta!}(\Delta\boldsymbol{p}_{n,ij})^{\alpha+\beta},$$

~对于任意的$|\alpha|=|\beta|=1$而言。根据函数$U(\boldsymbol{X};\ \boldsymbol{p})$与$V(\boldsymbol{p})$的表述形式，我们可以得到它们各偏导数的如下显式表达式:

$$\begin{aligned}\frac{\partial\widetilde{U}}{\partial\boldsymbol{p}^{(u)}}(\boldsymbol{X};\boldsymbol{p}) &= \exp((\bar{B}\sigma(\boldsymbol{p}))^{\top}\boldsymbol{X})\Big(\bar{B}\frac{\partial\sigma}{\partial\boldsymbol{p}^{(u)}}(\boldsymbol{p})\Big)^{\top}\boldsymbol{X}, \\ \frac{\partial^2\widetilde{U}}{\partial\boldsymbol{p}^{(u)}\partial\boldsymbol{p}^{(v)}}(\boldsymbol{X};\boldsymbol{p}) &= \exp((\bar{B}\sigma(\boldsymbol{p}))^{\top}\boldsymbol{X})\Big\{\Big(\bar{B}\frac{\partial^2\sigma}{\partial\boldsymbol{p}^{(u)}\partial\boldsymbol{p}^{(v)}}(\boldsymbol{p})\Big)^{\top}\boldsymbol{X} + \boldsymbol{X}^{\top}\Big(\bar{B}\frac{\partial\sigma}{\partial\boldsymbol{p}^{(u)}}(\boldsymbol{p})\Big)\Big(\bar{B}\frac{\partial\sigma}{\partial\boldsymbol{p}^{(v)}}(\boldsymbol{p})\Big)^{\top}\boldsymbol{X}\Big\}, \\ \frac{\partial\widetilde{V}}{\partial\boldsymbol{p}^{(u)}}(\boldsymbol{p}) &= \bar{C}\frac{\partial\sigma}{\partial\boldsymbol{p}^{(u)}}(\boldsymbol{p}), \\ \frac{\partial^2\widetilde{V}}{\partial\boldsymbol{p}^{(u)}\partial\boldsymbol{p}^{(v)}}(\boldsymbol{p}) &= \bar{C}\frac{\partial^2\sigma}{\partial\boldsymbol{p}^{(u)}\partial\boldsymbol{p}^{(v)}}(\boldsymbol{p}).\end{aligned}$$

~通过代入函数$U(\boldsymbol{X};\ \boldsymbol{p})$和$V(\boldsymbol{p})$的导数的这些显式表达式，我们能够将这些函数表示出来。

$\widetilde{A}_{n,1}(\boldsymbol{X})$ and $\widetilde{A}_{n,2}(\boldsymbol{X})$ as follows:

$$\widetilde{A}_{n,1}(\boldsymbol{X}) = \sum_{j:|\mathcal{C}_j|=1} \exp((\bar{B}\sigma(\boldsymbol{p}_{*,j}))^\top \boldsymbol{X})\big[\mathcal{L}_{1,n}(\boldsymbol{p}_{*,j}) + \mathcal{L}_{2,n}(\boldsymbol{p}_{*,j})^\top \bar{B}^\top \boldsymbol{X}\big) + \widetilde{R}_{n,1}(\boldsymbol{X}),$$

$$\widetilde{A}_{n,2}(\boldsymbol{X}) = \sum_{j:|\mathcal{C}_j|>1} \exp((\bar{B}\sigma(\boldsymbol{p}_{*,j}))^\top \boldsymbol{X})\big[\bar{\mathcal{L}}_{1,n}(\boldsymbol{p}_{*,j}) + \bar{\mathcal{L}}_{2,n}(\boldsymbol{p}_{*,j})^\top \bar{B}^\top \boldsymbol{X} + (\bar{B}^\top \boldsymbol{X})^\top \bar{\mathcal{L}}_{3,n}(\boldsymbol{p}_{*,j})\bar{B}^\top \boldsymbol{X}\big] + \widetilde{R}_{n,2}(\boldsymbol{X}),$$

where the functions $\mathcal{L}_{1,n}(\boldsymbol{p}), \mathcal{L}_{2,n}(\boldsymbol{p}), \bar{\mathcal{L}}_{1,n}(\boldsymbol{p}), \bar{\mathcal{L}}_{2,n}(\boldsymbol{p})$, and $\bar{\mathcal{L}}_{3,n}(\boldsymbol{p})$ are defined as follows:

$$\mathcal{L}_{1,n}(\boldsymbol{p}) = \sum_{u=1}^{d} \bar{M}_{n,j,1_u}\bar{C}\frac{\partial\sigma}{\partial \boldsymbol{p}^{(u)}}(\boldsymbol{p}),$$

$$\mathcal{L}_{2,n}(\boldsymbol{p}) = \sum_{u=1}^{d} \bar{M}_{n,j,1_u}\frac{\partial\sigma}{\partial \boldsymbol{p}^{(u)}}(\boldsymbol{p})\bar{C}\sigma(\boldsymbol{p}),$$

$$\bar{\mathcal{L}}_{1,n}(\boldsymbol{p}) = \sum_{1\le u,v\le d} \bar{M}_{n,j,1_{uv}}\bar{C}\frac{\partial^2\sigma}{\partial \boldsymbol{p}^{(u)}\partial \boldsymbol{p}^{(v)}}(\boldsymbol{p}) = \sum_{u=1}^{d} M_{n,j,1_{uu}}\bar{C}\frac{\partial^2\sigma}{\partial \boldsymbol{p}^{(u)}\partial \boldsymbol{p}^{(u)}}(\boldsymbol{p}),$$

$$\bar{\mathcal{L}}_{2,n}(\boldsymbol{p}) = \sum_{u=1}^{d} \bar{M}_{n,j,1_u}\frac{\partial\sigma}{\partial \boldsymbol{p}^{(u)}}(\boldsymbol{p})\bar{C}\sigma(\boldsymbol{p}) + \sum_{1\le u,v\le d}\big[\bar{M}_{n,j,1_v,1_u}\bar{C}\frac{\partial\sigma}{\partial \boldsymbol{p}^{(u)}}(\boldsymbol{p})\frac{\partial\sigma}{\partial \boldsymbol{p}^{(v)}}(\boldsymbol{p}) + \bar{M}_{n,j,1_{uv}}\frac{\partial^2\sigma}{\partial \boldsymbol{p}^{(u)}\partial \boldsymbol{p}^{(v)}}(\boldsymbol{p})\bar{C}\sigma(\boldsymbol{p})\big],$$

$$\bar{\mathcal{L}}_{3,n}(\boldsymbol{p}) = \sum_{1\le u,v\le d} \bar{M}_{n,j,1_{uv}}\frac{\partial\sigma}{\partial \boldsymbol{p}^{(u)}}(\boldsymbol{p})(\frac{\partial\sigma}{\partial \boldsymbol{p}^{(v)}}(\boldsymbol{p}))^\top\bar{C}\sigma(\boldsymbol{p}).$$

In these formulations, we use 1_u to denote the vector that its u-th element is 1 and its other elements are 0 for $1 \le u \le d$. Furthermore, we denote 1_{uv} as the matrix that its (u,v)-th element is 1 and its other elements are 0 for any $1 \le u, v \le d$.

Decomposition of the function $\widetilde{B}_n(\boldsymbol{X})$. Similar to the decomposition of the function $\widetilde{A}_n(\boldsymbol{X})$, we can decompose the function $\widetilde{B}_n(\boldsymbol{X})$ as follows:

$$\begin{aligned}\widetilde{B}_n(\boldsymbol{X}) &= \sum_{j:|\mathcal{C}_j|=1}\sum_{i\in\mathcal{C}_j}\tanh(\alpha_n)\exp(\bar{b}_{n,i})\Big[\widetilde{U}(\boldsymbol{X};\boldsymbol{p}_{n,i}) - \widetilde{U}(\boldsymbol{X};\boldsymbol{p}_{*,j})\Big]g_{G_n}(\boldsymbol{X})\\ &+ \sum_{j:|\mathcal{C}_j|>1}\sum_{i\in\mathcal{C}_j}\tanh(\alpha_n)\exp(\bar{b}_{n,i})\Big[\widetilde{U}(\boldsymbol{X};\boldsymbol{p}_{n,i}) - \widetilde{U}(\boldsymbol{X};\boldsymbol{p}_{*,j})\Big]g_{G_n}(\boldsymbol{X})\\ &:= \widetilde{B}_{n,1}(\boldsymbol{X}) + \widetilde{B}_{n,2}(\boldsymbol{X})\end{aligned}$$

where we denote $\widetilde{B}_{n,1}(\boldsymbol{X}) = \sum_{j:|\mathcal{C}_j|=1}\sum_{i\in\mathcal{C}_j}\tanh(\alpha_n)\exp(\bar{b}_{n,i})\Big[\widetilde{U}(\boldsymbol{X};\boldsymbol{p}_{n,i}) - \widetilde{U}(\boldsymbol{X};\boldsymbol{p}_{*,j})\Big]g_{G_n}(\boldsymbol{X})$ and $\bar{B}_{n,2}(\boldsymbol{X}) = \sum_{j:|\mathcal{C}_j|>1}\sum_{i\in\mathcal{C}_j}\tanh(\alpha_n)\exp(\bar{b}_{n,i})\Big[\widetilde{U}(\boldsymbol{X};\boldsymbol{p}_{n,i}) - \widetilde{U}(\boldsymbol{X};\boldsymbol{p}_{*,j})\Big]g_{G_n}(\boldsymbol{X})$.

Similar to the Taylor expansions for the functions $\widetilde{A}_{n,1}(\boldsymbol{X})$ and $\widetilde{A}_{n,2}(\boldsymbol{X})$, by using the first-order Taylor expansion to $\widetilde{B}_{n,1}(\boldsymbol{X})$ and the second-order Taylor expansion to $\widetilde{B}_{n,2}(\boldsymbol{X})$, we obtain that

$$\widetilde{B}_{n,1}(\boldsymbol{X}) = \sum_{j:|\mathcal{C}_j|=1}\sum_{|\alpha|=1}\bar{M}_{n,j,\alpha}\frac{\partial^{|\alpha|}\widetilde{U}}{\partial \boldsymbol{p}^{\alpha}}(\boldsymbol{X};\boldsymbol{p}_{*,j})g_{G_n}(\boldsymbol{X}) + \widetilde{R}_{n,3}(\boldsymbol{X}),$$

$$\widetilde{B}_{n,2}(\boldsymbol{X}) = \sum_{j:|\mathcal{C}_j|=1}\sum_{1\le|\alpha|\le2}\bar{M}_{n,j,\alpha}\frac{\partial^{|\alpha|}\widetilde{U}}{\partial \boldsymbol{p}^{\alpha}}(\boldsymbol{X};\boldsymbol{p}_{*,j})g_{G_n}(\boldsymbol{X}) + \widetilde{R}_{n,4}(\boldsymbol{X})$$

where the functions $\widetilde{R}_{n,3}(\boldsymbol{X}), \widetilde{R}_{n,4}(\boldsymbol{X})$ are Taylor remainders. Furthermore, they satisfy that $\widetilde{R}_{n,3}(\boldsymbol{X})/(\mathcal{D}_n+|\alpha_n-\alpha_*|) \to$

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达形式如下：$A_{n,\ 1}(\boldsymbol{X})$ 以及$A_{n,\ 2}(\boldsymbol{X})$ 的表

$$\widetilde{A}_{n,1}(\boldsymbol{X}) = \sum_{j:|\mathcal{C}_j|=1} \exp((\bar{B}\sigma(\boldsymbol{p}_{*,j}))^\top \boldsymbol{X})\big[\mathcal{L}_{1,n}(\boldsymbol{p}_{*,j}) + \mathcal{L}_{2,n}(\boldsymbol{p}_{*,j})^\top \bar{B}^\top \boldsymbol{X}\big) + \widetilde{R}_{n,1}(\boldsymbol{X}),$$

$$\widetilde{A}_{n,2}(\boldsymbol{X}) = \sum_{j:|\mathcal{C}_j|>1} \exp((\bar{B}\sigma(\boldsymbol{p}_{*,j}))^\top \boldsymbol{X})\big[\bar{\mathcal{L}}_{1,n}(\boldsymbol{p}_{*,j}) + \bar{\mathcal{L}}_{2,n}(\boldsymbol{p}_{*,j})^\top \bar{B}^\top \boldsymbol{X} + (\bar{B}^\top \boldsymbol{X})^\top \bar{\mathcal{L}}_{3,n}(\boldsymbol{p}_{*,j})\bar{B}^\top \boldsymbol{X}\big] + \widetilde{R}_{n,2}(\boldsymbol{X}),$$

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其中，函数 $\mathcal{L}_{1,n}(\boldsymbol{p})$ $\mathcal{L}_{2,n}(\boldsymbol{p})$ $\mathcal{L}_{1,n}(\boldsymbol{p})\mathcal{L}_{2,n}(\boldsymbol{p})$以及$\mathcal{L}_{3,n}(\boldsymbol{p})$的定义如下：

$$\mathcal{L}_{1,n}(\boldsymbol{p}) = \sum_{u=1}^{d} \bar{M}_{n,j,1_u}\bar{C}\frac{\partial\sigma}{\partial \boldsymbol{p}^{(u)}}(\boldsymbol{p}),$$

$$\mathcal{L}_{2,n}(\boldsymbol{p}) = \sum_{u=1}^{d} \bar{M}_{n,j,1_u}\frac{\partial\sigma}{\partial \boldsymbol{p}^{(u)}}(\boldsymbol{p})\bar{C}\sigma(\boldsymbol{p}),$$

$$\bar{\mathcal{L}}_{1,n}(\boldsymbol{p}) = \sum_{1\le u,v\le d} \bar{M}_{n,j,1_{uv}}\bar{C}\frac{\partial^2\sigma}{\partial \boldsymbol{p}^{(u)}\partial \boldsymbol{p}^{(v)}}(\boldsymbol{p}) = \sum_{u=1}^{d} M_{n,j,1_{uu}}\bar{C}\frac{\partial^2\sigma}{\partial \boldsymbol{p}^{(u)}\partial \boldsymbol{p}^{(u)}}(\boldsymbol{p}),$$

$$\bar{\mathcal{L}}_{2,n}(\boldsymbol{p}) = \sum_{u=1}^{d} \bar{M}_{n,j,1_u}\frac{\partial\sigma}{\partial \boldsymbol{p}^{(u)}}(\boldsymbol{p})\bar{C}\sigma(\boldsymbol{p}) + \sum_{1\le u,v\le d}\big[\bar{M}_{n,j,1_v,1_u}\bar{C}\frac{\partial\sigma}{\partial \boldsymbol{p}^{(u)}}(\boldsymbol{p})\frac{\partial\sigma}{\partial \boldsymbol{p}^{(v)}}(\boldsymbol{p}) + \bar{M}_{n,j,1_{uv}}\frac{\partial^2\sigma}{\partial \boldsymbol{p}^{(u)}\partial \boldsymbol{p}^{(v)}}(\boldsymbol{p})\bar{C}\sigma(\boldsymbol{p})\big],$$

$$\bar{\mathcal{L}}_{3,n}(\boldsymbol{p}) = \sum_{1\le u,v\le d} \bar{M}_{n,j,1_{uv}}\frac{\partial\sigma}{\partial \boldsymbol{p}^{(u)}}(\boldsymbol{p})(\frac{\partial\sigma}{\partial \boldsymbol{p}^{(v)}}(\boldsymbol{p}))^\top\bar{C}\sigma(\boldsymbol{p}).$$

在这些公式中，我们用 1_u 来表示这样一个向量：对于 $1 \le u \le d$而言，其第 u个元素为1，其余元素均为0。此外，我们用 1_{uv} 来表示这样一个矩阵：对于任意的 $1 \le u$，其(u,v)位置上的元素为1，其余元素均为0。

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函数 $B_n(\boldsymbol{X})$ 的分解。与函数 $A_n(\boldsymbol{X})$ 的分解方式类似，我们可以将函数 $B_n(\boldsymbol{X})$按如下方式分解：

$$\begin{aligned}\widetilde{B}_n(\boldsymbol{X}) &= \sum_{j:|\mathcal{C}_j|=1}\sum_{i\in\mathcal{C}_j}\tanh(\alpha_n)\exp(\bar{b}_{n,i})\Big[\widetilde{U}(\boldsymbol{X};\boldsymbol{p}_{n,i}) - \widetilde{U}(\boldsymbol{X};\boldsymbol{p}_{*,j})\Big]g_{G_n}(\boldsymbol{X})\\ &+ \sum_{j:|\mathcal{C}_j|>1}\sum_{i\in\mathcal{C}_j}\tanh(\alpha_n)\exp(\bar{b}_{n,i})\Big[\widetilde{U}(\boldsymbol{X};\boldsymbol{p}_{n,i}) - \widetilde{U}(\boldsymbol{X};\boldsymbol{p}_{*,j})\Big]g_{G_n}(\boldsymbol{X})\\ &:= \widetilde{B}_{n,1}(\boldsymbol{X}) + \widetilde{B}_{n,2}(\boldsymbol{X})\end{aligned}$$

其中，我们用 $\widetilde{B}_{n,1}(\boldsymbol{X}) = \sum_{j:|\mathcal{C}_j|=1}\sum_{i\in\mathcal{C}_j}\tanh(\alpha_n)\exp(\bar{b}_{n,i})\Big[\widetilde{U}(\boldsymbol{X};\boldsymbol{p}_{n,i}) - \widetilde{U}(\boldsymbol{X};\boldsymbol{p}_{*,j})\Big]g_{G_n}(\boldsymbol{X})$ 来表示，而用 $\bar{B}_{n,2}(\boldsymbol{X}) = \sum_{j:|\mathcal{C}_j|>1}\sum_{i\in\mathcal{C}_j}\tanh(\alpha_n)\exp(\bar{b}_{n,i})\Big[\widetilde{U}(\boldsymbol{X};\boldsymbol{p}_{n,i}) - \widetilde{U}(\boldsymbol{X};\boldsymbol{p}_{*,j})\Big]g_{G_n}(\boldsymbol{X})$ 来表示。

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与函数 $A_{n,\ 1}(\boldsymbol{X})$以及$A_{n,\ 2}(\boldsymbol{X})$的泰勒展开类似，通过运用一阶泰勒展开对
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$B_{n,\ 1}(\boldsymbol{X})$进行近似，再结合二阶泰勒展开对$B_{n,\ 2}(\boldsymbol{X})$进行近似，我们最终可以得到

$$\widetilde{B}_{n,1}(\boldsymbol{X}) = \sum_{j:|\mathcal{C}_j|=1}\sum_{|\alpha|=1}\bar{M}_{n,j,\alpha}\frac{\partial^{|\alpha|}\widetilde{U}}{\partial \boldsymbol{p}^{\alpha}}(\boldsymbol{X};\boldsymbol{p}_{*,j})g_{G_n}(\boldsymbol{X}) + \widetilde{R}_{n,3}(\boldsymbol{X}),$$

$$\widetilde{B}_{n,2}(\boldsymbol{X}) = \sum_{j:|\mathcal{C}_j|=1}\sum_{1\le|\alpha|\le2}\bar{M}_{n,j,\alpha}\frac{\partial^{|\alpha|}\widetilde{U}}{\partial \boldsymbol{p}^{\alpha}}(\boldsymbol{X};\boldsymbol{p}_{*,j})g_{G_n}(\boldsymbol{X}) + \widetilde{R}_{n,4}(\boldsymbol{X})$$

~ ~
其中函数 $R_{n,3}(\boldsymbol{X})R_{n,4}(\boldsymbol{X})$属于泰勒余项。此外，它们还满足 $R_{n,3}(\boldsymbol{X})/(\mathcal{D}_n+|\alpha_n-\alpha_*|) \to$

0 and $\widetilde{R}_{n,4}(\boldsymbol{X})/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0$ when $n \to \infty$. Given the explicit formulations of the derivatives of the functions $\widetilde{U}(\boldsymbol{X}; \boldsymbol{p})$ and $\widetilde{V}(\boldsymbol{p})$, the functions $\widetilde{B}_{n,1}(\boldsymbol{X})$ and $\widetilde{B}_{n,2}(\boldsymbol{X})$ can be then rewritten as follows:

$$\widetilde{B}_{n,1}(\boldsymbol{X}) = \sum_{j:|\mathcal{C}_j|=1} \exp((\bar{B}\sigma(\boldsymbol{p}_{*,j}))^\top \boldsymbol{X}) \mathcal{N}_{1,n}(\boldsymbol{p}_{*,j})^\top \boldsymbol{X} g_{G_n}(\boldsymbol{X}) + \widetilde{R}_{n,3}(\boldsymbol{X}),$$

$$\widetilde{B}_{n,2}(\boldsymbol{X}) = \sum_{j:|\mathcal{C}_j|>1} \exp((\bar{B}\sigma(\boldsymbol{p}_{*,j}))^\top \boldsymbol{X}) \big[\bar{\mathcal{N}}_{1,n}(\boldsymbol{p}_{*,j})^\top \bar{B}^\top \boldsymbol{X} + (\bar{B}^\top \boldsymbol{X})^\top \bar{\mathcal{N}}_{2,n}(\boldsymbol{p}_{*,j})(\bar{B}^\top \boldsymbol{X})\big] g_{G_n}(\boldsymbol{X}) + \widetilde{R}_{n,4}(\boldsymbol{X}).$$

Here, the functions $\mathcal{N}_{1,n}(\boldsymbol{X})$, $\bar{\mathcal{N}}_{1,n}(\boldsymbol{X})$, and $\bar{\mathcal{N}}_{2,n}(\boldsymbol{X})$ have the following formulations:

$$\mathcal{N}_{1,n}(\boldsymbol{p}) = \sum_{u=1}^{d} \bar{M}_{n,j,1_u} \frac{\partial \sigma}{\partial \boldsymbol{p}^{(u)}}(\boldsymbol{p}),$$

$$\bar{\mathcal{N}}_{1,n}(\boldsymbol{p}) = \sum_{u=1}^{d} \bar{M}_{n,j,1_u} \frac{\partial \sigma}{\partial \boldsymbol{p}^{(u)}}(\boldsymbol{p}) + \sum_{1\le u,v\le d} \bar{M}_{n,j,1_{uv}} \frac{\partial^2 \sigma}{\partial \boldsymbol{p}^{(u)} \partial \boldsymbol{p}^{(v)}}(\boldsymbol{p}),$$

$$\bar{\mathcal{N}}_{2,n}(\boldsymbol{p}) = \sum_{1\le u,v\le d} \bar{M}_{n,j,1_{uv}} \frac{\partial \sigma}{\partial \boldsymbol{p}^{(u)}}(\boldsymbol{p}) \frac{\partial \sigma}{\partial \boldsymbol{p}^{(v)}}(\boldsymbol{p})^\top.$$

Collecting all of the above results with the decomposition of the functions $\widetilde{A}_n(\boldsymbol{X})$ and $\widetilde{B}_n(\boldsymbol{X})$, we can represent the function $\widetilde{Q}_{n,1}(\boldsymbol{X})$ in equation (22) as follows:

$$\begin{aligned}
\widetilde{Q}_{n,1}(\boldsymbol{X}) = &\sum_{j:|\mathcal{C}_j|=1} \exp((\bar{B}\sigma(\boldsymbol{p}_{*,j}))^\top \boldsymbol{X}) \big[\mathcal{L}'_{1,n}(\boldsymbol{p}_{*,j}) + \mathcal{L}_{2,n}(\boldsymbol{p}_{*,j})^\top \bar{B}^\top \boldsymbol{X}\big) \\
&+ \sum_{j:|\mathcal{C}_j|>1} \exp((\bar{B}\sigma(\boldsymbol{p}_{*,j}))^\top \boldsymbol{X}) \big[\bar{\mathcal{L}}'_{1,n}(\boldsymbol{p}_{*,j}) + \bar{\mathcal{L}}_{2,n}(\boldsymbol{p}_{*,j})^\top \bar{B}^\top \boldsymbol{X} + (\bar{B}^\top \boldsymbol{X})^\top \bar{\mathcal{L}}_{3,n}(\boldsymbol{p}_{*,j}) \bar{B}^\top \boldsymbol{X}\big] \\
&- \sum_{j:|\mathcal{C}_j|=1} \exp((\bar{B}\sigma(\boldsymbol{p}_{*,j}))^\top \boldsymbol{X}) \big[\bar{M}_{n,j,0_d} + \mathcal{N}_{1,n}(\boldsymbol{p}_{*,j})^\top \bar{B}^\top \boldsymbol{X}\big] g_{G_n}(\boldsymbol{X}) \\
&- \sum_{j:|\mathcal{C}_j|>1} \exp((\bar{B}\sigma(\boldsymbol{p}_{*,j}))^\top \boldsymbol{X}) \big[\bar{M}_{n,j,0_d} + \bar{\mathcal{N}}_{1,n}(\boldsymbol{p}_{*,j})^\top \bar{B}^\top \boldsymbol{X} + (\bar{B}^\top \boldsymbol{X})^\top \bar{\mathcal{N}}_{2,n}(\boldsymbol{p}_{*,j}) \bar{B}^\top \boldsymbol{X}\big] g_{G_n}(\boldsymbol{X}) \\
&+ \widetilde{R}_{n,1}(\boldsymbol{X}) + \widetilde{R}_{n,2}(\boldsymbol{X}) - \widetilde{R}_{n,3}(\boldsymbol{X}) - \widetilde{R}_{n,4}(\boldsymbol{X})
\end{aligned} \tag{32}$$

where we define $\bar{M}_{n,j,0_d} = \tanh(\alpha_n)\Big(\sum_{i\in\mathcal{C}_j} \exp(\bar{b}_{n,i}) - \exp(\bar{b}_{*,j})\Big)$ for any index $1 \le j \le L$, $\mathcal{L}'_{1,n}(\boldsymbol{p}_{*,j}) = \mathcal{L}_{1,n}(\boldsymbol{p}_{*,j}) + \bar{M}_{n,j,0_d} \bar{C}\sigma(\boldsymbol{p}_{*,j})$, and $\bar{\mathcal{L}}'_{1,n}(\boldsymbol{p}_{*,j}) = \bar{\mathcal{L}}_{1,n}(\boldsymbol{p}_{*,j}) + \bar{M}_{n,j,0_d} \bar{C}\sigma(\boldsymbol{p}_{*,j})$.

Decomposition of the function $\widetilde{Q}_{n,2}(\boldsymbol{X})$. An application of the first-order Taylor expansion leads to the following expression for the function $\widetilde{Q}_{n,2}(\boldsymbol{X})$:

$$\begin{aligned}
\widetilde{Q}_{n,2}(\boldsymbol{X}) &= [\tanh(\alpha_n) - \tanh(\alpha_*)] \cdot \Big[\sum_{k'=1}^{L} \exp((\bar{B}\sigma(\boldsymbol{p}_{*,k'}))^\top \boldsymbol{X} + \bar{b}_{*,k'})\Big] \cdot g_{G_*}(\boldsymbol{X}) \\
&= (\alpha_n - \alpha_*)[1 - \tanh^2(\alpha_*)] \cdot \Big[\sum_{k'=1}^{L} \exp((\bar{B}\sigma(\boldsymbol{p}_{*,k'}))^\top \boldsymbol{X} + \bar{b}_{*,k'})\Big] \cdot g_{G_*}(\boldsymbol{X}) + \widetilde{R}_{n,5}(\boldsymbol{X}),
\end{aligned} \tag{33}$$

where the function $\widetilde{R}_{n,5}(\boldsymbol{X})$ is Taylor remainder and satisfies that $\widetilde{R}_{n,5}(\boldsymbol{X})/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0$ as $n \to \infty$.

Step 2 - Non-vanishing coefficients. From the results of equations (32) and (33), we can express $[\widetilde{Q}_n(\boldsymbol{X}) - \sum_{i=1}^{5} \widetilde{R}_{n,i}(\boldsymbol{X})]/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$ as a combination of the linearly independent terms

$$\exp((\bar{B}\sigma(\boldsymbol{p}_{*,j}))^\top \boldsymbol{X}),\ (\bar{B}^\top \boldsymbol{X})^{(u)} \exp((\bar{B}\sigma(\boldsymbol{p}_{*,j}))^\top \boldsymbol{X}),$$

~0与$\widetilde{R}_{n,\ 4}(\boldsymbol{X})/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0$ 在 $n \to \infty$时。鉴于函数~ ~ ~ ~ $\widetilde{U}(\boldsymbol{X};\ \boldsymbol{p})$以及 $\widetilde{V}(\boldsymbol{p})$的导数已给出明确表达式，函数$\widetilde{B}_{n,\ 1}(\boldsymbol{X})$与$\widetilde{B}_{n,\ 2}(\boldsymbol{X})$可被重新表示为如下形式：

$$\widetilde{B}_{n,1}(\boldsymbol{X}) = \sum_{j:|\mathcal{C}_j|=1} \exp((\bar{B}\sigma(\boldsymbol{p}_{*,j}))^\top \boldsymbol{X}) \mathcal{N}_{1,n}(\boldsymbol{p}_{*,j})^\top \boldsymbol{X} g_{G_n}(\boldsymbol{X}) + \widetilde{R}_{n,3}(\boldsymbol{X}),$$

$$\widetilde{B}_{n,2}(\boldsymbol{X}) = \sum_{j:|\mathcal{C}_j|>1} \exp((\bar{B}\sigma(\boldsymbol{p}_{*,j}))^\top \boldsymbol{X}) \big[\bar{\mathcal{N}}_{1,n}(\boldsymbol{p}_{*,j})^\top \bar{B}^\top \boldsymbol{X} + (\bar{B}^\top \boldsymbol{X})^\top \bar{\mathcal{N}}_{2,n}(\boldsymbol{p}_{*,j})(\bar{B}^\top \boldsymbol{X})\big] g_{G_n}(\boldsymbol{X}) + \widetilde{R}_{n,4}(\boldsymbol{X}).$$

此处，函数$\mathcal{N}_{1,n}(\boldsymbol{X})$、$\bar{\mathcal{N}}_{1,n}(\boldsymbol{X})$以及$\bar{\mathcal{N}}_{2,n}(\boldsymbol{X})$具有如下的表达形式：

$$\mathcal{N}_{1,n}(\boldsymbol{p}) = \sum_{u=1}^{d} \bar{M}_{n,j,1_u} \frac{\partial \sigma}{\partial \boldsymbol{p}^{(u)}}(\boldsymbol{p}),$$

$$\bar{\mathcal{N}}_{1,n}(\boldsymbol{p}) = \sum_{u=1}^{d} \bar{M}_{n,j,1_u} \frac{\partial \sigma}{\partial \boldsymbol{p}^{(u)}}(\boldsymbol{p}) + \sum_{1\le u,v\le d} \bar{M}_{n,j,1_{uv}} \frac{\partial^2 \sigma}{\partial \boldsymbol{p}^{(u)} \partial \boldsymbol{p}^{(v)}}(\boldsymbol{p}),$$

$$\bar{\mathcal{N}}_{2,n}(\boldsymbol{p}) = \sum_{1\le u,v\le d} \bar{M}_{n,j,1_{uv}} \frac{\partial \sigma}{\partial \boldsymbol{p}^{(u)}}(\boldsymbol{p}) \frac{\partial \sigma}{\partial \boldsymbol{p}^{(v)}}(\boldsymbol{p})^\top.$$

~ ~将上述所有结果与函数的分解相结合 $\widetilde{A}_n(\boldsymbol{X})$ 以及 $\widetilde{B}_n(\boldsymbol{X})$，我们就可以如下表示~该函数$\widetilde{Q}_{n,\ 1}(\boldsymbol{X})$ 不等式（22）：

$$\begin{aligned}
\widetilde{Q}_{n,1}(\boldsymbol{X}) = &\sum_{j:|\mathcal{C}_j|=1} \exp((\bar{B}\sigma(\boldsymbol{p}_{*,j}))^\top \boldsymbol{X}) \big[\mathcal{L}'_{1,n}(\boldsymbol{p}_{*,j}) + \mathcal{L}_{2,n}(\boldsymbol{p}_{*,j})^\top \bar{B}^\top \boldsymbol{X}\big) \\
&+ \sum_{j:|\mathcal{C}_j|>1} \exp((\bar{B}\sigma(\boldsymbol{p}_{*,j}))^\top \boldsymbol{X}) \big[\bar{\mathcal{L}}'_{1,n}(\boldsymbol{p}_{*,j}) + \bar{\mathcal{L}}_{2,n}(\boldsymbol{p}_{*,j})^\top \bar{B}^\top \boldsymbol{X} + (\bar{B}^\top \boldsymbol{X})^\top \bar{\mathcal{L}}_{3,n}(\boldsymbol{p}_{*,j}) \bar{B}^\top \boldsymbol{X}\big] \\
&- \sum_{j:|\mathcal{C}_j|=1} \exp((\bar{B}\sigma(\boldsymbol{p}_{*,j}))^\top \boldsymbol{X}) \big[\bar{M}_{n,j,0_d} + \mathcal{N}_{1,n}(\boldsymbol{p}_{*,j})^\top \bar{B}^\top \boldsymbol{X}\big] g_{G_n}(\boldsymbol{X}) \\
&- \sum_{j:|\mathcal{C}_j|>1} \exp((\bar{B}\sigma(\boldsymbol{p}_{*,j}))^\top \boldsymbol{X}) \big[\bar{M}_{n,j,0_d} + \bar{\mathcal{N}}_{1,n}(\boldsymbol{p}_{*,j})^\top \bar{B}^\top \boldsymbol{X} + (\bar{B}^\top \boldsymbol{X})^\top \bar{\mathcal{N}}_{2,n}(\boldsymbol{p}_{*,j}) \bar{B}^\top \boldsymbol{X}\big] g_{G_n}(\boldsymbol{X}) \\
&+ \widetilde{R}_{n,1}(\boldsymbol{X}) + \widetilde{R}_{n,2}(\boldsymbol{X}) - \widetilde{R}_{n,3}(\boldsymbol{X}) - \widetilde{R}_{n,4}(\boldsymbol{X})
\end{aligned} \tag{32}$$

在此我们定义 $\bar{M}_{n,j,0_d}$ = 为$\tanh(\alpha_n)\Big(\sum_{i\in\mathcal{C}_j} \exp(\bar{b}_{n,i})$, − 为$\exp(\bar{b}_{*,j})\Big)$，其中 $1 \le j \le L$, $\mathcal{L}'_{1,n}$对应$(\boldsymbol{p}_{*,j})$、 = − − − − − $\mathcal{L}_{1,n}$对应$(\boldsymbol{p}_{*,j})$+ $\bar{M}_{n,j,0_d} C\sigma$对应$(\boldsymbol{p}_{*,j})$, 而$\mathcal{L}'_{1,n}$对应$(\boldsymbol{p}_{*,j})$, = $\mathcal{L}_{1,n}$对应$(\boldsymbol{p}_{*,j})$+ $\bar{M}_{n,j,0_d} C\sigma$对应$(\boldsymbol{p}_{*,j})$。

~**该函数的可分解形式为**$\widetilde{Q}_{n,\ 2}(\boldsymbol{X})$**。通过一阶泰勒展开，可得到该函数的如下表达式：**

$$\begin{aligned}
\widetilde{Q}_{n,2}(\boldsymbol{X}) &= [\tanh(\alpha_n) - \tanh(\alpha_*)] \cdot \Big[\sum_{k'=1}^{L} \exp((\bar{B}\sigma(\boldsymbol{p}_{*,k'}))^\top \boldsymbol{X} + \bar{b}_{*,k'})\Big] \cdot g_{G_*}(\boldsymbol{X}) \\
&= (\alpha_n - \alpha_*)[1 - \tanh^2(\alpha_*)] \cdot \Big[\sum_{k'=1}^{L} \exp((\bar{B}\sigma(\boldsymbol{p}_{*,k'}))^\top \boldsymbol{X} + \bar{b}_{*,k'})\Big] \cdot g_{G_*}(\boldsymbol{X}) + \widetilde{R}_{n,5}(\boldsymbol{X}),
\end{aligned} \tag{33}$$

其中该函数 ~ ~n $\widetilde{R}_{n,\ 5}(\boldsymbol{X})$即为泰勒余项，且满足$\widetilde{R}_{n,\ 5}(\boldsymbol{X})/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0$ 作为 $n \to \infty$.

~**第二步——非消失系数**。根据方程(32)和(33)的结果，我们可以将 $[Q_n(\boldsymbol{X}) - \sum_{i=1}^{5} \widetilde{R}_{n,\ i}(\boldsymbol{X})]/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$表示为这些线性无关项的组合。

$$\exp((\bar{B}\sigma(\boldsymbol{p}_{*,j}))^\top \boldsymbol{X}),\ (\bar{B}^\top \boldsymbol{X})^{(u)} \exp((\bar{B}\sigma(\boldsymbol{p}_{*,j}))^\top \boldsymbol{X}),$$

$$(\bar{B}^\top \boldsymbol{X})^{(u)}(\bar{B}^\top \boldsymbol{X})^{(v)} \exp((\bar{B}\sigma(\boldsymbol{p}_{*,j}))^\top \boldsymbol{X}),\ \exp((\bar{B}\sigma(\boldsymbol{p}_{*,j}))^\top \boldsymbol{X}) g_{G_n}(\boldsymbol{X}),$$
$$(\bar{B}^\top \boldsymbol{X})^{(u)} \exp((\bar{B}\sigma(\boldsymbol{p}_{*,j}))^\top \boldsymbol{X}) g_{G_n}(\boldsymbol{X}),\ (\bar{B}^\top \boldsymbol{X})^{(u)}(\bar{B}^\top \boldsymbol{X})^{(v)} \exp((\bar{B}\sigma(\boldsymbol{p}_{*,j}))^\top \boldsymbol{X}) g_{G_n}(\boldsymbol{X}),$$
$$[1-\tanh^2(\alpha_*)] \cdot \Big[\sum_{k'=1}^{L} \exp((\bar{B}\sigma(\boldsymbol{p}_{*,k'}))^\top \boldsymbol{X} + \bar{b}_{*,k'})\Big] \cdot g_{G_*}(\boldsymbol{X}),$$

for any $1 \leq j \leq L$ and $1 \leq u, v \leq d$.

Our claim is that at least one of the coefficients of these linearly independent terms in the formulation of $[\widetilde{Q}_n(\boldsymbol{X}) - \sum_{i=1}^{5} R_{n,i}(\boldsymbol{X})]/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$ does not go to 0 as $n \to \infty$. Assume by contrary that this claim does not hold, which means that all the coefficients of these linearly independent terms go to 0 as $n \to \infty$. Therefore, as $n \to \infty$ we obtain that

$$\mathcal{L}_{1,n}(\boldsymbol{p}_{*,j})/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0,\ \mathcal{L}_{2,n}(\boldsymbol{p}_{*,j})^{(u)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0,\ \bar{\mathcal{L}}_{1,n}(\boldsymbol{p}_{*,j})/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0,$$
$$\bar{\mathcal{L}}_{2,n}(\boldsymbol{p}_{*,j})^{(u)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0,\ \bar{\mathcal{L}}_{2,n}(\boldsymbol{p}_{*,j})^{(u)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0,\ \bar{\mathcal{L}}_{2,n}(\boldsymbol{p}_{*,j})^{(u)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0,$$
$$\bar{\mathcal{L}}_{3,n}(\boldsymbol{p}_{*,j})^{(uv)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0,\ \mathcal{N}_{1,n}(\boldsymbol{p}_{*,j})/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0,\ \bar{\mathcal{N}}_{1,n}((\boldsymbol{p}_{*,j})^{(u)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0,$$
$$\bar{\mathcal{N}}_{2,n}(\boldsymbol{p}_{*,j})^{(uv)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0,\ \bar{M}_{n,j,0_d}/\mathcal{D}_n \to 0,\ (\alpha_n - \alpha_*)/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0$$

for any $1 \leq u, v \leq d$ and $1 \leq j \leq L$.

Since $(\alpha_n - \alpha_*)/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$, we obtain that

$$\frac{|\alpha_n - \alpha_*|}{(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to 0. \tag{34}$$

Furthermore, as $\alpha_n \not\to 0$ as $n \to \infty$, we have $1/\tanh(\alpha_n) \not\to \infty$. Given that $\bar{M}_{n,j,0_d}/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0$, it demonstrates that

$$\frac{|\sum_{i\in\mathcal{C}_j} \exp(\bar{b}_{n,i}) - \exp(\bar{b}_{*,j})|}{(\mathcal{D}_n + |\alpha_n - \alpha_*|)} = \frac{1}{\tanh(\alpha_n)} \cdot \frac{|\bar{M}_{n,j,0_d}|}{(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to 0,$$

for any $1 \leq j \leq L$. By varying the index j from 1 to L in these limits and summing them up, we achieve that

$$\frac{\sum_{j=1}^{L} |\sum_{i\in\mathcal{C}_j} \exp(\bar{b}_{n,i}) - \exp(\bar{b}_{*,j})|}{(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to 0. \tag{35}$$

Now, we first consider indices $j \in [L]$ such that its corresponding Voronoi cell $\mathcal{C}_j$ satisfying $|\mathcal{C}_j| = 1$. From the hypothesis, $\mathcal{L}_{2,n}(\boldsymbol{p}_{*,j})^{(u)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0$. Therefore, $\bar{M}_{n,j,1_u}/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0$, which leads to

$$\frac{\sum_{i\in\mathcal{C}_j} \exp(\bar{b}_{n,i})\|\Delta\boldsymbol{p}_{n,ij}\|}{(\mathcal{D}_n + |\alpha_n - \alpha_*|)} = \frac{\sum_{u=1}^{d} |\bar{M}_{n,j,1_u}|}{\tanh(\alpha_n)(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to 0.$$

The above limit indicates that

$$\frac{\sum_{j:|\mathcal{C}_j|=1}\sum_{i\in\mathcal{C}_j} \exp(\bar{b}_{n,i})\|\Delta\boldsymbol{p}_{n,ij}\|}{(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to 0. \tag{36}$$

Moving to indices $j \in [L]$ such that their corresponding Voronoi cells $\mathcal{C}_j$ satisfying $|\mathcal{C}_j| > 1$. The limit $\bar{\mathcal{L}}_{3,n}(\boldsymbol{p}_{*,j})^{(uu)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0$ leads to

$$\frac{\sum_{i\in\mathcal{C}_j} \exp(\bar{b}_{n,i})\|\Delta\boldsymbol{p}_{n,ij}\|^2}{(\mathcal{D}_n + |\alpha_n - \alpha_*|)} = \frac{\sum_{u=1}^{d} \bar{\mathcal{L}}_{3,n}(\boldsymbol{p}_{*,j})^{(uu)}}{\tanh(\alpha_n)(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to 0.$$

The above limit demonstrates that

$$\frac{\sum_{j:|\mathcal{C}_j|>1}\sum_{i\in\mathcal{C}_j} \exp(\bar{b}_{n,i})\|\Delta\boldsymbol{p}_{n,ij}\|^2}{(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to 0. \tag{37}$$

$$(\bar{B}^\top \boldsymbol{X})^{(u)}(\bar{B}^\top \boldsymbol{X})^{(v)} \exp((\bar{B}\sigma(\boldsymbol{p}_{*,j}))^\top \boldsymbol{X}),\ \exp((\bar{B}\sigma(\boldsymbol{p}_{*,j}))^\top \boldsymbol{X}) g_{G_n}(\boldsymbol{X}),$$
$$(\bar{B}^\top \boldsymbol{X})^{(u)} \exp((\bar{B}\sigma(\boldsymbol{p}_{*,j}))^\top \boldsymbol{X}) g_{G_n}(\boldsymbol{X}),\ (\bar{B}^\top \boldsymbol{X})^{(u)}(\bar{B}^\top \boldsymbol{X})^{(v)} \exp((\bar{B}\sigma(\boldsymbol{p}_{*,j}))^\top \boldsymbol{X}) g_{G_n}(\boldsymbol{X}),$$
$$[1-\tanh^2(\alpha_*)] \cdot \Big[\sum_{k'=1}^{L} \exp((\bar{B}\sigma(\boldsymbol{p}_{*,k'}))^\top \boldsymbol{X} + \bar{b}_{*,k'})\Big] \cdot g_{G_*}(\boldsymbol{X}),$$

对于任意的 $1 \leq j \leq L$ 和 $1 \leq u,\ v \leq d$。

~我们的论点是，在 $[Q_n(\boldsymbol{X}) - \sum_{i=1}^{5} R_{n,i}(\boldsymbol{X})]/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$的表述中，这些线性无关项的系数至少有一个不会像 $n \to \infty$那样趋于0。反过来说，假设这一论点不成立，那就意味着这些线性无关项的所有系数都会像 $n \to \infty$那样趋于0。因此，正如 $n \to \infty$ 所示，我们得出了这样的结论。

$$\mathcal{L}_{1,n}(\boldsymbol{p}_{*,j})/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0,\ \mathcal{L}_{2,n}(\boldsymbol{p}_{*,j})^{(u)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0,\ \bar{\mathcal{L}}_{1,n}(\boldsymbol{p}_{*,j})/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0,$$
$$\bar{\mathcal{L}}_{2,n}(\boldsymbol{p}_{*,j})^{(u)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0,\ \bar{\mathcal{L}}_{2,n}(\boldsymbol{p}_{*,j})^{(u)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0,\ \bar{\mathcal{L}}_{2,n}(\boldsymbol{p}_{*,j})^{(u)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0,$$
$$\bar{\mathcal{L}}_{3,n}(\boldsymbol{p}_{*,j})^{(uv)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0,\ \mathcal{N}_{1,n}(\boldsymbol{p}_{*,j})/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0,\ \bar{\mathcal{N}}_{1,n}((\boldsymbol{p}_{*,j})^{(u)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0,$$
$$\bar{\mathcal{N}}_{2,n}(\boldsymbol{p}_{*,j})^{(uv)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0,\ \bar{M}_{n,j,0_d}/\mathcal{D}_n \to 0,\ (\alpha_n - \alpha_*)/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0$$

对于任意的 $1 \leq u,\ v \leq d$ 以及 $1 \leq j \leq L$。

由于 $(\alpha_n - \alpha_*)/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$，我们由此得出

$$\frac{|\alpha_n - \alpha_*|}{(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to 0. \tag{34}$$

~此外，正如 $\alpha_n \not\to 0$ 以及 $n \to \infty$所指出的，我们有 $1/\tanh(\alpha_n) \not\to \infty$。考虑到$\bar{M}_{n,j,0_d}/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0$，这便证明了

$$\frac{|\sum_{i\in\mathcal{C}_j} \exp(\bar{b}_{n,i}) - \exp(\bar{b}_{*,j})|}{(\mathcal{D}_n + |\alpha_n - \alpha_*|)} = \frac{1}{\tanh(\alpha_n)} \cdot \frac{|\bar{M}_{n,j,0_d}|}{(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to 0,$$

对于任意的 $1 \leq j \leq L$。在这些范围内将索引 j 从1变化到 L 并求和，即可实现该目标

$$\frac{\sum_{j=1}^{L} |\sum_{i\in\mathcal{C}_j} \exp(\bar{b}_{n,i}) - \exp(\bar{b}_{*,j})|}{(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to 0. \tag{35}$$

现在，我们首先考虑满足条件 $|\mathcal{C}_j| = 1$的索引 $j \in [L]$ 及其对应的Voronoi单元 $\mathcal{C}_j$。根据假设，有$\mathcal{L}_{2,n}(\boldsymbol{p}_{*,j})^{(u)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0$。因此，可得$\bar{M}_{n,j,1_u}/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0$，进而推导出

$$\frac{\sum_{i\in\mathcal{C}_j} \exp(\bar{b}_{n,i})\|\Delta\boldsymbol{p}_{n,ij}\|}{(\mathcal{D}_n + |\alpha_n - \alpha_*|)} = \frac{\sum_{u=1}^{d} |\bar{M}_{n,j,1_u}|}{\tanh(\alpha_n)(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to 0.$$

上述极限表明

$$\frac{\sum_{j:|\mathcal{C}_j|=1}\sum_{i\in\mathcal{C}_j} \exp(\bar{b}_{n,i})\|\Delta\boldsymbol{p}_{n,ij}\|}{(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to 0. \tag{36}$$

转向满足 $|\mathcal{C}_j| > 1$条件的对应Voronoi单元 $\mathcal{C}_j$ 的索引 $j \in [L]$。极限$\bar{\mathcal{L}}_{3,n}(\boldsymbol{p}_{*,j})^{(uu)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0$ 将导致

$$\frac{\sum_{i\in\mathcal{C}_j} \exp(\bar{b}_{n,i})\|\Delta\boldsymbol{p}_{n,ij}\|^2}{(\mathcal{D}_n + |\alpha_n - \alpha_*|)} = \frac{\sum_{u=1}^{d} \bar{\mathcal{L}}_{3,n}(\boldsymbol{p}_{*,j})^{(uu)}}{\tanh(\alpha_n)(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to 0.$$

上述极限表明

$$\frac{\sum_{j:|\mathcal{C}_j|>1}\sum_{i\in\mathcal{C}_j} \exp(\bar{b}_{n,i})\|\Delta\boldsymbol{p}_{n,ij}\|^2}{(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to 0. \tag{37}$$

Collecting all the limits in equations (34), (35), (36), and (37), we obtain that

$$1 = \frac{\mathcal{D}_n + |\alpha_n - \alpha_*|}{\mathcal{D}_n + |\alpha_n - \alpha_*|} \to 0$$

as $n \to \infty$, which is a contradiction. As a consequence, not all of the coefficients of the linearly independent terms in $[\widetilde{Q}_n(\boldsymbol{X}) - \sum_{i=1}^{5} \widetilde{R}_{n,i}(\boldsymbol{X})]/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$ go to 0 as $n \to \infty$.

Step 3 - Application of the Fatou's lemma. We denote m_n as the maximum of the absolute values of $\mathcal{L}'_{1,n}(\boldsymbol{p}_{*,j})/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$, $\mathcal{L}_{2,n}(\boldsymbol{p}_{*,j})^{(u)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$, $\bar{\mathcal{L}}'_{1,n}(\boldsymbol{p}_{*,j})/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$, $\bar{\mathcal{L}}_{2,n}(\boldsymbol{p}_{*,j})^{(u)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$, $\bar{\mathcal{L}}_{3,n}(\boldsymbol{p}_{*,j})^{(uv)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$, $\mathcal{N}_{1,n}(\boldsymbol{p}_{*,j})/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$, $\bar{\mathcal{N}}_{1,n}((\boldsymbol{p}_{*,j})^{(u)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$, $\bar{\mathcal{N}}_{2,n}(\boldsymbol{p}_{*,j})^{(uv)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$, $\bar{M}_{n,j,0_d}/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$, and $(\alpha_n - \alpha_*)/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$ for all $1 \leq u, v \leq d$. From the result of Step 2 in the proof, we have $1/m_n \not\to \infty$ as $n \to \infty$.

Recall that $\|g_{G_n,\alpha_n} - g_{G_*,\alpha_*}\|_{L^2(\mu)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|) \to 0$ as $n \to \infty$, which indicates that $\|g_{G_n,\alpha_n} - g_{G_*,\alpha_*}\|_{L^2(\mu)}/(m_n(\mathcal{D}_n + |\alpha_n - \alpha_*|)) \to 0$. Furthermore, since the $L^2(\mu)$ norm is equivalent to the $L^1(\mu)$ norm, we have $\|g_{G_n,\alpha_n} - g_{G_*,\alpha_*}\|_{L^1(\mu)}/(m_n(\mathcal{D}_n + |\alpha_n - \alpha_*|)) \to 0$. An application of Fatou's lemma leads to

$$0 = \lim_{n\to\infty} \frac{\|g_{G_n,\alpha_n} - g_{G_*,\alpha_*}\|_{L^1(\mu)}}{m_n(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \geq \int \liminf_{n\to\infty} \frac{\|g_{G_n,\alpha_n}(\boldsymbol{X}) - g_{G_*,\alpha_*}(\boldsymbol{X})\|_1}{m_n(\mathcal{D}_n + |\alpha_n - \alpha_*|)} d\mu(\boldsymbol{X}) \geq 0.$$

It indicates that $\liminf_{n\to\infty} \frac{\|g_{G_n,\alpha_n}(\boldsymbol{X}) - g_{G_*,\alpha_*}(\boldsymbol{X})\|_1}{m_n(\mathcal{D}_n + |\alpha_n - \alpha_*|)} = 0$ for almost surely $\boldsymbol{X}$. As $n \to \infty$, we denote

$$\begin{aligned}
&\frac{\mathcal{L}'_{1,n}(\boldsymbol{p}_{*,j})}{m_n(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to \widehat{\alpha}_j, \quad \frac{\mathcal{L}_{2,n}(\boldsymbol{p}_{*,j})}{m_n(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to \beta_j, \\
&\frac{\bar{\mathcal{L}}'_{1,n}(\boldsymbol{p}_{*,j})}{m_n(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to \bar{\alpha}_j, \quad \frac{\bar{\mathcal{L}}_{2,n}(\boldsymbol{p}_{*,j})}{m_n(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to \bar{\beta}_j, \quad \frac{\bar{\mathcal{L}}_{3,n}(\boldsymbol{p}_{*,j})}{m_n(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to \bar{\gamma}_j, \\
&\frac{\bar{M}_{n,j,0_d}}{m_n(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to \tilde{\alpha}_j, \quad \frac{\mathcal{N}_{1,n}(\boldsymbol{p}_{*,j})}{m_n(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to \tilde{\beta}_j, \\
&\frac{\bar{\mathcal{N}}_{1,n}(\boldsymbol{p}_{*,j})}{m_n(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to \widehat{\beta}_j, \quad \frac{\bar{\mathcal{N}}_{2,n}(\boldsymbol{p}_{*,j})}{m_n(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to \widehat{\gamma}_j, \quad \frac{\alpha_n - \alpha_*}{m_n(\mathcal{D}_n + |\alpha_n - \alpha_*|)} \to \tau
\end{aligned}$$

for any $1 \leq j \leq L$. Here, from the definition of m_n, at least one coefficient among $\{\widehat{\alpha}_j, \beta_j, \tilde{\alpha}_j, \tilde{\beta}_j\}_{j:|\mathcal{C}_j|=1}$, $\{\bar{\alpha}_j, \bar{\beta}_j, \bar{\gamma}_j, \tilde{\alpha}_j, \widehat{\beta}_j, \widehat{\gamma}_j\}_{j:|\mathcal{C}_j|>1}$, and τ is different from 0. Then, the equation

$$\liminf_{n\to\infty} \frac{\|\widetilde{Q}_n(\boldsymbol{X})\|_1}{m_n(\mathcal{D}_n + |\alpha_n - \alpha_*|)} = \Big[\sum_{k'=1}^{L} \exp((\bar{B}\sigma(\boldsymbol{p}_{*,k'}))^\top \boldsymbol{X} + \bar{b}_{*,k'})\Big] \cdot \liminf_{n\to\infty} \frac{\|g_{G_n,\alpha_n}(\boldsymbol{X}) - g_{G_*,\alpha_*}(\boldsymbol{X})\|_1}{m_n(\mathcal{D}_n + |\alpha_n - \alpha_*|)} = 0$$

leads to

$$\begin{aligned}
&\sum_{j:|\mathcal{C}_j|=1} \exp((\bar{B}\sigma(\boldsymbol{p}_{*,j}))^\top \boldsymbol{X})(\alpha_j + \beta_j^\top(\bar{B}^\top \boldsymbol{X})) \\
&+ \sum_{j:|\mathcal{C}_j|>1} \exp((\bar{B}\sigma(\boldsymbol{p}_{*,j}))^\top \boldsymbol{X})\big[\bar{\alpha}_j + \bar{\beta}_j^\top(B^\top \boldsymbol{X}) + (\bar{B}^\top \boldsymbol{X})^\top \bar{\gamma}_j(\bar{B}^\top \boldsymbol{X})\big] \\
&- \sum_{j:|\mathcal{C}_j|=1} \exp((\bar{B}\sigma(\boldsymbol{p}_{*,j}))^\top \boldsymbol{X})(\tilde{\alpha}_j + \tilde{\beta}_j^\top(\bar{B}^\top \boldsymbol{X}))f_{G_*}(\boldsymbol{X}) \\
&- \sum_{j:|\mathcal{C}_j|>1} \exp((\bar{B}\sigma(\boldsymbol{p}_{*,j}))^\top \boldsymbol{X})\big[\tilde{\alpha}_j + \widehat{\beta}_j^\top(\bar{B}^\top \boldsymbol{X}) + (\bar{B}^\top \boldsymbol{X})^\top \widehat{\gamma}_j \bar{B}^\top \boldsymbol{X}\big] g_{G_*}(\boldsymbol{X}) \\
&+ \tau[1 - \tanh^2(\alpha_*)] \cdot \Big[\sum_{k'=1}^{L} \exp((\bar{B}\sigma(\boldsymbol{p}_{*,k'}))^\top \boldsymbol{X} + \bar{b}_{*,k'})\Big] \cdot g_{G_*}(\boldsymbol{X}) = \mathbf{0}_d
\end{aligned}$$

for almost surely $\boldsymbol{X}$. That equation only holds if and only if all the coefficients $\{\widehat{\alpha}_j, \beta_j, \tilde{\alpha}_j, \tilde{\beta}_j\}_{j:|\mathcal{C}_j|=1}$,

将方程中的所有极限值（34）、（35）、（36）以及（37）汇总起来后，我们可以得出

$$1 = \frac{\mathcal{D}_n + |\alpha_n - \alpha_*|}{\mathcal{D}_n + |\alpha_n - \alpha_*|} \to 0$$

这与 $n \to \infty$的情况相矛盾。因此，$[\widetilde{Q}_n(\boldsymbol{X}) - \sum_{i=1}^{5} \widetilde{R}_{n,\ i}(\boldsymbol{X})]/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$中那些线性无关项的系数并非都会在 $n \to \infty$时变为0。

Step 3 - Application of the Fatou's lemma. We denote m_n as the maximum of the absolute values of $\mathcal{L}'_{1,n}(\boldsymbol{p}_{*,j})/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$, $\mathcal{L}_{2,n}(\boldsymbol{p}_{*,j})^{(u)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$, $\bar{\mathcal{L}}'_{1,n}(\boldsymbol{p}_{*,j})/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$, $\bar{\mathcal{L}}_{2,n}(\boldsymbol{p}_{*,j})^{(u)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$, $\bar{\mathcal{L}}_{3,n}(\boldsymbol{p}_{*,j})^{(uv)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$, $\mathcal{N}_{1,n}(\boldsymbol{p}_{*,j})/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$, $\bar{\mathcal{N}}_{1,n}((\boldsymbol{p}_{*,j})^{(u)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$, $\bar{\mathcal{N}}_{2,n}(\boldsymbol{p}_{*,j})^{(uv)}/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$, $\bar{M}_{n,j,0_d}/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$, and $(\alpha_n - \alpha_*)/(\mathcal{D}_n + |\alpha_n - \alpha_*|)$ for all $1 \leq u, v \leq d$. From the result of Step 2 in the proof, we have $1/m_n \not\to \infty$ as $n \to \infty$.

请记住，$\|g_{G_n,\alpha_n} - g_{G_*,\alpha_*}\|_{L^2(\ \ \ \

$\{\bar{\alpha}_j, \bar{\beta}_j, \bar{\gamma}_j, \tilde{\alpha}_j, \widehat{\beta}_j, \widehat{\gamma}_j\}_{j:|\mathcal{C}_j|>1}$, and τ are 0, which is a contradiction.

It indicates that we indeed have the conclusion of the local part, namely,

$$\lim_{\varepsilon \to 0} \inf_{G \in \mathcal{G}_{L'}(\Omega):(\mathcal{D}(G,G_*)+|\alpha-\alpha_*|)\leq \varepsilon} \|g_{G,\alpha} - g_{G_*,\alpha_*}\|_{L^2(\mu)} / (\mathcal{D}(G,G_*) + |\alpha - \alpha_*|) > 0.$$

As a consequence, we obtain the conclusion of the theorem.

Proof for the identifiability property. We now demonstrate that if $g_{\bar{G},\bar{\alpha}}(\boldsymbol{X}) = g_{\bar{G}_*,\bar{\alpha}_*}(\boldsymbol{X})$ for almost every $\boldsymbol{X}$, then we obtain that $(\bar{G}, \bar{\alpha}) \equiv (\bar{G}_*, \bar{\alpha}_*)$.

To ease the presentation we denote the following notations:

$$\begin{aligned}
\mathrm{softmax}^{\text{Pretrain}}_{\bar{G}}(u) &= \frac{\exp(u)}{\sum_{k=1}^{N} \exp(\boldsymbol{X}^\top \bar{A}_k^0 \boldsymbol{X} + \bar{a}_k^0)}, \\
\mathrm{softmax}^{\text{Prompt}}_{\bar{G}}(u') &= \frac{\exp(u)}{\sum_{j'=1}^{L'} \exp((\bar{B}\sigma(\boldsymbol{p}_{j'}))^\top \boldsymbol{X} + \bar{b}_{j'})}, \\
\mathrm{softmax}^{\text{Pretrain}}_{\bar{G}_*}(u_*) &= \frac{\exp(u_*)}{\sum_{k=1}^{N} \exp(\boldsymbol{X}^\top \bar{A}_k^0 \boldsymbol{X} + \bar{a}_k^0)}, \\
\mathrm{softmax}^{\text{Prompt}}_{\bar{G}_*}(u'_*) &= \frac{\exp(u_*)}{\sum_{j'=1}^{L} \exp((\bar{B}\sigma(\boldsymbol{p}_{*,j'}))^\top \boldsymbol{X} + \bar{b}_{*,j'})}.
\end{aligned}$$

Here, u, u', u_*, and u'_* in these equations satisfy:

$$u \in \{\boldsymbol{X}^\top \bar{A}_j^0 \boldsymbol{X} + \bar{a}_j^0 : j \in [N]\},\ u' \in \{(\bar{B}\sigma(\boldsymbol{p}_{j'}))^\top \boldsymbol{X} + \bar{b}_{j'} : j' \in [L']\}$$
$$u_* \in \{\boldsymbol{X}^\top \bar{A}_j^0 \boldsymbol{X} + \bar{a}_j^0 : j \in [N]\},\ u'_* \in \{(\bar{B}\sigma(\boldsymbol{p}_{*,j'}))^\top \boldsymbol{X} + \bar{b}_{*,j'} : j' \in [L]\}.$$

The equation $g_{\bar{G},\bar{\alpha}}(\boldsymbol{X}) = g_{\bar{G}_*,\bar{\alpha}_*}(\boldsymbol{X})$ for almost every $\boldsymbol{X}$ indicates that

$$\begin{aligned}
&\sum_{j=1}^{N} \mathrm{softmax}^{\text{Pretrain}}_{\bar{G}}(\boldsymbol{X}^\top \bar{A}_j^0 \boldsymbol{X} + \bar{a}_j^0)) h(\boldsymbol{X}, \bar{\eta}_j^0) + \tanh(\bar{\alpha}) \sum_{j'=1}^{L'} \mathrm{softmax}^{\text{Prompt}}_{\bar{G}}((\bar{B}\sigma(\boldsymbol{p}_{j'}))^\top \boldsymbol{X} + \bar{b}_{j'}) \bar{C}\sigma(\boldsymbol{p}_{j'}) \\
&= \sum_{j=1}^{N} \mathrm{softmax}^{\text{Pretrain}}_{\bar{G}_*}(\boldsymbol{X}^\top \bar{A}_j^0 \boldsymbol{X} + \bar{a}_j^0)) h(\boldsymbol{X}, \bar{\eta}_j^0) + \tanh(\bar{\alpha}_*) \sum_{j'=1}^{L} \mathrm{softmax}^{\text{Prompt}}_{\bar{G}_*}((\bar{B}\sigma(\boldsymbol{p}_{*,j'}))^\top \boldsymbol{X} + \bar{b}_{*,j'}) \bar{C}\sigma(\boldsymbol{p}_{*,j'}).
\end{aligned} \tag{38}$$

The above equation only holds when $L = L'$. Furthermore, we also have that

$$\{\mathrm{softmax}^{\text{Prompt}}_{\bar{G}}((\bar{B}\sigma(\boldsymbol{p}_{j'}))^\top \boldsymbol{X} + \bar{b}_{j'}) : j' \in [L']\} = \{\mathrm{softmax}^{\text{Prompt}}_{\bar{G}_*}((\bar{B}\sigma(\boldsymbol{p}_{*,j'}))^\top \boldsymbol{X} + \bar{b}_{*,j'}) : j' \in [L]\},$$

for almost every $\boldsymbol{X}$. By relabelling the indices, we can assume that

$$\mathrm{softmax}^{\text{Prompt}}_{\bar{G}}((\bar{B}\sigma(\boldsymbol{p}_{j'}))^\top \boldsymbol{X} + \bar{b}_{j'}) = \mathrm{softmax}^{\text{Prompt}}_{\bar{G}_*}((\bar{B}\sigma(\boldsymbol{p}_{*,j'}))^\top \boldsymbol{X} + \bar{b}_{*,j'}),$$

for almost every $\boldsymbol{X}$ and any $j' \in [L]$. From the translation invariant property of the softmax function, the above equations only hold when $\bar{b}_{j'} = \bar{b}_{*,j'} + \bar{r}$ for some $\bar{r} \in \mathbb{R}$ and any $j' \in [L]$. Given these results, equation (38) leads to

$$\tanh(\bar{\alpha}) \sum_{j=1}^{L} \exp(\bar{b}_j) \exp((\bar{B}\sigma(\boldsymbol{p}_j))^\top \boldsymbol{X}) \bar{C}\sigma(\boldsymbol{p}_j) = \tanh(\bar{\alpha}_*) \sum_{j=1}^{L} \exp(\bar{b}_{*,j}) \exp((\bar{B}\sigma(\boldsymbol{p}_{*,j}))^\top \boldsymbol{X}) \bar{C}\sigma(\boldsymbol{p}_{*,j}), \tag{39}$$

for almost surely $\boldsymbol{X}$.

Now, we partition the set $\{1, 2, \ldots, L\}$ into m subsets $\bar{K}_1, \bar{K}_2, \ldots, \bar{K}_m$ where $m \leq L$, such that $\exp(\bar{b}_j) = \exp(\bar{b}_{*,j'})$ for any $j, j' \in \bar{K}_i$ and $i \in [m]$. It is clear that $\exp(\bar{b}_j) \neq \exp(\bar{b}_{*,j'})$ when j, j' belong to different subsets $\bar{K}_i$. Collecting these

– $\{\bar{\alpha}_j, \bar{\beta}_j, \bar{\gamma}_j, \tilde{\alpha}_j, \widehat{\beta}_j, \widehat{\gamma}_j\}_{j:|\mathcal{C}_j|>1}$，同时 τ的值为0，这显然存在矛盾。

这表明我们确实得到了局部部分的结论，即：

$$\lim_{\varepsilon \to 0} \inf_{G \in \mathcal{G}_{L'}(\Omega):(\mathcal{D}(G,G_*)+|\alpha-\alpha_*|)\leq \varepsilon} \|g_{G,\alpha} - g_{G_*,\alpha_*}\|_{L^2(\mu)} / (\mathcal{D}(G,G_*) + |\alpha - \alpha_*|) > 0.$$

由此，我们得出了该定理的结论。

关于可识别性性质的证明。我们现在要证明：若对于几乎所有的 $\boldsymbol{X}$，都有$g_{\bar{G}、\bar{\alpha}}(\boldsymbol{X}) = g_{\bar{G}_*、\bar{\alpha}_*}(\boldsymbol{X})$成立，那么我们能够得出$(\bar{G}、\bar{\alpha}) \equiv (\bar{G}_*、\bar{\alpha}_*)$这一结论。

为便于表述，我们定义如下符号：

$$\begin{aligned}
\mathrm{softmax}^{\text{Pretrain}}_{\bar{G}}(u) &= \frac{\exp(u)}{\sum_{k=1}^{N} \exp(\boldsymbol{X}^\top \bar{A}_k^0 \boldsymbol{X} + \bar{a}_k^0)}, \\
\mathrm{softmax}^{\text{Prompt}}_{\bar{G}}(u') &= \frac{\exp(u)}{\sum_{j'=1}^{L'} \exp((\bar{B}\sigma(\boldsymbol{p}_{j'}))^\top \boldsymbol{X} + \bar{b}_{j'})}, \\
\mathrm{softmax}^{\text{Pretrain}}_{\bar{G}_*}(u_*) &= \frac{\exp(u_*)}{\sum_{k=1}^{N} \exp(\boldsymbol{X}^\top \bar{A}_k^0 \boldsymbol{X} + \bar{a}_k^0)}, \\
\mathrm{softmax}^{\text{Prompt}}_{\bar{G}_*}(u'_*) &= \frac{\exp(u_*)}{\sum_{j'=1}^{L} \exp((\bar{B}\sigma(\boldsymbol{p}_{*,j'}))^\top \boldsymbol{X} + \bar{b}_{*,j'})}.
\end{aligned}$$

在此，这些方程式中的 u、 u'、 u_*和 u'_* 满足以下条件：

$$u \in \{\boldsymbol{X}^\top \bar{A}_j^0 \boldsymbol{X} + \bar{a}_j^0 : j \in [N]\},\ u' \in \{(\bar{B}\sigma(\boldsymbol{p}_{j'}))^\top \boldsymbol{X} + \bar{b}_{j'} : j' \in [L']\}$$
$$u_* \in \{\boldsymbol{X}^\top \bar{A}_j^0 \boldsymbol{X} + \bar{a}_j^0 : j \in [N]\},\ u'_* \in \{(\bar{B}\sigma(\boldsymbol{p}_{*,j'}))^\top \boldsymbol{X} + \bar{b}_{*,j'} : j' \in [L]\}.$$

方程 $g_{\bar{G},\bar{\alpha}}(\boldsymbol{X}) = g_{\bar{G}_*,\bar{\alpha}_*}(\boldsymbol{X})$ 对于几乎所有的 $\boldsymbol{X}$ 而言都表明

$$\begin{aligned}
&\sum_{j=1}^{N} \mathrm{softmax}^{\text{Pretrain}}_{\bar{G}}(\boldsymbol{X}^\top \bar{A}_j^0 \boldsymbol{X} + \bar{a}_j^0)) h(\boldsymbol{X}, \bar{\eta}_j^0) + \tanh(\bar{\alpha}) \sum_{j'=1}^{L'} \mathrm{softmax}^{\text{Prompt}}_{\bar{G}}((\bar{B}\sigma(\boldsymbol{p}_{j'}))^\top \boldsymbol{X} + \bar{b}_{j'}) \bar{C}\sigma(\boldsymbol{p}_{j'}) \\
&= \sum_{j=1}^{N} \mathrm{softmax}^{\text{Pretrain}}_{\bar{G}_*}(\boldsymbol{X}^\top \bar{A}_j^0 \boldsymbol{X} + \bar{a}_j^0)) h(\boldsymbol{X}, \bar{\eta}_j^0) + \tanh(\bar{\alpha}_*) \sum_{j'=1}^{L} \mathrm{softmax}^{\text{Prompt}}_{\bar{G}_*}((\bar{B}\sigma(\boldsymbol{p}_{*,j'}))^\top \boldsymbol{X} + \bar{b}_{*,j'}) \bar{C}\sigma(\boldsymbol{p}_{*,j'}).
\end{aligned} \tag{38}$$

上述方程仅在 $L = L'$时成立。此外，我们还有

$$\{\mathrm{softmax}^{\text{Prompt}}_{\bar{G}}((\bar{B}\sigma(\boldsymbol{p}_{j'}))^\top \boldsymbol{X} + \bar{b}_{j'}) : j' \in [L']\} = \{\mathrm{softmax}^{\text{Prompt}}_{\bar{G}_*}((\bar{B}\sigma(\boldsymbol{p}_{*,j'}))^\top \boldsymbol{X} + \bar{b}_{*,j'}) : j' \in [L]\},$$

对于几乎所有的 $\boldsymbol{X}$而言。通过重新标记索引，我们可以假设

$$\mathrm{softmax}^{\text{Prompt}}_{\bar{G}}((\bar{B}\sigma(\boldsymbol{p}_{j'}))^\top \boldsymbol{X} + \bar{b}_{j'}) = \mathrm{softmax}^{\text{Prompt}}_{\bar{G}_*}((\bar{B}\sigma(\boldsymbol{p}_{*,j'}))^\top \boldsymbol{X} + \bar{b}_{*,j'}),$$

对于几乎所有的 $\boldsymbol{X}$ 以及任意的 $j' \in [L]$。根据softmax函数的平移不变性，上述方程式仅在 $b_{j'} = b_{*}$的情况下成立，$j' + \bar{r}$ 对于某些 $\bar{r} \in \mathbb{R}$ 和任意的 $j' \in [L]$。基于这些结果，方程（38）可推导出

$$\tanh(\bar{\alpha}) \sum_{j=1}^{L} \exp(\bar{b}_j) \exp((\bar{B}\sigma(\boldsymbol{p}_j))^\top \boldsymbol{X}) \bar{C}\sigma(\boldsymbol{p}_j) = \tanh(\bar{\alpha}_*) \sum_{j=1}^{L} \exp(\bar{b}_{*,j}) \exp((\bar{B}\sigma(\boldsymbol{p}_{*,j}))^\top \boldsymbol{X}) \bar{C}\sigma(\boldsymbol{p}_{*,j}), \tag{39}$$

对于几乎必然成立的情况 $\boldsymbol{X}$。

xp($b_{*、 j'}$)对任意有exp(b_j) ≠ exp($b_{*、 j'}$)。将所有此类情况汇总起来，

现在，我们将集合 $\{1, 2, \ldots, L\}$ 划分为 m 个子集K_1、 K_2、.....、 K_m，其中 $m \leq L$满足指数函数exp(b_j) = ej、 $j' \in K_i$ 以及 $i \in [m]$均成立。显然，当 j、 j'属于不同的子集 K_i时，

results, equation (30) can be rewritten as follows:

$$\tanh(\bar{\alpha})\sum_{i=1}^{m}\sum_{j\in\bar{K}_i}\exp(\bar{b}_j)\exp((\bar{B}\sigma(\boldsymbol{p}_j))^\top\boldsymbol{X})\bar{C}\sigma(\boldsymbol{p}_j) = \tanh(\bar{\alpha}_*)\sum_{i=1}^{m}\sum_{j\in\bar{K}_i}\exp(\bar{b}_{*,j})\exp((\bar{B}\sigma(\boldsymbol{p}_{*,j}))^\top\boldsymbol{X})\bar{C}\sigma(\boldsymbol{p}_{*,j}),$$

for almost surely $\boldsymbol{X}$. Hence, we achieve that

$$\{((\bar{B}\sigma(\boldsymbol{p}_j))^\top,\boldsymbol{p}_j): j\in\bar{K}_i\} = \{((\bar{B}\sigma(\boldsymbol{p}_{*,j}))^\top,\boldsymbol{p}_{*,j}): j\in\bar{K}_i\} \text{ and } \tanh(\bar{\alpha})=\tanh(\bar{\alpha}_*).$$

It naturally leads to

$$\{\sigma(\boldsymbol{p}_j): j\in\bar{K}_i\} = \{\sigma(\boldsymbol{p}_{*,j}): j\in\bar{K}_i\}.$$

Without loss of generality, $\boldsymbol{p}_j = \boldsymbol{p}_{*,j}$ for all $j\in\bar{K}_i$. As a consequence, we obtain that $\bar{\alpha}=\bar{\alpha}_*$ and

$$\sum_{i=1}^{m}\sum_{j\in\bar{K}_i}\exp(\bar{b}_j)\delta_{\boldsymbol{p}_j} = \sum_{i=1}^{m}\sum_{j\in\bar{K}_i}\exp(\bar{b}_{*,j})\delta_{\boldsymbol{p}_{*,j}}.$$

It is equivalent to $(\bar{G},\bar{\alpha})\equiv(\bar{G}_*,\bar{\alpha}_*)$. We achieve the conclusion of the identifiability claim.

A.3. Proof of Proposition 4.1

Recall that the i.i.d sample $(\boldsymbol{X}_1,Y_1),(\boldsymbol{X}_2,Y_2),\ldots,(\boldsymbol{X}_n,Y_n)\in\mathbb{R}^d\times\mathbb{R}^{d'}$ are generated from the model:

$$Y_i = f_{G_*,\alpha_*}(\boldsymbol{X}_i)+\varepsilon_i,\quad i=1,2,\ldots,n,$$

where $\varepsilon_1,\ldots,\varepsilon_n$ are independent Gaussian noise variables such that $\mathbb{E}[\varepsilon_i|\boldsymbol{X}_i]=0$ and $\mathrm{Var}(\varepsilon_i|\boldsymbol{X}_i)=\sigma^2 I_{d'}$ for all $1\le i\le n$. Since $\varepsilon_i|\boldsymbol{X}_i\sim\mathcal{N}(\boldsymbol{0}_d,\sigma^2 I_{d'})$, a least square estimator $(\widehat{G}_n,\widehat{\alpha}_n)$ defined as

$$(\widehat{G}_n,\widehat{\alpha}_n) := \mathop{\arg\min}_{G\in\mathcal{G}_{L'}(\Theta),\alpha\in\Omega}\sum_{i=1}^{n}\|Y_i - f_{G,\alpha}(\boldsymbol{X}_i)\|^2,$$

is exactly a maximum likelihood estimator given by

$$(\widehat{G}_n,\widehat{\alpha}_n)\in\mathop{\arg\max}_{G\in\mathcal{G}_{L'}(\Theta),\alpha\in\Omega}\frac{1}{n}\sum_{i=1}^{n}\log(\pi(Y_i|f_{G,\alpha}(\boldsymbol{X}_i),\sigma^2 I_{d'})),$$

where $\pi(Y_i|f_{G,\alpha}(\boldsymbol{X}_i),\sigma^2 I_{d'})$ denotes the probability density function of the multivariate Gaussian distribution with mean vector $f_{G,\alpha}(\boldsymbol{X})$ and covariance matrix $\sigma^2 I_{d'}$. Furthermore, it follows from the result in (van de Geer, 2000) that

$$h(\pi(Y|f_{\widehat{G}_n,\widehat{\alpha}_n}(\boldsymbol{X}),\sigma^2 I_{d'}),\pi(Y|f_{G_*,\alpha_*}(\boldsymbol{X}),\sigma^2 I_{d'})) = \mathcal{O}_P(\sqrt{\log(n)/n}).$$

According to Pardo et al. (Pardo, 2018), we have

$$h^2(\pi(Y|\theta_1,\Sigma_1),\pi(Y|\theta_2,\Sigma_2)) = 1-\frac{\det(\Sigma_1)^{1/4}\det(\Sigma_2)^{1/4}}{\det\left(\frac{1}{2}\Sigma_1+\frac{1}{2}\Sigma_2\right)^{1/2}}\exp\left\{-\frac{1}{8}(\theta_1-\theta_2)^\top\left(\frac{1}{2}\Sigma_1+\frac{1}{2}\Sigma_2\right)^{-1}(\theta_1-\theta_2)\right\}.$$

Therefore, we deduce that

$$h^2(\pi(Y|f_{\widehat{G}_n,\widehat{\alpha}_n}(\boldsymbol{X}),\sigma^2 I_{d'}),\pi(Y|f_{G_*,\alpha_*}(\boldsymbol{X}),\sigma^2 I_{d'})) = 1-\exp\left\{-\frac{1}{8\sigma^2}\|f_{\widehat{G}_n,\widehat{\alpha}_n}(\boldsymbol{X})-f_{G_*,\alpha_*}(\boldsymbol{X})\|^2\right\}.$$

As a result, it follows that

$$1-\exp\left\{-\frac{1}{8\sigma^2}\|f_{\widehat{G}_n,\widehat{\alpha}_n}(\boldsymbol{X})-f_{G_*,\alpha_*}(\boldsymbol{X})\|^2\right\} = \mathcal{O}_P(\log(n)/n).$$

其结果，即方程（30），可改写为如下形式：

$$\tanh(\bar{\alpha})\sum_{i=1}^{m}\sum_{j\in\bar{K}_i}\exp(\bar{b}_j)\exp((\bar{B}\sigma(\boldsymbol{p}_j))^\top\boldsymbol{X})\bar{C}\sigma(\boldsymbol{p}_j) = \tanh(\bar{\alpha}_*)\sum_{i=1}^{m}\sum_{j\in\bar{K}_i}\exp(\bar{b}_{*,j})\exp((\bar{B}\sigma(\boldsymbol{p}_{*,j}))^\top\boldsymbol{X})\bar{C}\sigma(\boldsymbol{p}_{*,j}),$$

几乎可以肯定地成立 $\boldsymbol{X}$。因此，我们便能够实现这一目标。

$$\{((\bar{B}\sigma(\boldsymbol{p}_j))^\top,\boldsymbol{p}_j): j\in\bar{K}_i\} = \{((\bar{B}\sigma(\boldsymbol{p}_{*,j}))^\top,\boldsymbol{p}_{*,j}): j\in\bar{K}_i\} \text{ and } \tanh(\bar{\alpha})=\tanh(\bar{\alpha}_*).$$

这自然会引出后续结果。

$$\{\sigma(\boldsymbol{p}_j): j\in\bar{K}_i\} = \{\sigma(\boldsymbol{p}_{*,j}): j\in\bar{K}_i\}.$$

¯在不失一般性的前提下，$\boldsymbol{p}_j=\boldsymbol{p}_{*,\,j}$ 对所有的 $j\in K_i$均成立。由此我们可以得出 $\bar{\alpha}=\bar{\alpha}_*$，同时还有……

$$\sum_{i=1}^{m}\sum_{j\in\bar{K}_i}\exp(\bar{b}_j)\delta_{\boldsymbol{p}_j} = \sum_{i=1}^{m}\sum_{j\in\bar{K}_i}\exp(\bar{b}_{*,j})\delta_{\boldsymbol{p}_{*,j}}.$$

¯ ¯其等价于（G，$\bar{\alpha}$）≡（G_*，$\bar{\alpha}_*$）。这样我们就得出了关于可识别性结论的证明。

A.3. 命题的证明4.1

请回想一下 独立同分布样本$(\boldsymbol{X}_1,Y_1),(\boldsymbol{X}_2,Y_2)\ldots,(\boldsymbol{X}_n,Y_n)\in\mathbb{R}^d\times\mathbb{R}^{d'}$ 是从模型中生成的。

$$Y_i = f_{G_*,\alpha_*}(\boldsymbol{X}_i)+\varepsilon_i,\quad i=1,2,\ldots,n,$$

其中 ε_1、.....、 ε_n 为相互独立的高斯噪声变量，且满足对所有$1\le i\le n$均有 $\mathbb{E}[\varepsilon_i|\boldsymbol{X}_i]=0$ 以及$\mathrm{Var}(\varepsilon_i|\boldsymbol{X}_i)$ $=\sigma^2 I_{d'}$。由于 $\varepsilon_i|\boldsymbol{X}_i\sim\mathcal{N}(\boldsymbol{0}_d$、 $\sigma^2 I_{d'})$，因此可定义一个最小二乘估计量（$\widehat{G}_n$、 $\widehat{\alpha}_n$）

$$(\widehat{G}_n,\widehat{\alpha}_n) := \mathop{\arg\min}_{G\in\mathcal{G}_{L'}(\Theta),\alpha\in\Omega}\sum_{i=1}^{n}\|Y_i - f_{G,\alpha}(\boldsymbol{X}_i)\|^2,$$

正是由最大似然估计法给出的

$$(\widehat{G}_n,\widehat{\alpha}_n)\in\mathop{\arg\max}_{G\in\mathcal{G}_{L'}(\Theta),\alpha\in\Omega}\frac{1}{n}\sum_{i=1}^{n}\log(\pi(Y_i|f_{G,\alpha}(\boldsymbol{X}_i),\sigma^2 I_{d'})),$$

where $\pi(Y_i|f_{G,\alpha}(\boldsymbol{X}_i)$ $\sigma^2 I_{d'})$ denotes the probability density function of the multivariate Gaussian distribution with mean vector $f_{G,\alpha}(\boldsymbol{X})$ and covariance matrix $\sigma^2 I_{d'}$. Furthermore, it follows from the result in (van de Geer, 2000) that

$$h(\pi(Y|f_{\widehat{G}_n,\widehat{\alpha}_n}(\boldsymbol{X}),\sigma^2 I_{d'}),\pi(Y|f_{G_*,\alpha_*}(\boldsymbol{X}),\sigma^2 I_{d'})) = \mathcal{O}_P(\sqrt{\log(n)/n}).$$

根据Pardo等人（Pardo、2018）的研究，我们有

$$h^2(\pi(Y|\theta_1,\Sigma_1),\pi(Y|\theta_2,\Sigma_2)) = 1-\frac{\det(\Sigma_1)^{1/4}\det(\Sigma_2)^{1/4}}{\det\left(\frac{1}{2}\Sigma_1+\frac{1}{2}\Sigma_2\right)^{1/2}}\exp\left\{-\frac{1}{8}(\theta_1-\theta_2)^\top\left(\frac{1}{2}\Sigma_1+\frac{1}{2}\Sigma_2\right)^{-1}(\theta_1-\theta_2)\right\}.$$

因此，我们由此推断出

$$h^2(\pi(Y|f_{\widehat{G}_n,\widehat{\alpha}_n}(\boldsymbol{X}),\sigma^2 I_{d'}),\pi(Y|f_{G_*,\alpha_*}(\boldsymbol{X}),\sigma^2 I_{d'})) = 1-\exp\left\{-\frac{1}{8\sigma^2}\|f_{\widehat{G}_n,\widehat{\alpha}_n}(\boldsymbol{X})-f_{G_*,\alpha_*}(\boldsymbol{X})\|^2\right\}.$$

因此，可以得出如下结论。

$$1-\exp\left\{-\frac{1}{8\sigma^2}\|f_{\widehat{G}_n,\widehat{\alpha}_n}(\boldsymbol{X})-f_{G_*,\alpha_*}(\boldsymbol{X})\|^2\right\} = \mathcal{O}_P(\log(n)/n).$$

This means that with probability one, there exists some constant $C > 0$ and a natural number n_0 such that

$$1 - C\log(n_0)/n_0 \leq \exp\Big\{-\frac{1}{8\sigma^2}\|f_{\widehat{G}_n,\widehat{\alpha}_n}(\boldsymbol{X}) - f_{G_*,\alpha_*}(\boldsymbol{X})\|^2\Big\}.$$

Assume that the natural number n_0 is large enough so that $1 - C\log(n)/n \geq 1/2$ is true for all $n \geq n_0$. Then, the above inequality is equivalent to

$$\begin{aligned}\|f_{\widehat{G}_n,\widehat{\alpha}_n}(\boldsymbol{X}) - f_{G_*,\alpha_*}(\boldsymbol{X})\|^2 &\leq 8\sigma^2\log\Big(\frac{1}{1 - C\log(n)/n}\Big)\\ &= 8\sigma^2\log\Big(1 + \frac{C\log(n)/n}{1 - C\log(n)/n}\Big)\\ &\leq 8\sigma^2\cdot\frac{C\log(n)/n}{1 - C\log(n)/n}\\ &\leq 16\sigma^2 C\log(n)/n,\end{aligned}$$

which implies that

$$\|f_{\widehat{G}_n,\widehat{\alpha}_n}(\boldsymbol{X}) - f_{G_*,\alpha_*}(\boldsymbol{X})\| = \mathcal{O}_P(\sqrt{\log(n)/n}).$$

Consequently, we have

$$\|f_{\widehat{G}_n,\widehat{\alpha}_n} - f_{G_*,\alpha_*}\|_{L^2(\mu)} = \Big(\int_{\mathcal{X}}\|f_{\widehat{G}_n,\widehat{\alpha}_n}(\boldsymbol{X}) - f_{G_*,\alpha_*}(\boldsymbol{X})\|^2\mathrm{d}\mu(\boldsymbol{X})\Big)^{1/2} = \mathcal{O}_P(\sqrt{\log(n)/n}).$$

Hence, the proof is completed.

B. Additional Related Works

Theory of Mixture of Experts.

While there has been a surge of interest in using Mixture of Experts (MoE) to scale model capacity in large-scale systems, its theoretical foundations remain underdeveloped. From a probabilistic perspective, a series of works on Gaussian MoE (Ho et al., 2022; Nguyen et al., 2023; 2024c; Yan et al., 2025) have analyzed the convergence behavior of maximum likelihood estimation, providing insights into optimal expert structures for MoE training. In a related line, (Nguyen et al., 2024f) studied a regression framework where the true regression function follows an MoE form, showing that sigmoid gating yields faster convergence rates for expert estimation compared to softmax gating. Theoretical analyses of MoE for classification tasks were further developed in (Chen et al., 2022; Nguyen et al., 2024d), where each expert was modeled as a classifier.

In the context of deep learning, Han et al. (2024) and Nguyen et al. (2024e) incorporated MoE and its hierarchical variants into multi-modal learning, training each expert to specialize in processing specific data modalities such as time series, text, or images. Pham et al. (2024) proposed a competition-based routing strategy to effectively train sparse MoE models for language modeling, supported by theoretical justification. Additional theoretical insights into MoE have been explored for domain adaptation (Nguyen et al., 2025) and continual learning (Li et al., 2024b). However, to the best of our knowledge, the theoretical analysis of MoE in the context of zero-initialized attention remains unexplored.

C. Additional Experimental Details

C.1. Datasets Description

We first fine-tune LLaMA-Adapter with all prefix-tuning settings on Alpaca dataset, which include 52K instruction-following data for training. Then, we evaluate our experiments on LLM benchmarks, including AI2 Reasoning Challenge (ARC), HellaSwag, MMLU, and TruthfulQA. The statistics of 4 datasets about testing subset are summarized in detail in Table 4.

- ARC dataset is a multiple-choice question-answering dataset which contain science questions in exams from grade 3 to grade 9. It has two types: Easy and Challenge. In this paper, we report performance on both types and average accuracy.

这意味着 即，以概率1存在某个常数 $C > 0$ 以及一个自然数 n_0，使得…… t

$$1 - C\log(n_0)/n_0 \leq \exp\Big\{-\frac{1}{8\sigma^2}\|f_{\widehat{G}_n,\widehat{\alpha}_n}(\boldsymbol{X}) - f_{G_*,\alpha_*}(\boldsymbol{X})\|^2\Big\}.$$

假设自然数 n_0 足够大，以至于对所有的 $n \geq n_0$而言都有 $1 - C\log(n)/n \geq 1/2$ 成立。此时，上述不等式等价于

$$\begin{aligned}\|f_{\widehat{G}_n,\widehat{\alpha}_n}(\boldsymbol{X}) - f_{G_*,\alpha_*}(\boldsymbol{X})\|^2 &\leq 8\sigma^2\log\Big(\frac{1}{1 - C\log(n)/n}\Big)\\ &= 8\sigma^2\log\Big(1 + \frac{C\log(n)/n}{1 - C\log(n)/n}\Big)\\ &\leq 8\sigma^2\cdot\frac{C\log(n)/n}{1 - C\log(n)/n}\\ &\leq 16\sigma^2 C\log(n)/n,\end{aligned}$$

这进而表明

$$\|f_{\widehat{G}_n,\widehat{\alpha}_n}(\boldsymbol{X}) - f_{G_*,\alpha_*}(\boldsymbol{X})\| = \mathcal{O}_P(\sqrt{\log(n)/n}).$$

因此，我们得到了

$$\|f_{\widehat{G}_n,\widehat{\alpha}_n} - f_{G_*,\alpha_*}\|_{L^2(\mu)} = \Big(\int_{\mathcal{X}}\|f_{\widehat{G}_n,\widehat{\alpha}_n}(\boldsymbol{X}) - f_{G_*,\alpha_*}(\boldsymbol{X})\|^2\mathrm{d}\mu(\boldsymbol{X})\Big)^{1/2} = \mathcal{O}_P(\sqrt{\log(n)/n}).$$

至此，该证明便已完成。

B. 其他相关研究

专家混合模型理论。

While there has been a surge of interest in using Mixture of Experts (MoE) to scale model capacity in large-scale systems, its theoretical foundations remain underdeveloped. From a probabilistic perspective, a series of works on Gaussian MoE (Ho et al., 2022; Nguyen et al., 2023; 2024c; Yan et al., 2025) have analyzed the convergence behavior of maximum likelihood estimation, providing insights into optimal expert structures for MoE training. In a related line, (Nguyen et al., 2024f) studied a regression framework where the true regression function follows an MoE form, showing that sigmoid gating yields faster convergence rates for expert estimation compared to softmax gating. Theoretical analyses of MoE for classification tasks were further developed in (Chen et al., 2022; Nguyen et al., 2024d), where each expert was modeled as a classifier.

In the context of deep learning, Han et al. (2024) and Nguyen et al. (2024e) incorporated MoE and its hierarchical variants into multi-modal learning, training each expert to specialize in processing specific data modalities such as time series, text, or images. Pham et al. (2024) proposed a competition-based routing strategy to effectively train sparse MoE models for language modeling, supported by theoretical justification. Additional theoretical insights into MoE have been explored for domain adaptation (Nguyen et al., 2025) and continual learning (Li et al., 2024b). However, to the best of our knowledge, the theoretical analysis of MoE in the context of zero-initialized attention remains unexplored.

C. 其他实验细节

C.1. 数据集说明

我们首先在Alpaca数据集上使用所有前缀微调参数对LLaMA-Adapter进行训练，该数据集包含用于训练的52K条指令遵循数据。随后，我们在各类大语言模型基准测试上评估我们的实验结果，这些基准测试包括AI2推理挑战赛（ARC）、HellaSwag、MMLU测试以及TruthfulQA测试。关于这4个数据集在测试子集上的统计信息已在表格中详细汇总4。

- ARC数据集是一组用于选择题答题测试的数据集，其中包含从三年级到九年级考试中的科学题目。该数据集分为简单型和挑战型两种类型。在本文中，我们报告了这两种类型下的模型表现以及整体准确率。

- HellaSwag provides multiple-choice questions to evaluate commonsense NLI of LLMs. Given a paragraph which is incomplete, the model need to find the suitable option to complete it.
- MMLU is a benchmark that covers 57 subjects across STEM, the social sciences, humanities, and more through multiple-choice questions. This dataset test the model on both world knowledge and problem solving ability. Its subjects range from traditional areas such as mathematics to more specialized areas like law and ethics.
- TruthfulQA measure whether the LLMs is truthful in generating answers given questions. It comprises 817 questions with each question has two types, generative questions and multiple-choice questions. In this paper, we evaluate all settings on multiple-choice questions.

Table 4. Statistics of 4 LLM benchmarks about the testing subset.

	ARC (Easy)	ARC (Challenge)	HellaSwag	MMLU	TruthfulQA
Testing	2376	1172	10042	14042	817

C.2. Prompt Templates

We also provide the prompt templates that we use for all settings to evaluate on ARC, MMLU, and TruthfulQA benchmarks in Figure 4, 5, and 6, respectively. These templates are based on the fact that Alpaca Fully Fine-tuning and LLaMA-Adapter both use this prompt template structure in training and we custom a little bit for each dataset.

ARC Prompt Template

```
Below is an instruction that describes a task.  Write a response that
appropriately completes the multiple-choice question.

### Instruction:

Question:  <question>

Options:

  - Option 1
  - Option 2
  - Option 3
  - Option 4
  - ...

### Response:
```

Figure 4. Prompt templates for ARC dataset

MMLU Prompt Template

```
Below is an instruction that describes a task.  Write a response that
appropriately completes the multiple-choice question about {task}.

### Instruction:

Question:  <question>

Options:

  - Option 1
  - Option 2
  - Option 3
  - Option 4
  - ...

### Response:
```

Figure 5. Prompt template for MMLU dataset.

• HellaSwag提供了大量选择题，用于评估大语言模型的常识类自然语言推理能力。给定一段不完整的文本后，模型需要找出合适的选项来补全它。

• MMLU测试是一种通过选择题形式，涵盖科学、技术、工程与数学领域、社会科学、人文科学等共57个学科的基准测试。该数据集用于评估模型在世界知识以及问题解决能力方面的表现，其测试主题既包括数学这类传统学科，也涉及法律和伦理学这类更为专业的领域。

• TruthfulQA用于检测大语言模型在回答问题时是否保持真实性。该测试包含817道题目，每道题目分为两种类型：生成式问题和选择题。在本文中，我们针对所有配置仅对选择题进行了评估。

表4。 4个LLM基准测试在测试子集上的统计数据。

	ARC（简单版）	ARC测试（挑战）	HellaSwag	MMLU	TruthfulQA
Testing	2376	1172	10042	14042	817

C.2. 提示词模板

我们还在图中分别提供了用于在ARC测试、MMLU测试以及TruthfulQA测试中评估所有设置所使用的提示词模板，对应编号为 4、5以及6。这些模板的设计基于这样一个事实：Alpaca模型经过全微调后以及LLaMA-Adapter在训练时都会采用这种提示词模板结构，同时我们会针对不同的数据集稍作定制。

ARC提示词模板

```
以下是一条用于描述具体任务的指令。                    编写一个响应内容
该回应需准确填补选择题中的空白内容。

### 指令:

问题:      <问题>

选项:
  - 选项 1
  - 选项 2
  - 选项 3
  - 选项 4
  - ……

### 回复:
```

图*4.* ARC数据集的提示词模板

MMLU提示词模板

```
以下是用于描述某项任务的指令。                        请撰写一段回应,
能够恰当地完成关于 {任务}的多项选择题。

### 指令:

问题:      <问题>

选项:
  - 选项 1
  - 选项 2
  - 选项 3
  - 第4种选项
  - ……

### 回答:
```

图*5*。*MMLU*数据集的提示词模板。

Table 5. Visualize some question-answering outputs with different prompt-tuning methods.

Question: Which best explains what scientists are referring to when they use the term conservation? **(a)** nonliving parts of the environment. **(b)** living organisms in the environment. **(c)** health of the living organisms in the environment. **(d)** protection, management, and renewal of resources.	**Question:** A pitcher throws a 0.15 kg baseball at 43 40 m/s towards the catcher. What is the momentum of the baseball while moving at 40 m/s? **(a)** 0.025 kg x m/s. **(b)** 3.8 kg x m/s. **(c)** 6.0 kg x m/s. **(d)** 270 kg x m/s.	**Question:** Part of the east coast of South America and the west coast of Africa have matching fossils within the same series of rock layers. This provides evidence that these two continents were once: **(a)** separated by a much larger ocean. **(b)** joined together as one landmass. **(c)** located near the North Pole. **(d)** in a different hemisphere.
Ground Truth: d.	**Ground Truth:** c.	**Ground Truth:** b.
Non-Linear prompt: The term conservation refers to the protection, management, and renewal of resources.	**Non-Linear prompt:** 6.0 kg x m/s	**Non-Linear prompt:** joined together as one landmass
Linear prompt: protection, management, and renewal of resources.	**Linear prompt:** 3.8 kg x m/s	**Linear prompt:** The two continents were once joined together as one landmass.
Random-Init: protection, management, and renewal of resources.	**Random-Init:** The momentum of the baseball while moving at 40 m/s is 0.025 kg x m/s.	**Random-Init:** located near the North Pole.

TruthfulQA Prompt Template

```
Below is an instruction that describes a task.  Write a response that
appropriately completes the multiple-choice question about {task}.

### Instruction:
Interpret each multiple-choice question literally, and as a multiple-choice
question about the real world; check carefully to see if the question misses
the concept or not and research each option and only pick one most suitable
option, without falling prey to any common mythss.

Question:  <question>

Options:

 - Option 1
 - Option 2
 - Option 3
 - Option 4
 - ...

### Response:
```

Figure 6. Prompt templates for TruthfulQA datasets.

C.3. Visualize question-answering

We provide in Table 5 illustrations on output of LLaMA-Adapter trained with different prompt-tuning strategy.

表*5.*展示了采用不同提示词调优方法后得到的部分问答系统输出结果。

问题: 哪一种方法最能解释科学家们所提到的内容背后的含义。 他们会使用这个术语。 资源保护? **(a)** 环境中的非生物部分。 该环境。 **(b)** 环境中的生物体。 该环境。 **(c)** 其中生物体的健康状况 包括在其中的生物体健康状况 环境。 **(d)** 此外还包括资源的保护、管理以及更新工作。 资源的保护、管理以及更新。 资源的保护、管理及更新。	**问题:** 一名投手抛出一颗质量为0.15公斤的棒球，速度为43 40米/秒，朝 接球手方向抛去。 其速度是多少? 棒球的动量 在以40米/秒的速度运动时的动量? **(a)** 0.025千克·米/秒。 **(b)** 3.8千克·米/秒。 **(c)** 6.0千克·米/秒。 **(d)** 270千克·米/秒。	**问题:** 其中一部分存在相同的化石。 美国与西方世界 非洲海岸地区 匹配其中的化石 同一系列的岩石层。 这便提供了 层。 即这两块大陆曾经是相连的。 这两块大陆曾经处于同一位置。 (a) **它们曾被较大的距离分隔开**。 更大的海洋。 **(b)** 这些区域相互连接形成一块陆地。 **(c)** 它们位于靠近北极的地方。 **(d)** 不同的区域中。 半球。
真实值: d.	**真实值:** c.	**真实值:** b.
非线性提示词: “资源保护”这一术语指的是……保护、管理以及更新。 资源。	**非线性提示词:** 6.0 千克·米/秒	**非线性提示词:** 这些陆地连为一体形成了一个巨大的大陆。
线性提示词: 包括保护、管理以及资源的更新。 资源。	**线性提示词:** 3.8 千克·米/秒	**线性提示词:** 这两块大陆曾经连为一体。 它们曾共同构成一片巨大的陆地。
随机初始化提示: 资源的保护、管理以及更新。 资源。	**随机初始化提示:** 棒球运动时的动量 随机策略：位于北极附近的区域。	**随机初始化提示:** TruthfulQA提示词模板:

以下是用于描述具体任务的指令。

```
下方是一则描述任务的指令。                                    请撰写一段回复，
能够恰当地完成关于 {task}的多项选择题。

### 指导要求:
请严格按照字面意思理解每一道选择题，并将其视为一道选择题来作答。
并对每个选项进行逐一分析，最终仅选出最合适的答案。
先考虑是否采用该概念，逐一研究各种选项，最终仅选择最合适的一个。
选项，且不会陷入任何常见的误区。

问题:          <问题>

选项:
 - 选项1
 - 选项2
 - 第3项选项
 - 第4项选项
 - …

### 回答:
```

图*6*。*TruthfulQA*数据集的提示词模板。

C.3. 可视化问答过程

我们在此提供相关内容。 表 5展示了使用不同提示词优化策略训练得到的LLaMA-Adapter的输出结果 y.